Description of identification numbers

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1 IDN - Identification numbers

1.1 Introduction

All parameters are assigned to IDNs.

Every parameter consists of elements. Elements are used to supply additional information, which is required to allow the display and input of data and the use of universal routines by means of the control terminal. This additional information is necessary for handling arbitrary slave-related data. With this information, anonymous parameter can be interpreted by the user interface.

The parameter structure shall be as shown in Table 1. In a parameter, elements 1, 3, and 7 are mandatory and shall always be present. Elements 2, 4, 5, and 6 are optional and may be supported depending on configuration. Elements 5 and 6 are mandatory for cycle time parameters only. The appropriate elements of the parameters shall be selected via the service channel control bits.

Table 1: Parameter structure

element No.	Description	Requirement
1	IDN	mandatory
2	Name	optional
3	Attribute	mandatory
4	Unit	optional
5	Minimum input value	optional
6	Maximum input value	optional
7	Operation data	mandatory
NOTE Elements 5 and 6 are mandatory for cycle time parameters (S-0-1050.x.10, S-0-1002).		

1.2 Element 1: structure of IDN

If written and read via the service channels, the appropriate data shall be addressed by means of the IDNs. Beyond that, operation data within the configurable part of the connections of the AT and MDT shall be as defined by means of the IDNs.

IDN numbering shall have a range of 2^{32}, which are subdivided as follow:

1. Two ranges shall be available for standard IDNs and product-specific IDNs. Product-specific IDNs are out of the scope of standardization.
2. Every range shall be subdivided into eight parameter sets.
3. Each set shall thus have up to 4 095 data block numbers or function groups.
4. Each IDN may have up to 256 structure instances and up to 256 structure elements.

IDNs shall be transferred in telegrams as 32-bit binary numbers.

Table 2 describe the structure of IDNs.

Table 2: IDN structure

Bit No.	Value	Description	Comments
31-24		Structure instance (SI)	
	0-255	Number of structure instance (SI)	
23-16		Structure element (SE)	
	0-127 (bit 23 = 0)	Standard SE	determined by SERCOS (bit 15 = 0)
	128-255 (bit 23 = 1)	Product specific SE	determined by manufacturer (bit 15 = 0)
15		Standard or Product specific IDN (S or P)	

	0	Standard IDN (S-0-nnnn)	SE (0-127), SI and data block number determined by SERCOS
	1	Product specific IDN (P-0-nnnn)	Bits 31 to 0 determined by manufacturer
14-12		Parameter sets	
	0-7	Parameter set	SERCOS specifies IDNs with parameter set 0 only.
11-0		Data block or Function group	
	0-4095	Data block number (if SI = SE = 0); Function group (if SI or SE is not 0)	

The notation of an IDN shall be as follows:

- If no SI or SE exist it shall be: S/P - Parameter Set - Data block number (e.g. S-0-1002)
- If SI and SE exist it shall be: S/P - Parameter Set - Function group . SI . SE (e.g. S-0-1300.0.2)

1.3 Element 2: structure of name

The name shall consist of 64 octets maximum. It shall have two length specifications of two octets each and a character string of maximum 60 UTF8 characters (60 octets). Octets 1 and 2 of the name shall specify the current text length in octets. Octets 3 and 4 of the name shall indicate the maximum number of characters available for text in a slave if the name is changeable. Text longer than that specified by these octets cannot be stored in the slaves. Length specifications of the initial four octets shall be coded for hexadecimal digits.

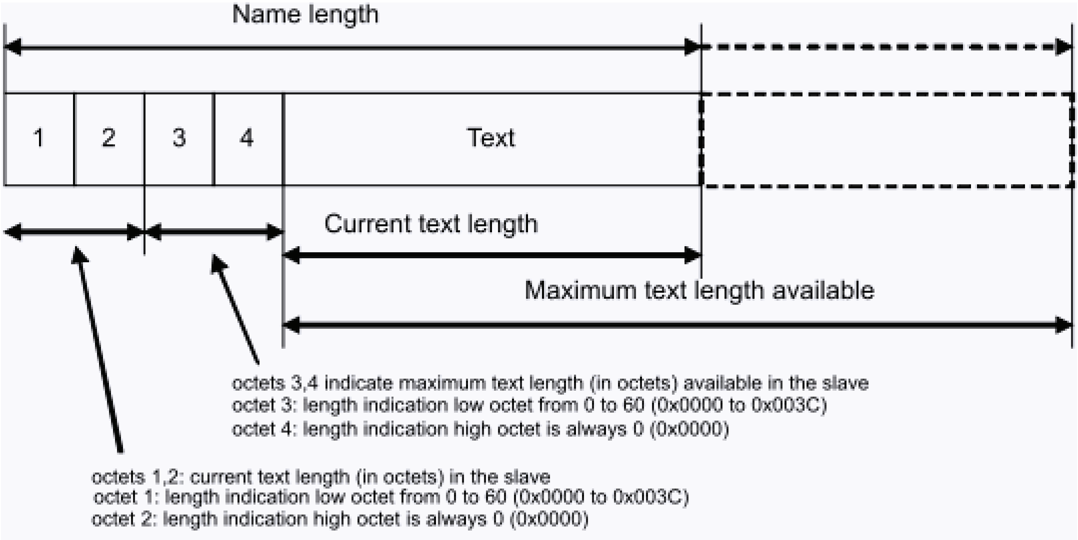


Figure 1: Name structure

If the current text has a length of 0, only the two length indications shall be transmitted. Octets 1 and 2 shall then contain the value 0.

Reading: In order to complete a read command in the service channel, the master shall require octets 1 and 2. Octets 3 and 4 shall only be read by the master to prevent writing text which is too long.

Writing: When writing a name, the master shall set octets 1 and 2 according to the current text length. The text shall not be longer than specified in octets 3 and 4. During writing the slave shall ignore octets 3 and 4 and insert its available length during reading.

1.4 Element 3: structure of attribute

Every parameter shall have an attribute which allows for an intelligible representation of various operation data by means of universal routines. The attribute shall contain all information which is needed to display operation data intelligibly. The attribute makes it possible to convert the transferred operation data into intelligible display data and vice versa. The conversion shall have no impact on the data itself. If data needs to be scaled, specific scaling parameters shall be supplied. Every scaling modification needs a change in the attributes of the affected data. It is recommended to write protect the attribute. See Table 3.

Table 3: Element 3: attribute

Bit No.	Value	Description	Comments
31		Reserved	
30-28		Write protection: The write protection concerns only access over the SVC or the S/IP (NRT channel). Cyclically configured data (RTD) shall be write-	

		protected via the SVC and S/IP in the slave. For the adjusting of the write protection the bits 30-28 are fixed in the attribute.	
30		SCP: Write protected in CP4 non SCP: Write protected in operating level (OL)	
	0	Operation data is writable	
	1	Operation data is write protected	
29		SCP: Write protected in CP3 non SCP: Write protected in parameterization level (PL)	
	0	Operation data is writable	
	1	Operation data is write protected	
28		SCP: Write protected in CP2 non SCP: reserved	
	0	Operation data is writable	
	1	Operation data is write protected	
27-24		Decimal point: This is a additional display information. Places after the decimal point indicates the position of the decimal point for the display and input of appropriate operation data. Decimal point is used for displaying of signed and unsigned decimal. For all other display formats the decimal point shall be = 0	
	0000-1111	No place to 15 places after decimal point (maximum)	
23		Reserved	
22-20		Data type and display format: Data type and display format are used to convert the operation data and the minimum and maximum input value to the correct display format.	
	000	Data type: Binary number Display format: Binary	
	001	Data type: Unsigned integer Display format: Unsigned decimal	
	010	Data type: Integer Display format: Signed decimal	
	011	Data type: Unsigned integer Display format: Hexadecimal	
	100	Data type: Extended character set Display format: Text (UTF8)	
	101	Data type: Unsigned integer Display format: IDN	
	110	Data type: Floating-point number	

		Display format: Signed decimal with exponent (float) Single or double precision, according to ANSI/IEEE 752-1995	
	111	Data type: Date and time Display format: according to IEC 61588 4 octets date & 4 octets time in nano seconds since 1.1.1970	
19		Parameter is:	
	0	not a procedure command	
	1	a procedure command	
18-16		Data length: Data length is required so that the control unit is able to complete service channel data transfers correctly	
	000	Reserved	
	001	Operation data is two octets long	
	010	Operation data is four octets long	
	011	Operation data is eight octets long	
	100	Variable length with one-octet data strings	
	101	Variable length with two-octet data strings	
	110	Variable length with four-octet data strings	
	111	Variable length with eight-octet data strings	
15-0		Conversion factor: the conversion factor is an unsigned integer used to convert numeric data to display format. The conversion factor shall be set to a value of 1 when it is not needed for data display (e.g., for binary number, character string or floating-point number etc.)	

The display formats and data length shall have any of the valid combinations (“yes” marked) in Table 4.

Table 4: Valid combinations of the display formats

Data length	Binary	Unsigned decimal	Signed decimal	Hex	Text	IDN	Float	Time
2 octet	yes	yes	yes	yes				
4 octet	yes	yes	yes	yes		yes	yes	
8 octet	yes	yes	yes	yes			yes	yes
1 octet list		yes		yes	yes			
2 octet list	yes	yes	yes	yes				
4 octet list	yes	yes	yes	yes		yes	yes	
8 octet list	yes	yes	yes	yes			yes	yes

1.5 Element 4: structure of unit

The unit shall consist of 16 octets maximum. It shall have two length specifications of two octets each and an UTF8 character string of 12 characters maximum (12 octets). Octets 1 and 2 of the unit shall specify the length in the current text in octets. Octets 3 and 4 of the unit shall indicate the maximum number of characters available for text in a slave if it is changeable. Text longer than that specified by these octets may not be stored in the slaves. Length specifications of the initial four octets shall be coded for hexadecimal digits. Parameter shall not have any unit if the data type is either a binary number or a character string. See Figure 2

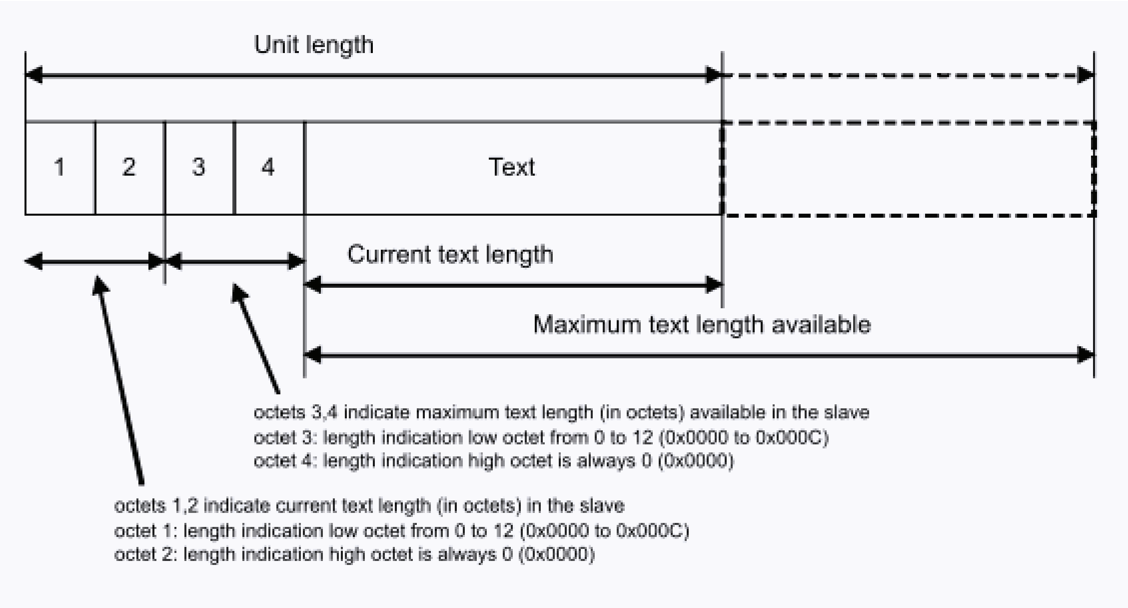


Figure 2: Unit structure

If the current text has the length 0, only the two length indications shall be transmitted. Octets 1 and 2 shall then contain the value 0.

Reading: In order to complete a read command in the service channel, octets 1 and 2 shall be required by the master. Octets 3 and 4 shall only be read by the master to prevent writing text which is too long.

Writing: When writing a unit, the master shall set octets 1 and 2 according to the current text length. The text shall not be longer than specified in octets 3 and 4. During writing the slave shall ignore octets 3 and 4 and insert its available length during reading.

1.6 Element 5: structure of minimum value

The minimum input value shall be the smallest numerical value for the operation data which the slave is able to process and have the same length as operation data.

If, in a write request of the operation data, the value is lower than the minimum input value, the operation data shall not be changed.

The following data types do not have minimum input value:

- binary number
- character string
- IDN.

The minimum input value shall be displayed like the operation data.

It is recommended to write protect the minimum input value.

1.7 Element 6: structure of maximum value

The maximum input value shall be the largest numerical value for the operation data which the slave is able to process and has the same length as operation data.

If, in a write request of the operation data, the maximum input value is exceeded, the operation data shall not be changed.

If the operation data is a binary number, then the supported bits are set in the maximum input value. The master therefore recognizes which bits are supported by the slave in this parameter.

The operation data shall have no maximum input value if a character string or an IDN is used.

The maximum input value shall be displayed like the operation data.

It is recommended to write protect the maximum input value.

1.8 Element 7: structure of operation data

The operation data shall have any one of following lengths:

- fixed length with two octets;
- fixed length with four octets;
- fixed length with eight octets;
- variable length up to 65 532 octets;

Length specifications for the variable length only shall be coded in the initial four octets for hexadecimal digits.

Structure of operation data with variable length shall be as shown in Figure 3

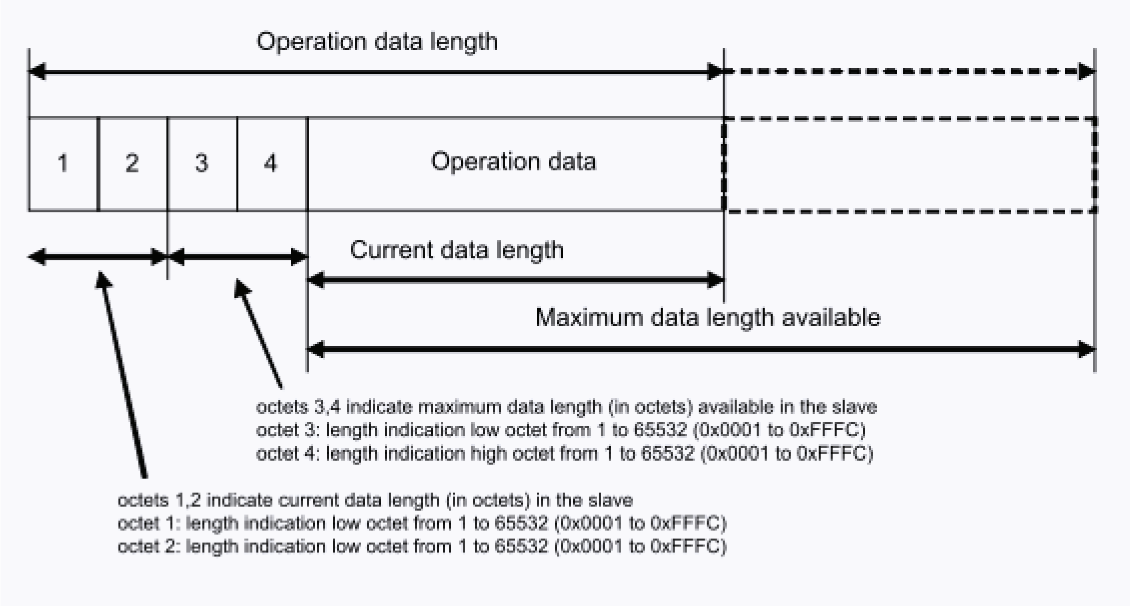


Figure 3: Structure of operation data with variable length

Operation data with variable length shall consist of length indicators in the initial four octets, followed by the programmed operation data.

Files or tables shall be loaded from the control unit to the slaves or vice versa by means of the transfer of operation data with variable length (e.g., the IDN-list of all operation data in a slave).

If the operation data has the length 0, only the two length indications shall be transmitted. Octets 1 and 2 shall then contain the value 0.

Reading: In order to complete a read command in the service channel correctly, the master shall require octets 1 and 2. Octets 3 and 4 shall only be read by the master to prevent writing operation data which is too long.

Writing: When writing operation data, the master shall set octets 1 and 2 according to the current data length. The data shall not be longer than specified in octets 3 and 4. During writing, the slave shall ignore octets 3 and 4 and insert its available length during reading.

Sorting of the IDNs in an IDN list: The IDNs in an sorted IDN list e.g. S-0-0017 shall be sorted in the following order,

1. Function group or data block number,
2. Structure instance,
3. Structure element.

Table 5 shows the structure of an IDN list and the sorting of the IDNs.

Table 5: Example of the structure of an IDN-list

1. 2.	3. 4.	<-- Length of list -->						
1C 00	1C 00							
		S-0-1000	S-0-1300.0.2	S-0-1300.0.4	S-0-1502.0.1	S-0-1502.0.2	S-0-1502.1.1	S-0-1502.1.2
		list element #0	list element #1	list element #2	list element #3	list element #4	list element #5	list element #6
	Octets 3 and 4 indicates the maximum length for operation data available in the slave; example length = 28 octets 0x001C							
Octets 1 and 2 indicates the current length for programmed operation data; example length = 28 octets 0x001C								

1.9 Structure of Data status

The content of “data status” shall be related to the entire parameter. “Data status” shall contain conditions which change dynamically. When opening the service channel via an IDN, the current data status shall be transferred automatically to the master. This enables the control unit to respond to procedure command acknowledgments during transmission of a procedure command. The data status (procedure command acknowledgment) shall be reset by the device during every renewed initialization.

Bits 3-0 shall only be present for procedure commands (procedure command acknowledgment).

Changes in the procedure command acknowledgment by:

- bit 3-0: procedure command executed correctly (0111-->0011 = positive acknowledgment), or
- bit 3-0: procedure command execution is impossible (0111-->1111 = negative acknowledgment)

shall lead to setting the procedure command change bit in the device status.

Bit 8 shall be set by the device if the parameter is recognized as invalid, such as if the data memory is checked for data loss and a checksum error is set.

The structure of the data status is shown in Table 6.

Table 6: Data status structure

Bit No.	Value	Description	Comments
15-9		Reserved	

8		Validity of operation data	
	0	Operation data is valid	
	1	Operation data is invalid	
7-4		Reserved	
3-0		Procedure command acknowledgment	
	0000	Procedure command not activated	
	0001	Procedure command is set and the execution is interrupted	
	0011	Procedure command executed correctly (positive acknowledgment), procedure command change bit in device status is set to 1	
	0111	Procedure command not yet executed (processing)	
	1111	Error, procedure command execution is impossible (negative acknowledgment), procedure command change bit in device status is set to 1	
		All other codings are reserved	

S-0-0011 Class 1 diagnostic (C1D)

- **Description**

Drive shut-down error:

A drive error situation of C1D leads to the following:

1. A best case deceleration followed by torque disable at nmin.
2. The drive shut-down error bit for C1D is set to '1' in the drive status (bit 13). The error bit is reset to '0' by the drive only when no errors of C1D exists and after the procedure command 'reset class 1 diagnostic' (S-0-0099) has been received by the drive via the service channel.

- SERCOS II: communication error sets bit 12 of C1D to 1
- SERCOS III: communication error sets bit 15 of device status (S-Dev, S-0-1045) to 1. Bit 12 of C1D is not used.

Structure of C1D:

Bit No.	Value	Description	Comments
15		manufacturer-specific error	see S-0-0129
	0	no error	
	1	error	

14		reserved	
13		over travel limit is exceeded	see S-0-0049, S-0-0050
	0	no error	
	1	error	
12		communication error	SERCOS II see S-0-0014 SERCOS III: not used
	0	no error	
	1	error	
11		excessive position deviation	see S-0-0159 or S-0-0391
	0	no error	
	1	error	
10		power supply phase error	
	0	no error	
	1	error	
9		under voltage error	less mains voltage or DC Bus voltage, the limit is defined by the manufacturer
	0	no error	
	1	error	
8		over voltage error	high mains voltage or DC Bus voltage, the limit is defined by the manufacturer
	0	no error	
	1	error	
7		over current error	
	0	no error	
	1	error	
6		error in the „commutation“ system	
	0	no error	
	1	error	
5		feedback error	
	0	no error	
	1	error	
4		control voltage error	
	0	no error	
	1	error	

3		cooling error shut-down	see S-0-0205
	0	no error	
	1	error	
2		motor over temperature shut-down	see S-0-0204
	0	no error	
	1	error	
1		amplifier over temperature shut-down	see S-0-0203
	0	no error	
	1	error	
0		overload shut-down	see S-0-0114
	0	no error	
	1	error	

Name		Class 1 diagnostic (C1D)
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0014 Interface Status

- **Description**

If an error is set in the interface status, then the communication error is set in C1D (S-0-0011). The setting of bits 2–0 does not signify an error. If there are no communication errors present, the current communication phase is contained in the interface status. If an error has occurred, the error and the current CP will be stored simultaneously.

The error bits of the interface status are reset to '0' by the sub-device only if no errors of interface status exists and after the procedure command 'reset class 1 diagnostic' (S-0-0099) has been received by the sub-device via the service channel.

Structure of interface status			
Bit No.	**Value**	**Description**	**Comments**

15-14		reserved	
13		phase switching without CPS bit	SERCOS III only
	0	no error	
	1	error	CPS is not set to 1
12		timeout phase-switching	SERCOS III only
	0	no error	
	1	error	timeout occurred
11		IPO-SYNC error	SERCOS II only
	0	no error	
	1	error	
10		devices with the same address in the ring	SERCOS II only
	0	no error	
	1	error	
9		switching to uninitialized operation mode	SERCOS II only
	0	no error	
	1	error	
8		phase switching without ready acknowledge	S-0-0127 S-0-0128
	0	no error	
	1	error	
7		error during phase downshift	not to CP0
	0	no error	
	1	error	
6		error during phase upshift	invalid sequence
	0	no error	
	1	error	
5		invalid phase	SERCOS II: CP > 6 SERCOS III: CP > 4
	0	no error	
	1	error	
4		MDT failure	SERCOS II only S-0-0029
	0	no error	
	1	error	
3		MST failure	SERCOS II: S-0-0028

			SERCOS III: S-0-1003
	0	no error	
	1	error	
2-0		Communication phase	see MST_Info_Field
	000 ... 100	CP0 ... CP4	
	101 110	CP5 CP6	SERCOS II only
	111	Reserved	

Name		Interface status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0015 Telegram type

- **Description**

The telegram type allows selection between standard telegrams and application telegrams (bits 2-0). The position feedback 1 or 2 is programmed in bit 3. The extension of the service channel INFO is programmed in the telegram type in bit 8 - 11.

- SERCOS II:
  - Service channel: The extended service channel INFO is only active in CP3 and CP4. In CP2 the service channel INFO stays unchanged (2 octets).
  - Application telegram: The telegram configuration uses S-0-0016 and S-0-0024
  - Standard telegrams 0 to 6: The IDNs of drive control (S-0-0134) and drive status (S-0-0135) are not used. These IDNs are fix configured.
- SERCOS III V1.1.1:
  - Service channel: The service channel INFO is always 4 octets long.
  - The connection data are configured with S-0-1050.x.1

Structure of telegram type:			
Bit No.	**Value**	**Description**	**Comments**

15-12		Reserved	
11-10		Length of AT service channel (SVC)	
	00	2 octets, Slave SVC INFO	(Standard, CP2/3/4), SERCOS II
	01	4 octets, Slave SVC INFO	(CP3/4), SERCOS II and SERCOS III
	10	6 octets, Slave SVC INFO	(CP3/4), SERCOS II
	11	8 octets, Slave SVC INFO	(CP3/4), SERCOS II
9-8		Length of MDT service channel (SVC)	
	00	2 octets, Master SVC INFO	(Standard, CP2/3/4), SERCOS II
	01	4 octets, Master SVC INFO	(CP3/4), SERCOS II and SERCOS III
	10	6 octets, Master SVC INFO	(CP3/4), SERCOS II
	11	8 octets, Master SVC INFO	(CP3/4), SERCOS II
7-5		Reserved	
4		Feedback value	only valid for standard telegrams 3, 4 and 5
	0	Configured position feedback value	
	1	Active position feedback value (S-0-0386)	
3		Position feedback value	only valid for standard telegrams 3, 4 and 5
	0	Position feedback value 1 (S-0-0051)	motor feedback
	1	Position feedback value 2 (S-0-0053)	external feedback
2-0		Standard telegrams	SERCOS II: S-0-0134 and S-0-0135 are not used SERCOS III: see defined contents and sequence in MDT and AT
	000	Standard telegram 0	MDT: S-0-0134 AT: S-0-0135
	001	Standard telegram 1	MDT: S-0-0134 + **S-0-0080** AT: S-0-0135
	010	Standard telegram 2	MDT: S-0-0134 + **S-0-0036** AT: S-0-0135 + **S-0-0040**
	011	Standard telegram 3	MDT: S-0-0134 + **S-0-0036** AT: S-0-0135 + **(S-0-0051 or S-0-0053)**
	100	Standard telegram 4	MDT: S-0-0134 + **S-0-0047** AT: S-0-0135 + **(S-0-0051 or S-0-0053)**
	101	Standard telegram 5	MDT: S-0-0134 + **S-0-0047 + S-0-0036**

			AT: S-0-0135 + **(S-0-0051 or S-0-0053) + S-0-0040**
	110	Standard telegram 6	MDT: S-0-0134 + **S-0-0036** AT: S-0-0135
	111	Application telegram	SERCOS II and SERCOS III V1.0 uses S-0-0016 and S-0-0024 SERCOS III V1.1.1 uses the configuration with S-0-1050.x.6

Name		Telegram type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0017 IDN-list of all operation data

- **Description**

All IDNs of all procedure commands and parameters of the sub-device are stored in this IDN list.

Name		IDN-list of all operation data
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0021 IDN-list of invalid operation data for CP2

- **Description**

IDNs which are considered invalid by the slave when performing the CP3 transition check are stored in this IDN-list.

- Case 1: procedure command S-0-0127 is performed correctly; the IDN-list (S-0-0021) contains no IDNs.
- Case 2: procedure command S-0-0127 results in an error; the IDN-list (S-0-0021) contains all IDNs of invalid operation data.

The presence of this parameter is only necessary if this slave have the ability for errors during phase switching (S-0-0127).

Name		IDN-list of invalid operation data for CP2
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0022 IDN-list of invalid operation data for CP3

- **Description**

IDNs which are considered invalid by the slave when performing the CP4 transition check are stored in this IDN-list.

- Case 1: procedure command S-0-0128 is performed correctly; the IDN-list (S-0-0022) contains no IDNs.
- Case 2: procedure command S-0-0128 results in an error; the IDN-list (S-0-0022) contains all IDNs of invalid operation data.

The presence of this parameter is only necessary if this slave have the ability for errors during phase switching (S-0-0128).

Name		IDN-list of invalid operation data for CP3
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0025 IDN-list of all procedure commands

- **Description**

The IDNs of all procedure commands are stored in this IDN-list.

Name		IDN-list of all procedure commands
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0026 Configuration list for signal status word

- **Description**

Bits in the signal status word (S-0-0144) are definable by means of the configuration list of the signal status word represented in this IDN. The sequence of the IDNs in the configuration list determines the bit numbering scheme in the signal status word. The first IDN of the configuration list defines bit 0, the last IDN defines bit 15 of the signal status word. If IDN S-0-0328 is not supported by the slave, bit 0 of all the configured IDNs are used, else S-0-0328 defines for each IDN which bit is used.

Name		Configuration list for signal status word
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0027 Configuration list for signal control word

- **Description**

Bits in the signal control word (S-0-0145) are definable by means of the configuration list of the signal control word represented in this IDN. The sequence of the IDNs in the configuration list determines the bit numbering scheme in the signal control word. The first IDN of the configuration list defines bit 0, the last IDN defines bit 15 of the signal control word. If IDN S-0-0329 is not supported by the slave, bit 0 of all the configured IDNs are used, else S-0-0329 defines for each IDN which bit is used.

Name		Configuration list for signal control word
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0032 Primary operation mode

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Primary operation mode
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0033 Secondary operation mode 1

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 1
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0034 Secondary operation mode 2

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 2
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0035 Secondary operation mode 3

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 3
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0036 Velocity command value

- **Description**

During the velocity control drive operation mode, the velocity command values are consumed by the drive according to the time pattern after the following conditions.

- With SERCOS II in the time pattern of the control unit cycle time (S-0-0001).
- With SERCOS III in the time pattern of the producer cycle time (S-0-1050.x.10).

Name		Velocity command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0037 Additive velocity command value

- **Description**

The additive velocity command value is an additional velocity offset which is to be added to the velocity command value (S-0-0036).

Name		Additive velocity command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0044

S-0-0038 Positive velocity limit value

- **Description**

The positive velocity limit value describes the maximum allowable velocity in the positive direction. If the velocity limit value is exceeded, the drive responds by setting the status 'ncommand > nlimit' (see S-0-0335).

Name		Positive velocity limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0039 Negative velocity limit value

- **Description**

The negative velocity limit value describes the maximum allowable velocity in the negative direction. If the velocity limit value is exceeded, the drive responds by setting the status 'ncommand > nlimit' (see S-0-0335).

Name		Negative velocity limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		0
Scaling / Resolution of LSB		see S-0-0044

S-0-0040 Velocity feedback value 1

- **Description**

The velocity feedback value 1 is transferred from the drive e.g. to the control unit in order to allow the control unit to periodically display the velocity value. The velocity feedback value 1 refers to the motor encoder.

Name		Velocity feedback value 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0041 Homing velocity

- **Description**

The homing velocity is used during the procedure command 'drive controlled homing' (S-0-0148) when activated. The drive performs its own homing control.

Name		Homing velocity
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0042 Homing acceleration

- **Description**

The homing acceleration is needed by the drive if the procedure command 'drive controlled homing' (S-0-0148) is activated.

Name		Homing acceleration
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160

Unit	see S-0-0160
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0160

S-0-0043 Velocity polarity parameter

- **Description**

This parameter is used to switch polarities of velocity data for specific applications. Polarities are not switched internally but externally (on the input and output) of a closed loop system. The motor shaft turns clockwise when there is a positive velocity command difference and no inversion is programmed. For more details see clause 1.2.

Table 1: Structure of velocity polarity parameter

Bit No.	Value	Description	Comments
15-4		reserved	
3		Velocity feedback value 2	S-0-0156
	0	non inverted	
	1	inverted	
2		Velocity feedback value 1	S-0-0040
	0	non inverted	
	1	inverted	
1		Additive velocity command value1	S-0-0037
	0	non inverted	
	1	inverted	
0		Velocity command value1	S-0-0036
	0	non inverted	
	1	inverted	

Name		Velocity polarity parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0044 Velocity data scaling type

- **Description**

A variety of scaling methods can be selected by means of the scaling type parameter. Bit 5 is set to 'minute' for preferred data (see also Scaling of operation data)

Table 2: Structure of velocity data scaling type

Bit No.	Value	Description	Comments
15-7		reserved	
6		Data reference	
	0	at the motor shaft	
	1	at the load	uses S-0-0121, S-0-0122, S-0-0123
5		Time units	
	0	minutes [min]	
	1	seconds [s]	
4		Units for linear scaling	
	0	meters [m]	
	1	inches [in]	optional
4		Units for rotational scaling	
	0	revolutions [Rev]	
	1	reserved	
3		Scaling type	
	0	preferred scaling	
	1	parameter scaling	uses S-0-0045 and S-0-0046
2–0		Scaling method	
	000	no scaling	
	001	linear scaling	
	010	rotational scaling	
	011-111	reserved	

Name		Velocity data scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-0045 Velocity data scaling factor

- **Description**

This parameter defines the scaling factor for all velocity data in a sub-device (see Scaling of operation data).

Name		Velocity data scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0046 Velocity data scaling exponent

- **Description**

This parameter defines the scaling exponent for all velocity data in a sub-device (see Scaling of operation data).

Table 3: Structure of the scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Velocity data scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0047 Position command value

- **Description**

During the position control drive operation mode, the position command values are consumed by the drive according to the time pattern after the following conditions.

- With SERCOS II in the time pattern of the control unit cycle time (S-0-0001).
- With SERCOS III in the time pattern of the producer cycle time (S-0-1050.x.10).

Name		Position command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0048 Additive position command value

- **Description**

This IDN is used if an additional position offset is required during position control operation mode in the drive. The additive position command value is added to the position command value S-0-0047, when the bit "SYNC output data" (drive control) toggles.

Name		Additive position command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0049 Positive position limit value

- **Description**

The positive position limit value describes the maximum allowed distance in the positive direction. The positive position limit value is only activated when

- the position limit values are enabled in the position polarity parameter (see IDN/S-0-0055) and
- all position data are based on the machine zero point.

When the positive position limit value is exceeded and the reaction is error, the drive

- sets an error bit in C1D (S-0-0011) and
- sets the diagnostic number (S-0-0390) to 0xC00F6029.

When the positive position limit value is exceeded and the reaction is warning, the drive

- sets the diagnostic number to 0xC00E6029.

Name		Positive position limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0050 Negative position limit value

- **Description**

The negative position limit value describes the maximum allowed distance in the negative direction. The negative position limit value is only activated when

- the position limit values are enabled in the position polarity parameter (see IDN/S-0-0055) and
- all position data are based on the machine zero point.

When the negative position limit value is exceeded and the reaction is error, the drive

- sets an error bit in C1D (S-0-0011) and
- sets the diagnostic number (S-0-0390) to 0xC00F6030.

When the negative position limit value is exceeded and the reaction is warning, the drive

- sets the diagnostic number to 0xC00E6030.

Name		Negative position limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076

	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0051 Position feedback value 1 (motor feedback)

- **Description**

The position feedback value 1 is transferred from the drive e.g. to the control unit so that it is possible for the control unit to perform block stepping and display position information if necessary.

Name		Position feedback value 1 (motor feedback)
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	always
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0052 Reference distance 1

- **Description**

This parameter describes the distance between the machine zero point and the reference point related to the motor feedback. After the homing procedure, the position feedback value 1 is calculated by:

- reference distance 1
- reference offset 1 S-0-0150
- marker position A/B S-0-0173/S-0-0174.

For absolute homing the reference distance 1 is the absolute position of the axis upon completion of the homing sequence with procedure command S-0-0447.

Name		Reference distance 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076

	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0053 Position feedback value 2 (external feedback)

- **Description**

The position feedback value 2 is transferred from the drive e.g. to the control unit so that it is possible for the control unit to perform block stepping and display position information if necessary.

Name		Position feedback value 2 (external feedback)
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	always
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0054 Reference distance 2

- **Description**

This parameter describes the distance between the machine zero point and the reference point related to the external feedback. After the homing procedure, the position feedback value 2 is calculated by:

- reference distance 2;
- reference offset 2 (S-0-0151);
- marker position A/B (S-0-0173/S-0-0174).

For absolute homing the reference distance 2 is the absolute position of the axis upon completion of the homing sequence with procedure command S-0-0447.

Name		Reference distance 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never

	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0055 Position polarity parameter

- **Description**

This parameter is used to switch polarities of reported position data for specific applications. Polarities are switched outside (i.e., on the input and output) of a closed loop system. The motor shaft turns clockwise (as viewed from the output shaft) when there is a positive position command difference and no inversion is programmed. For more details see clause 1.2.

Table 4: Structure of the Position polarity parameter

Bit No.	Value	Description	Comments
15-6		reserved	
5		Underflow / Overflow threshold	S-0-0280, S-0-0281
	0	disabled	
	1	enabled	
4		Position limit values	S-0-0049, S-0-0050
	0	disabled	
	1	enabled	
3		Position feedback value 2	S-0-0053
	0	Non-inverted	
	1	Inverted	
2		Position feedback value 1	S-0-0051
	0	Non-inverted	
	1	Inverted	
1		Additive position command value	S-0-0048
	0	Non-inverted	
	1	Inverted	
0		Position command value	S-0-0047
	0	Non-inverted	
	1	Inverted	

Name		Position polarity parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary

	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0057 Position window

- **Description**

When the difference between the accumulated position command value and the position feedback value is within the range of the position window, then the drive sets the status "in position" (S-0-0336). If needed, the status 'in position' may be assigned to a real-time bit.

Name		Position window
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0058 Reversal clearance

- **Description**

The reversal clearance describes the amount of backlash between motor and load during reversal, relative to the position data.

Name		Reversal clearance
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0059 Position switch flag

- **Description**

1. If the function of the position switch flag is deactivated in the position switch control (S-0-0476, bit 0 to 15), then the corresponding position switch flag is set to 0.
    - Position switch flag 0 uses position switch point on 1 (S-0-0060) and position switch point off 1 (S-0-0460).
    - ...
    - Position switch flag 15 uses position switch point on 16 (S-0-0075) and position switch point off 16 (S-0-0475).
2. **Position switch mode**
    - is activated in the position switch control (S-0-0476, bit 0 to 15) and uses
    - position switch point on 1 to 16 (S-0-0060 to S-0-0075) only.
3. **Cam switch mode**
    - is activated in the position switch control (S-0-0476, bit 0 to 31) and uses
    - Position switch point on 1 to 16 (S-0-0060 to S-0-0075),
    - Position switch point off 1 to 16 (S-0-0460 to S-0-0475) and the
    - position overflow must be taken into account at modulo format.

Structure of the position switch flag:

Bit No.	Value	Description	Comments
15-0 (n)		Position switch flag	n = 0...15
	0	Position feedback value < position switch point on (n+1)	see S-0-0060 to S-0-0075
	1	Position feedback value ≥ position switch point on (n+1)	see S-0-0060 to S-0-0075
	1	Position feedback value < position switch point off (n+1)	see S-0-0460 to S-0-0475
	0	Position feedback value ≥ position switch point off (n+1)	see S-0-0460 to S-0-0475

Name		Position switch flag
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0060 Position switch point 1

- **Description**

The position switches consist of a "position switch point on" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point on", the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the "position switch point on", the appropriate position switch flag is set to 1.

Name		Position switch point 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0061 Position switch point 2

- **Description**

The position switches consist of a "position switch point on" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point on", the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the "position switch point on", the appropriate position switch flag is set to 1.

Name		Position switch point 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0062 Position switch point 3

- **Description**

The position switches consist of a "position switch point on" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point on", the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the "position switch point on", the appropriate position switch flag is set to 1.

Name		Position switch point 3
Attribute	Length (Octet)	4

	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0063 Position switch point 4

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 4
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0064 Position switch point 5

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 5
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0076

S-0-0065 Position switch point 6

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 6
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0066 Position switch point 7

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 7
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0067 Position switch point 8

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 8
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0068 Position switch point 9

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 9
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0069 Position switch point 10

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 10
Attribute	Length (Octet)	4

	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0070 Position switch point 11

- **Description**

The position switches consist of a "position switch point on" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point on", the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the "position switch point on", the appropriate position switch flag is set to 1.

Name		Position switch point 11
Attribute	Length (Octet)	{{{Length}}}
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0071 Position switch point 12

- **Description**

The position switches consist of a "position switch point on" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point on", the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the "position switch point on", the appropriate position switch flag is set to 1.

Name		Position switch point 12
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0076

S-0-0072 Position switch point 13

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 13
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0073 Position switch point 14

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 14
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0074 Position switch point 15

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 15
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0075 Position switch point 16

- **Description**

The position switches consist of a “position switch point on” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point on”, the appropriate position switch flag is set to 0. If the position feedback value is equal to or greater than the “position switch point on”, the appropriate position switch flag is set to 1.

Name		Position switch point 16
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0076 Position data scaling type

- **Description**

A variety of scaling methods can be selected by means of the scaling type parameter (see Scaling of operation data).

Table 5: Structure of position data scaling type

Bit No.	Value	Description	Comments

15-8		reserved	
7		Processing format	
	0	absolute format	
	1	modulo format	see S-0-0103
6		Data reference	
	0	at the motor shaft	
	1	at the load	uses S-0-0121, S-0-0122, S-0-0123
5		(reserved)	
4		Units for linear scaling	
	0	metre [m]	
	1	inches [in]	optional
4		Units for rotational scaling	
	0	degrees	
	1	(reserved)	
3			
	0	preferred scaling	
	1	parameter scaling	uses S-0-0077, S-0-0078 and S-0-0079
2–0		Scaling method	
	000	no scaling	
	001	linear scaling	
	010	rotational scaling	
	100-111	reserved	

Name		Position data scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0077 Linear position data scaling factor

- **Description**

This parameter defines the scaling factor for all position data in a sub-device (see Scaling of operation data).

Name		Linear position data scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0078 Linear position data scaling exponent

- **Description**

This parameter defines the scaling exponent for all position data in a sub-device (see Scaling of operation data).

Table 6: Structure of the scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Linear position data scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0079 Rotational position resolution

- **Description**

This parameter defines the rotational position resolution for all position data in a sub-device (see Scaling of operation data).

Name		Rotational position resolution
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0080 Torque command value

- **Description**

During the torque control drive operation mode, the torque (force) command values are consumed by the drive according to the time pattern after the following conditions.

- With SERCOS II in the time pattern of the control unit cycle time (S-0-0001).
- With SERCOS III in the time pattern of the producer cycle time (S-0-1050.x.10).

Name		Torque command value
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0081 Additive torque command value

- **Description**

The additive torque (force) command value is an additional offset which is to be added to the torque (force) command value (S-0-0080).

Name		Additive torque command value
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0086

S-0-0082 Positive torque limit value

- **Description**

The positive torque limit value limits the maximum torque in the positive direction. If the torque limit value is exceeded, the drive sets the status 'T ≥ Tlimit' (S-0-0334).

Name		Positive torque limit value
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0083 Negative torque limit value

- **Description**

The negative torque limit value limits the maximum torque in the negative direction. If the torque limit value is exceeded, the drive sets the status 'T ≥ Tlimit' (S-0-0334).

Name		Negative torque limit value
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0084 Torque feedback value

- **Description**

The torque feedback value is transferred from the drive e.g. to the control unit.

Name	Torque feedback value

Attribute	Length (Octet)	see S-0-0086
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0085 Torque polarity parameter

- **Description**

This parameter is used to switch polarities of reported torque data for specific applications. Polarities are not switched internally but externally (on the input and output) of a closed loop system. The motor shaft turns clockwise when there is a positive torque command difference and no inversion. For more details see clause 1.2.

Table 7: Structure of torque polarity parameter

Bit No.	Value	Description	Comments
15-3		reserved	
2		Torque feedback value	S-0-0084
	0	non-inverted	
	1	inverted	
1		Additive torque command value	S-0-0081
	0	non-inverted	
	1	inverted	
0		Torque command value	S-0-0080
	0	non-inverted	
	1	inverted	

Name		Torque polarity parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0086 Torque/force data scaling type

- **Description**

A variety of scaling methods can be selected by means of this scaling type parameter (see Scaling of operation data).

Table 8: Structure of torque/force data scaling type

Bit No.	Value	Description	Comments
15-9		reserved	
8		Selection of data length	with S III V1.1..2
	0	torque/force data are 2 octets	
	1	torque/force data are 4 octets	
7		reserved	
6		Data reference	
	0	at the motor shaft	
	1	at the load	uses S-0-0121, S-0-0122, S-0-0123
5		(reserved)	
4		Units for force	
	0	newton [N]	
	1	pound force [lbf]	optional
4		Units for torque	
	0	newton metre [Nm]	
	1	inch pound force [in lbf]	optional
3			
	0	preferred scaling	
	1	parameter scaling	uses S-0-0093 and S-0-0094
2–0		Scaling method	
	000	percentage scaling	
	001	linear scaling	force
	010	rotational scaling	torque

Name		Torque/force data scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0091 Bipolar velocity limit value

- **Description**

The bipolar velocity limit value describes the maximum allowable velocity in both directions. If the velocity limit value is exceeded, the drive responds by setting the status 'ncommand > nlimit' (see S-0-0335).

Name		Bipolar velocity limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0092 Bipolar torque limit value

- **Description**

The bipolar torque limit value limits the maximum torque symmetrically in both directions. If the torque limit value is exceeded, the drive sets the status '$T \geq T_{limit}$' (S-0-0334).

Name		Bipolar torque limit value
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0093 Torque/force data scaling factor

- **Description**

This parameter defines the scaling factor for all torque/force data in a sub-device (see Scaling of operation data).

Name		Torque/force data scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0094 Torque/force data scaling exponent

- **Description**

This parameter defines the scaling exponent for all torque/force data in a sub-device (see Scaling of operation data).

Table 9: Structure of the torque/force data scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Torque/force data scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0095 Diagnostic message

- **Description**

The currently relevant operating status is being monitored with diagnostic messages. The diagnostic messages are generated by the slave as a text and stored in the operation data of this IDN.

Name		Diagnostic message
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0097 Mask class 2 diagnostic

- **Description**

Using this mask, warnings in class 2 diagnostic can be masked with respect to their effect on the change bit in drive status. When changing masked warnings, the change bit for class 2 diagnostic is not set in the drive status. The mask does not affect the operation data of class 2 diagnostic S-0-0012.

Structure of Mask C2D			
Bit No.	**Value**	**Description**	**Comments**
15-0		Mask C2D	
	all 0s	masked warning	
	all 1s	unmasked warning	

Name		Mask class 2 diagnostic
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0099 Reset class 1 diagnostic

- **Description**

If this procedure command is received by the sub-device via the service channel, then the errors, the error bits and the shut-down mechanism are deleted. This concerns the parameter C1D (S-0-0011), the interface status (S-0-0014) and the manufacturer's C1D (S-0-0129). The sub-device shut-down error (resource status bit 13) and the sub-device shut-down mechanism in the sub-device are deleted also.

This procedure command is not interruptible and generates no negative acknowledgment even if an error can not be deleted or no error exists in the sub-device.

Structure of procedure command control see Table Command Control

Structure of procedure command acknowledgment see Table Command Acknowledgment

Name		Reset class 1 diagnostic
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0100 Velocity loop proportional gain

- **Description**

The scaling and operational characteristic of the velocity loop proportional gain are defined by the drive manufacturer. The 4 byte data length is preferred for new implementations. The control unit has to check the data length before access the data.

Name		Velocity loop proportional gain
Attribute	Length (Octet)	SERCOS III: 4 SERCOS II: 2 or 4 (manufacturer specific)
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0101 Velocity loop integral action time

- **Description**

The operational characteristic of the velocity loop integral action time is defined by the drive manufacturer. With the maximum value of 2^{16}-1, the integrator in the velocity loop regulator is switched off.

Name		Velocity loop integral action time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0102 Velocity loop differential time

- **Description**

The operational characteristic of the velocity loop differential time is defined by the drive manufacturer.

Name		Velocity loop differential time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1 ms

S-0-0103 Modulo value

- **Description**

If the Modulo format is selected in the position data scaling factor (S-0-0076), the Modulo value defines the range that the drive and control unit shall implement.

- When extrapolation is used for missing command values, the position command difference between two consecutive cycles may not exceed half the Modulo value.

- If extrapolation is not used for missing command values, the position command difference between two consecutive cycles may not exceed one fourth the Modulo value.

This reduces the maximum possible velocity due to the position command difference. The spindle angle position in addition always is related on the physical modulo value to reach short positioning times. To compensate this disadvantage, the divider modulo value (S-0-0294) is programmed.

Example 1:		
	NC cycle time (SERCOS II) producer cycle time (SERCOS III)	4 ms
	Modulo value	360°
	Divider modulo value	1
	Vmax = (360°/2) / 4ms	45000°/s (7500 rpm)
Example 2:		
	NC cycle time (SERCOS II) producer cycle time (SERCOS III)	4 ms
	Modulo value	1080° (3*360°)
	Divider modulo value	3
	Vmax = (1080°/2) / 4ms	135000°/s (22500 rpm)

Name		Modulo value
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0104 Position loop KV -factor

- **Description**

The KV-factor determines the gain of the position loop regulator throughout the entire velocity range.

Name		Position loop KV -factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	2
	Write Protection	never
	Conversion Factor	1
Unit		(m/min)/mm
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,01(m/min)/mm

S-0-0105 Position loop integral action time

- **Description**

The method of operation of the position loop integral action time is defined by the manufacturer.

Name		Position loop integral action time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0106 Current loop proportional gain 1

- **Description**

The current loop proportional gain 1 influences the torque/force-producing current. The scaling and mode of operation is determined by the drive manufacturer.

The 4 octet data length is preferred for new implementations. The control unit has to check the data length before access the data.

Name	Current loop proportional gain 1

Attribute	Length (Octet)	SERCOS III: 4 SERCOS II: 2 or 4 (manufacturer specific)
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0107 Current loop integral action time 1

- **Description**

The current integral action time 1 influences the torque/force-producing current. The mode of operation is determined by the drive manufacturer.

Name		Current loop integral action time 1
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1µs

S-0-0108 Feedrate override

- **Description**

The feedrate override is activated only with drive controlled procedure commands. In such a case, the velocity command value is calculated internally by the drive. The feedrate override has multiplying effects on the velocity command value.

Name		Feedrate override
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	2
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	0,01%

S-0-0109 Motor peak current

- **Description**

With the motor peak current (or pressure) the motor produces the peak torque/force (S-0-0534) according to the motor spec sheet. If the motor peak current is less than that of the amplifier, the amplifier is automatically limited to the value of the motor peak current.

Name		Motor peak current
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0110 Amplifier peak current

- **Description**

The amplifier peak current is limited by the hardware, which means that the current for the maximum attainable torque limit value is fixed as well.

Name		Amplifier peak current
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0111 Motor continuous stall current

- **Description**

The motor continuous stall current is the current at which the motor produces the continuous stall torque/force (S-0-0533) according to the motor spec sheet. For all motors except for asynchronous motors, this parameter is used as a reference for the scaling of all torque/force data.

Name		Motor continuous stall current
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0112 Amplifier rated current

- **Description**

The amplifier rated current is equal to the allowable continuous current of the drive unit.

Name		Amplifier rated current
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0113 Maximum motor speed

- **Description**

The maximum motor speed is listed in the motor spec sheet provided by the manufacturer

Name		Maximum motor speed
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0114 Load limit of the motor

- **Description**

When the load limit is exceeded, the drive sets the overload warning bit in C2D (see S-0-0310). After a time period specified by the manufacturer, the overload shut-down bit is set in C1D (S-0-0011).

Name		Load limit of the motor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1%

S-0-0115 Position feedback 2 type

- **Description**

The position feedback 2 type parameter refers only to an external feedback. This parameter is programmed to define the corresponding conditions which apply to the external feedback (see also S-0-0277).

Structure of Position Feedback 1 and 2

Bit No.	Value	Description	Comments
15-10		reserved	
9		cyclic marker evaluation	
	0	not active	
	1	active	for incremental measuring system (no distance coded) with a cyclic marker pulse use the function "cyclic marker pulse detection".
8		reserved	
7,6		Type of measuring system and evaluation	
	x0	relative	(incremental) measuring system, no absolute evaluation
	01	absolute measuring system	Absolute evaluation of the measuring system. The initialization of the position is checked by the drive. Set

			absolute position S-0-0447 has to be done for homing.
	11	incremental use of a absolute measuring system	No monitoring for the initialization of the position. Set absolute position S-0-0447 is not available. For homing the axis use S-0-0148 or S-0-0146
5		Structure of distance coded feedback	
	0	counting positive with positive direction	
	1	counting negative with positive direction	
4		Marker pulse quantity	
	0	only one reference marker pulse	
	1	multiple cyclic reference marker pulses	
3		Direction polarity	
	0	not inverted	
	1	inverted	
2		Feedback resolution	Feedback 1 see S-0-0116 Feedback 2 see S-0-0117
	0	resolution is metric	
	1	resolution is inches	
1		Distance coded feedback	
	0	no distance coded reference marks	
	1	distance coded reference marks	see S-0-0165, S-0-0166
0		Feedback type	
	0	rotational feedback	
	1	linear feedback	

Name		Position feedback 2 type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0116 Resolution of feedback 1

- **Description**

The parameter resolution of feedback 1 (motor feedback) contains,

- for a rotary feedback, the cycles per revolution of the motor (see also S-0-0256 Multiplication factor 1),
- for a linear feedback the grid constant is entered.

Name		Resolution of feedback 1
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	OL
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0117 Resolution of feedback 2

- **Description**

The parameter resolution of feedback 2 (external feedback) contains,

- for a rotary feedback, the cycles per revolution (see also S-0-0257 Multiplication factor 2),
- for a linear feedback the grid constant is entered.

Name		Resolution of feedback 2
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	OL
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0119 Current loop proportional gain 2

- **Description**

The current loop proportional gain 2 influences the flux producing current. The scaling and mode of operation is determined by the drive manufacturer. The 4 octet data length is preferred for new implementations. The control unit has to check the data length before access the data.

Name		Current loop proportional gain 2
Attribute	Length (Octet)	SERCOS III: 4 SERCOS II: 2 or 4 (manufacturer specific)
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0120 Current loop integral action time 2

- **Description**

The current loop integral action time 2 influences the flux–producing current. The mode of operation is determined by the drive manufacturer.

Name		Current loop integral action time 2
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1µs

S-0-0121 Input revolutions of load gear

- **Description**

Input revolution values must be entered as integers.

Name		Input revolutions of load gear
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		[input revolution]
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0122 Output revolutions of load gear

- **Description**

Output revolution values must be entered as integers.

Name		Output revolutions of load gear
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		[output revolution]
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0123 Feed constant

- **Description**

The feed constant describes the machine element which converts a rotational motion into a linear motion. The feed constant indicates the linear distance during one revolution of the feed spindle.

Name		Feed constant
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0124 Standstill window

- **Description**

The standstill window describes the amount of the deviation of the velocity from 0. If the velocity feedback value is within the standstill window the drive sets the status n feedback = 0 (S-0-0331).

Name		Standstill window
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0125 Velocity threshold (nx)

- **Description**

If the velocity feedback value falls below the velocity threshold nx, the drive sets the status **nfeedback < nx** (´S-0-0332).

Name		Velocity threshold (nx)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0126 Torque threshold (Tx)

- **Description**

If the torque feedback value exceeds the torque threshold Tx, the drive sets the status $T \geq T_x$ (see S-0-0333).

Name		Torque threshold (Tx)
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086

Unit	see S-0-0086
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0086

S-0-0127 CP3 transition check

- **Description**

The master uses this procedure command to instruct the slave to check that all necessary parameters have been transferred for CP3. Otherwise, this procedure command results in an error (see S-0-0021). After the procedure command is performed correctly, the master has to cancel the procedure command. The master can then activate CP3 in the MST.

Structure of procedure command control see Table Command Control

Structure of procedure command acknowledgment see Table Command Acknowledgment

Name		CP3 transition check
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0128 CP4 transition check

- **Description**

The master uses this procedure command to instruct the slave to check that all necessary parameters have been transferred for CP4. Otherwise, this procedure command results in an error (see S-0-0022). After the procedure command is performed correctly, the master has to cancel the procedure command. The master can then activate CP4 in the MST.

Structure of procedure command control see Table Command Control.

Structure of procedure command acknowledgment see Table Command Acknowledgment.

Name		CP4 transition check
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never

	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0129 Manufacturer class 1 diagnostic

- **Description**

The drive manufacturer can define additional shut-down errors in manufacturer class 1 diagnostic. If an error is set in the manufacturer class 1 diagnostic, the manufacturer-specific error bit in class 1 diagnostic (see S-0-0011) is set as well. The drive cancels the manufacturer-specific error and resets to '0' only if the error in manufacturer class 1 diagnostic has been eliminated upon receiving the command 'reset class 1 diagnostic' (see S-0-0099) via the service channel.

Structure of manufacturer Class 1 diagnostic:

Bit No.	Value	Description	Comments
15-0		Manufacturer class 1 diagnostic	
	0	no error	all bits are set to 0
	1	error	one or more bits are set to 1

Name		Manufacturer class 1 diagnostic
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0130 Probe value 1 positive edge

- **Description**

If the positive edge of the input signal of probe 1 (see S-0-0401) is activated and enabled, the drive stores the measuring data allocated in S-0-0426 in this parameter. The control unit has the possibility to configure the 'probe value 1 positive edge' in the AT real-time data or read this parameter via the service channel.

Name		Probe value 1 positive edge
Attribute	Length (Octet)	4

	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0131 Probe value 1 negative edge

- **Description**

If the negative edge of the input signal of probe 1 (see S-0-0401) is activated and enabled, the drive stores the measuring data allocated in S-0-0426 in this parameter. The control unit has the possibility to configure the 'probe value 1 negative edge' in the AT real-time data or read this parameter via the service channel.

Name		Probe value 1 negative edge
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0132 Probe value 2 positive edge

- **Description**

If the positive edge of the input signal of probe 2 (see S-0-0402) is activated and enabled, the drive stores the measuring data allocated in S-0-0427 in this parameter. The control unit has the possibility to configure the 'probe value 2 positive edge' in the AT real-time data or read this parameter via the service channel.

Name		Probe value 2 positive edge
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0133 Probe value 2 negative edge

- **Description**

If the negative edge of the input signal of probe 2 (see S-0-0402) is activated and enabled, the drive stores the measuring data allocated in S-0-0427 in this parameter. The control unit has the possibility to configure the 'probe value 2 negative edge' in the AT real-time data or read this parameter via the service channel.

Name		Probe value 2 negative edge
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see Variant
	Write Protection	never
	Conversion Factor	see Variant
Unit		see Variant
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see Variant

S-0-0134 Drive control

- **Description**

- SERCOS III: The content of drive control field shall be as specified in Table 10.
- SERCOS II: The content of drive control field shall be as specified in Table 11.

For more details see clause 1.12.

Table 10: Structure of drive control (SERCOS III)

Bit No.	Value	Description	Comments
15-13			Drive ON, Drive enable, Drive restart
	111		Drive should follow the command values
15		Drive ON/OFF	Condition: bit 14 = 1
	0	Drive OFF	when changing from 1 -> 0: the „maximum drive off delay time“ (S-0-0273) is started, drive decelerates as best as possible limited by the „emergency stop deceleration“ (S-0-0429), followed by disabling of the torque at "nmin", after the „drive off delay time“ (S-0-0207). The power stage can remain in an

			activated state. After the „maximum drive off delay time“ (S-0-0273) is elapsed, the locking of brake is initiated and the torque is disabled.
	1	Drive ON	when changing from 0 -> 1: drive follows the command values of the control unit after the "drive on delay time" (S-0-0206).
14		Drive enable	Condition: independent of bits 15 and 13
	0	Drive disable	when changing from 1 –> 0: torque is immediately disabled and the power stage pulses are blocked.
	1	Drive enable	when changing from 0 –> 1: the enable is delayed in the drive by the "drive enable delay time" (S-0-0295). The enable delay is required at use of a contactor in the motor cable.
13		Drive Halt	this bit is used to stop or restart the drive regardless of the presently active control unit function Condition: bits 15 and 14 are set to 1
	0	Drive halt	when changing from 1 –> 0: • drive internal interpolator is inactive: drive is halted according to the „drive halt acceleration bipolar“ (S-0-0372) and the control loop remains closed. • drive internal interpolator is active: drive is halted according to the active parameters of the interpolator and the control loop remains closed.
	1	Drive restart	when changing from 0 –> 1: drive activates the selected operation mode. In position control the control unit has to set the position command value to the position feedback value before bit 13 is set to 1.
12		reserved	
11		reserved	
10-8		Selection of Operation mode	
	000	Primary operation mode	S-0-0032
	001	Secondary operation mode 1	S-0-0033
	010	Secondary operation mode 2	S-0-0034
	011	Secondary operation mode 3	S-0-0035
	100	Secondary	S-0-0284

		operation mode 4	
	101	Secondary operation mode 5	S-0-0285
	110	Secondary operation mode 6	S-0-0286
	111	Secondary operation mode 7	S-0-0287
7-0		reserved	

Table 11: Structure of drive control (SERCOS II)

Bit No.	Value	Description	Comments
15-13			Drive ON, Drive enable, Drive restart
	111		Drive should follow the command values
15		Drive ON/OFF	Condition: bit 14 = 1
	0	Drive OFF	when changing from 1 -> 0: the „maximum drive off delay time“ (S-0-0273) is started, drive decelerates as best as possible limited by the „emergency stop deceleration“ (S-0-0429), followed by disabling of the torque at "nmin", after the „drive off delay time“ (S-0-0207). The power stage can remain in an activated state. After the „maximum drive off delay time“ (S-0-0273) is elapsed, the locking of brake is initiated and the torque is disabled.
	1	Drive ON	when changing from 0 -> 1: drive follows the command values of the control unit after the "drive on delay time" (S-0-0206).
14		Drive enable	Condition: independent of bits 15 and 13
	0	Drive disable	when changing from 1 –> 0: torque is immediately disabled and the power stage pulses are blocked.
	1	Drive enable	when changing from 0 –> 1: the enable is delayed in the drive by the "drive enable delay time" (S-0-0295). The enable delay is required at use of a contactor in the motor cable.
13		Drive Halt	this bit is used to stop or restart the drive regardless of the presently active control unit function Condition: bits 15 and 14 are set to 1

	0	Drive halt	when changing from 1 –> 0: • drive internal interpolator is inactive: drive is halted according to the „drive halt acceleration bipolar“ (S-0-0372) and the control loop remains closed. • drive internal interpolator is active: drive is halted according to the active parameters of the interpolator and the control loop remains closed.
	1	Drive restart	when changing from 0 –> 1: drive activates the selected operation mode. In position control the control unit has to set the position command value to the position feedback value before bit 13 is set to 1.
12		reserved	
10		IPOSYNC	Control unit synchronization bit
	0/1	toggle	This bit is used to synchronize the interpolation in the control unit with the fine interpolator in the drive and is initially set to 0. It becomes valid in CP3 and shall remain valid during drive-controlled functions. This bit is toggled with the control unit cycle time (t Ncyc) indicating the update of the command values
11,9,8		Selection of Operation mode	
	000	Primary operation mode	S-0-0032
	001	Secondary operation mode 1	S-0-0033
	010	Secondary operation mode 2	S-0-0034
	011	Secondary operation mode 3	S-0-0035
	100	Secondary operation mode 4	S-0-0284
	101	Secondary operation mode 5	S-0-0285
	110	Secondary operation mode 6	S-0-0286
	111	Secondary operation mode 7	S-0-0287
7		Real-time control bit 2	S-0-0302
6		Real-time control bit	S-0-0300

		1	
5-3		Data block element	Service channel function
	000	Service channel not active, close service channel or break a transmission in progress	
	001	IDN of the operation data	The service channel is closed for the previous IDN and opened for a new IDN
	010	Name of the operation data	
	011	Attribute of the operation data	
	100	Unit of the operation data	
	101	Minimum value of the operation data	
	110	Maximum value of the operation data	
	111	Operation data	
2		Last transmission	Service channel function
	0	Transmission in progress	
	1	Last transmission	
1		Read/Write (R/W)	Service channel function
	0	Read AT service INFO	
	1	Write MDT service INFO	
0		MHS	Service channel function
	0/1	toggle	Service transport handshake of the master

Name		Drive control
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-0135 Drive status

- **Description**

- SERCOS III: The content of drive status field shall be as specified in Table 12.
- SERCOS II: The content of drive status field shall be as specified in Table 13.

For more details see clause 1.12.

Table 12: Structure of drive status (SERCOS III)

Bit No.	Value	Description	Comments
15-14		Ready to operate	
	00	Drive not ready	internal checks not yet concluded successfully, CP<4 or PL is active
	01	Drive ready for main power on	internal checks are concluded successfully, CP = 4 and OL (operating level) is active, main power is off
	10	Drive ready and main power applied	drive is free of torque, power stage pulses are blocked
	11	Drive enabled	drive enable and drive ON are set and active, power stage is active
13		Drive shut-down error in C1D (S-0-0011)	A drive error situation of C1D leads to the following: 1. A best case deceleration followed by torque disable at nmin. 2. The error bit is reset to '0' by the drive if the procedure command 'reset class 1 diagnostic' (S-0-0099) is activated.
	0	no error	
	1	Error	Drive is shut-down due to error
12		Warning in C2D (S-0-0012)	When a warning occurs this bit is set to 1. In case of no warning this bit is set to 0.
	0	no warning	
	1	warning	
11		reserved	
10-8		Current Operation mode	
	000	Primary operation mode	S-0-0032
	001	Secondary operation	S-0-0033

		mode 1	
	010	Secondary operation mode 2	S-0-0034
	011	Secondary operation mode 3	S-0-0035
	100	Secondary operation mode 4	S-0-0284
	101	Secondary operation mode 5	S-0-0285
	110	Secondary operation mode 6	S-0-0286
	111	Secondary operation mode 7	S-0-0287
7		Emergency stop	external input signal
	0	Emergency stop is not active	
	1	Emergency stop is active	bit 15 and 14 of drive control is ignored
6		reserved	
5		Position feedback value status	S-0-0403, bit 0
4		Drive halt	
	0	Drive halt is not active	bit 13 of drive control is set to 1
	1	Drive halt is active	bit 13 of drive control is set to 0 and the velocity feedback value is within the standstill window (S-0-0124)
3		Status command value processing	
	0	Drive ignores the command values	e.g., during drive halt, drive controlled functions, programmed delay times
	1	Drive follows the command values	
2-0		reserved	

Table 13: Structure of drive status (SERCOS II)

Bit No.	Value	Description	Comments
15-		Ready to operate	

14	00	Drive not ready	internal checks not yet concluded successfully, CP<4 or PL is active
	01	Drive ready for main power on	internal checks are concluded successfully, CP = 4 and OL (operating level) is active, main power is off
	10	Drive ready and main power applied	drive is free of torque, power stage pulses are blocked
	11	Drive enabled	drive enable and drive ON are set and active, power stage is active
13		Drive shut-down error in C1D (S-0-0011)	A drive error situation of C1D leads to the following: 1. A best case deceleration followed by torque disable at nmin. 2. The error bit is reset to '0' by the drive if the procedure command 'reset class 1 diagnostic' (S-0-0099) is activated.
	0	no error	
	1	Error	Drive is shut-down due to error
12		Warning in C2D (S-0-0012)	When a warning toggles in C2D this bit is set to 1. It is set to 0 if C2D is read via the SVC
	0	no warning	
	1	warning	warning has changed in C2D
11		Change bit for C3D (S-0-0013)	When a status message toggles in C3D this bit is set to 1. It is set to 0 if C3D is read via the SVC
	0	no change	
	1	change	status message has changed in C3D
10-8		Current Operation mode	
	000	Primary operation mode	S-0-0032
	001	Secondary operation mode 1	S-0-0033
	010	Secondary operation mode 2	S-0-0034
	011	Secondary operation mode 3	S-0-0035
	100	Secondary operation mode 4	S-0-0284
	101	Secondary operation mode 5	S-0-0285
	110	Secondary operation mode 6	S-0-0286
	111	Secondary operation mode 7	S-0-0287
7		Real-time status bit 2	See S-0-0306
6		Real-time status bit 1	See S-0-0304
5		Procedure command change bit	

	0	No change in procedure command acknowledgement	
	1	Changing procedure command acknowledgment	
4		Status Parameterization level (PL)	
	0	PL is not active	
	1	PL is active	
3		Status command value processing	
	0	Drive ignores the command values	e.g., during drive halt, drive controlled functions, programmed delay times
	1	Drive follows the command values	
2		Error in service channel	
	0	no error	
	1	Error in service channel, error message in AT service INFO	
1		BUSY	
	0	Step finished, ready for new step	
	1	Step in progress, new step not allowed	
0		AHS	
	0/1	Service transport handshake of the slave	

Name		Drive status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0136 Positive acceleration limit value

- **Description**

The positive acceleration parameter limits the maximum acceleration ability of the drive to the programmed value. If the velocity increases it is said to be a positive acceleration.

Name		Positive acceleration limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0137 Negative acceleration limit value

- **Description**

The negative acceleration parameter limits the maximum acceleration ability of the drive to the programmed value. If the velocity decreases it is said to be a negative acceleration.

Name		Negative acceleration limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0138 Bipolar acceleration limit value

- **Description**

The bipolar acceleration parameter limits the maximum acceleration ability of the drive symmetrically to the programmed value in both directions.

Name		Bipolar acceleration limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0160

S-0-0139 Park axis

- **Description**

Setting and enabling the park axis procedure command causes all monitors associated with the feedback sensing system to shut down. This affects the position control, the transducer monitoring circuit (feedback hardware), and the monitoring of the position window (S-0-0057). While the procedure command is active, the drive does not report a C1D error (S-0-0011). The position feedback value status (S-0-0403) is reset by the drive.

The procedure command is positively acknowledged when the monitoring system mentioned above is turned off.

When the procedure command is canceled, all monitors mentioned above are turned on again. In order to relate the position feedback values to the reference point again, the control unit shall start a homing procedure.

Name		Park axis
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0144 Signal status word

- **Description**

Signals can be transmitted in real-time from the slaves to the master by means of the signal status word. For this purpose, the signal status word needs to be integrated in the AT. Bits in the signal status word are definable by means of the configuration list of the signal status word (see S-0-0026 and S-0-0328).

Structure of signal status word:

Bit No.	Value	Description	Comments
15-0		are defined in configuration list S-0-0026 and S-0-0328 if supported	

Name		Signal status word
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0145 Signal control word

- **Description**

Signals can be transmitted in real-time from the master to the slaves by means of the signal control word. For this purpose, the signal control word needs to be integrated in the MDT. Bits in the signal control word are definable by means of the configuration list of the signal control word (see S-0-0027 and S-0-0329).

Structure of signal control word:			
Bit No.	**Value**	**Description**	**Comments**
15-0		are defined in configuration list S-0-0027 and S-0-0329 if supported	

Name		Signal control word
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0146 Control unit controlled homing procedure command

- **Description**

When the master sets and enables the control unit controlled homing procedure command, the drive has to react on the programmed or assigned signals (homing enable S-0-0407, home switch S-0-0400, reference marker pulse of the feedback system).

When it reaches the appropriate marker pulse of the feedback system, the drive has to store the position feedback value in the corresponding marker position (S-0-0173 and S-0-0174). Furthermore, the drive has to set the signal "reference marker pulse registered" (S-0-0408). Afterwards, the drive acknowledges the procedure command as performed correctly.

When an error of C1D occurs, the procedure command results in an error in the procedure command acknowledgment.

Name		Control unit controlled homing procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0147 Homing parameter

- **Description**

This parameter is used to align sequences during the Homing procedure with the installation of the machine, the control unit or the drive.

Structure of homing parameter:

Bit No.	Value	Description	Comments
15-11		reserved	
10		Homing with Positive Stop	
	0	without positive stop	
	1	with positive stop (bit 5 = 1, bit 9 = 0)	see S-0-0530 Clamping torque
9		Homing with Limit switch	
	0	without limit switch	
	1	with limit switch (bit 5 = 1, bit 10 = 0]	
8		Drive controlled homing with homing distance	
	0	Homing distance is not selected	
	1	Homing distance is selected	see S-0-0297

7		Position after drive controlled homing	
	0	drive is positioned at an arbitrary position	
	1	drive is positioned at the Reference position	see S-0-0052 or S-0-0054
6		Evaluation of position feedback marker pulse	
	0	marker pulse is evaluated	
	1	marker pulse is not evaluated	
5		Evaluation of home switch	
	0	home switch is evaluated	
	1	home switch is not evaluated	
4		Interpretation in the drive	
	0	home switch and homing enable	see S-0-0407
	1	homing enable only	
3		Homing	
	0	using motor feedback	see S-0-0277
	1	using external feedback	see S-0-0115
2		Home switch (S-0-0400)	
	0	connected to the control unit	
	1	connected to the drive	
1		Position feedback marker pulse	
	0	first marker pulse after the positive edge of the home switch	see S-0-0400
	1	first marker pulse after the negative edge of the home switch	see S-0-0400
0		Homing direction	
	0	positive: increasing position values	
	1	negative: decreasing position values	

Name		Homing parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0148 Drive controlled homing procedure command

- **Description**

When the master sets and enables the Drive Controlled homing procedure command, the drive automatically activates the drive internal position control and accelerates to the homing velocity (S-0-0041) taking the Homing acceleration (S-0-0042) into account. The drive resets the bit "position feedback value status" (S-0-0403). Further options for the homing procedure are programmed in the "homing parameter" (S-0-0147). All changes of the cyclic command values are ignored as long as the procedure command is activated.

After passing over the reference marker pulse, the drive decelerates to standstill, taking the homing acceleration into account, or travels to the reference position. The procedure command "drive controlled homing" is successfully completed when the drive has stopped and the position feedback value is referred to the reference point of the machine. The drive announces this by setting the bit "position feedback value status" (S-0-0403).

The drive internally calculates the commanded position value (S-0-0047) relationship to the reference mark and adjusts S-0-0047 accordingly. The control unit must then either read the "position command value" (S-0-0047) of the drive via the service channel and sets it's position command value to this position command value, or the control sets its position command to the reference distance (S-0-0052, S-0-0054), (S-0-0147 bit 7 must be set to 1). Afterwards, the procedure command is canceled by the control unit and the drive once again follows the command values of the control unit.

An interrupt of this procedure command will result in the position feedback value not being referenced to the position feedback reference mark. Also the 'position feedback status value' bit will not be set.

When an error of C1D occurs, the procedure command results in an error in the procedure command acknowledgment.

Name		Drive controlled homing procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0149 Positive stop drive procedure command

- **Description**

The positive stop drive procedure command results in all feedback monitors being shut off which otherwise would result in a C1D shut-down error due to drive locking during the positive stop. Shutting off the feedback monitoring system applies to all drive operation modes. The sequence of the positive stop drive procedure command is identical in both operation modes; velocity control and position control. The command is positively acknowledged as soon as:

- monitoring of the feedback system is turned off;
- $|T| >= |Tlimit|$ or $|clampingtorque|$ (S-0-0530); and
- $nfeedback = 0$ is true.

Name		Positive stop drive procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0150 Reference offset 1

- **Description**

This parameter describes the distance between the reference marker pulse of position feedback 1 and the reference point.

Name		Reference offset 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0151 Reference offset 2

- **Description**

This parameter describes the distance between the reference marker pulse of position feedback 2 and the reference point.

Name	Reference offset 2

Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0152 Position spindle procedure command

- **Description**

This procedure command automatically switches the drive to internal position loop control, below the spindle positioning speed (S-0-0222), and references the spindle, if necessary.

While this command is active, all changes to cyclic command values are ignored.

Additionally, depending on the Spindle position parameter (S-0-0154), the drive positions the spindle absolute to the programmed angle position (S-0-0153) or rotates the spindle relative (incrementally) (S-0-0180). When the drive interpolator reaches the selected command value, the drive sets the status ‘Target position attained’ (S-0-0342). The status ‘In coarse position’ (S-0-0341) or ‘In Position’ (S-0-0336) are updated by the drive.

While this procedure command is active, the drive maintains the position control and adjusts to every new command value (S-0-0153 or S-0-0180) which is transferred through the service channel.

When the control unit cancels this command, the drive switches over to the mode of operation set in the control word.

Name		Position spindle procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0153 Spindle angle position

- **Description**

This parameter is the absolute spindle angle position relative to the reference point. The parameter is enabled only in connection with the position spindle procedure command (see S-0-0152) or the drive-controlled synchronous operation procedure command (see S-0-0223).

Name		Spindle angle position
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0154 Spindle positioning parameter

- **Description**

When the velocity command value is equal to zero and the position spindle procedure command is active, the turning direction for reaching the spindle angle position can be given here. If the velocity command value is not equal to zero, the current turning direction is maintained in order to reach the spindle angle position.

Structure of spindle position parameter:

Bit No.	Value	Description	Comments
15-5		reserved	
4		homing	
	0	once	a not referenced spindle activates once the homing. At every further start of the position spindle procedure command the spindle is positioning on the spindle angle position only
	1	always	at every start of the position spindle procedure command the spindle activates the homing (with home switch or encoder reference) after this the spindle is positioning on the spindle angle position.
3		spindle feedback	
	0	motor feedback	
	1	external feedback	
2			
	0	spindle angle	see S-0-0153

		position	
	1	spindle relative offset	see S-0-0180
0-1		direction	
	00	rotate clockwise	
	01	rotate counter-clockwise	
	10	take shortest path	
	11	last active rotational direction	

Name		Spindle positioning parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0155 Friction torque compensation

- **Description**

The friction torque compensation is overlaid additively to the torque command value. During addition, the friction torque compensation and torque command value need to have the same sign. The inclusion of friction torque compensation helps compensate for the frictional grip during acceleration from standstill, and during reversals.

Name		Friction torque compensation
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0156 Velocity feedback value 2

- **Description**

The velocity feedback value 2 is transferred from the drive e.g. to the control unit in order to allow the control unit to display the velocity periodically. The velocity feedback value 2 refers to the external encoder.

Name		Velocity feedback value 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	always
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0157 Velocity window

- **Description**

The velocity window" relates the current velocity to the sum of the velocity command values. If the current velocity feedback value falls within the calculated velocity window, the drive sets the status 'n feedback = n command' (S-0-0330). See clause 1.9

Name		Velocity window
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0158 Power threshold (Px)

- **Description**

The power threshold (Px) parameter determines at which power level the drive generates the status 'P ≥ Px' (see S-0-0337).

Name		Power threshold (Px)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		W
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1W

S-0-0159 Monitoring window

- **Description**

By means of the monitoring window, the maximum position deviation, as referenced to the current position feedback value, can be defined for the position feedback value. When the following distance (S-0-0189) exceeds the maximum value of the monitoring window, the drive sets the

- error for excessive position deviation in C1D (S-0-0011), and
- sets the diagnostic number (S-0-0390) to 0xC00F2028.

Name		Monitoring window
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0160 Acceleration data scaling type

- **Description**

A variety of scaling methods can be selected by means of the acceleration data scaling type parameter (see Scaling of operation data).

Table 14: Structure of the acceleration data scaling type

Bit No.	Value	Description	Comments
15-7		reserved	
6		Data reference	
	0	at the motor shaft	
	1	at the load	uses S-0-0121, S-0-0122, S-0-0123

5		Time units	
	0	seconds [s]	
	1	reserved	
4		Units for linear scaling	
	0	meters [m]	
	1	inches [in]	optional
4		Units for rotational scaling	
	0	radian [rad]	
	1	reserved	
3			
	0	preferred scaling	
	1	parameter scaling	uses S-0-0161 and S-0-0162
2–0		Scaling method	
	000	no scaling	
	001	linear scaling	
	010	rotational scaling	
	011	ramp time	see S-0-0446 Ramp reference velocity

Name		Acceleration data scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0161 Acceleration data scaling factor

- **Description**

This parameter defines the scaling factor for all acceleration data in a sub-device (see Scaling of operation data).

Name		Acceleration data scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0

	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0162 Acceleration data scaling exponent

- **Description**

This parameter defines the scaling exponent for all acceleration data in a sub-device (see Scaling of operation data).

Table 15: Structure of the scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Acceleration data scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0163 Weight counterbalance

- **Description**

This parameter is used to program the counterbalance (torque) of vertically positioned (hanging) axes in the positive or negative effective direction.

Name		Weight counterbalance
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086

Unit	see S-0-0086
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0086

S-0-0164 Acceleration feedback value 1

- **Description**

The acceleration feedback value 1 is the velocity change refered to the motor encoder. Velocity increase is described as positive acceleration. Velocity decrease is described as negative acceleration.

Name		Acceleration feedback value 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0165 Distance-coded reference marks A

- **Description**

When a measuring system with distance-coded reference marks is being used, this parameter contains the larger periodic distance between two of the reference marks.

Name		Distance-coded reference marks A
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0166 Distance-coded reference marks B

- **Description**

When a measuring system with distance-coded reference marks is being used, this parameter contains the smaller periodic distance between two of the reference marks.

Name		Distance-coded reference marks B
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0167 Frequency limit of feedback 1

- **Description**

The frequency limit of feedback 1 is the maximum frequency of the motor feedback signal, i.e., the maximum working frequency which the electronics can output in pulses per second.

If this frequency is exceeded, the drive would loose its reference to the machine zero point and bit 0 of the position feedback value status (S-0-0403) shall be reset.

Name		Frequency limit of feedback 1
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		Hz
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1Hz

S-0-0168 Frequency limit of feedback 2

- **Description**

The frequency limit of feedback 2 is the maximum frequency of the external feedback signal, i.e., the maximum working frequency which the electronics can output in pulses per second.

If this frequency is exceeded, the drive would loose its reference to the machine zero point and bit 0 of the position feedback value status (S-0-0403) shall be reset.

Name		Frequency limit of feedback 2
Attribute	Length (Octet)	4

	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		Hz
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1Hz

S-0-0169 Probe control

- **Description**

This parameter defines which probes and which edges are activated for the measuring function. With bit 5 and bit 6 either the single measuring or the continuous measuring is configured. The automatic activation of the measuring function is configured with bit 8.

This parameter shall be programmed according to the application before the measuring function is activated. For more details see clause Measuring functions.

Structure of probe control:

Bit No.	Value	Description	Comments
15-9		reserved	
8		Auto-activation	
	0	The measuring function shall be activated with the probing cycle procedure command	see S-0-0170.
	1	The measuring function is activated automatically in the drive at the phase switching from CP3 to CP4 or when leaving the parameterization level	see S-0-0422.
7		reserved	
6		Mode Probe 2	
	0	Single measuring:	the enable probe 2 shall be disabled and set once more to enable a further measuring.
	1	Continuous measuring:	measuring is processed as long as the enable probe 2 is set to 1
5		Mode Probe 1	
	0	Single measuring:	the enable probe 1 shall be

			disabled and set once more to enable a further measuring.
	1	Continuous measuring:	measuring is processed as long as the enable probe 1 is set to 1
4		reserved	
3		Probe 2 negative edge	
	0	negative edge is not active	
	1	negative edge is active	
2		Probe 2 positive edge	
	0	positive edge is not active	
	1	positive edge is active	
1		Probe 1 negative edge	
	0	negative edge is not active	
	1	negative edge is active	
0		Probe 1 positive edge	
	0	positive edge is not active	
	1	positive edge is active	

Name		Probe control
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0170 Probing cycle procedure command

- **Description**

When the master sets and enables the probing cycle procedure command, the drive reacts on the following parameters:

- probe 1/2 enable (S-0-0405, S-0-0406); and
- probe 1/2 (S-0-0401, S-0-0402) as programmed in the probe control parameter (S-0-0169).

While the procedure command is activated, the control unit can start multiple measurements.

If the control unit does not want any more measurements, the control unit cancels the procedure command.

For more details see clause 1.8.

Name		Probing cycle procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0171 Calculate displacement procedure command

- **Description**

When the master sets and enables the procedure command "calculate displacement" the drive takes the parameters:

- reference distance 1/2 (S-0-0052/S-0-0054);
- reference Offset 1/2 (S-0-0150/S-0-0151);
- marker position A and marker position B (S-0-0173/S-0-0174);

into account to calculate the displacement between the old and the new (referenced) command/feedback system.

The calculated displacement is stored in the parameters

- displacement parameter 1 (S-0-0175, motor feedback);
- displacement parameter 2 (S-0-0176, external feedback).

The feedback system for which the displacement has to be calculated is selected in the homing parameter (S-0-0147, bit 3).

When the drive recognizes the displacement as invalid, the procedure command results in an error in the procedure command acknowledgment.

For more details see clause Homing.

Name		Calculate displacement procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never

	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0172 Displacement to the referenced system procedure command

- **Description**

When the master sets and enables the procedure command "displacement to the referenced system", the drive switches to the referenced position feedback system and marks this by simultaneously setting the bit "position feedback value status" (S-0-0403).

To inform the control unit about the switching in real-time, the bit "position feedback value status" has to be assigned to a real-time status bit.

While the procedure command is activated, the control unit switches to the referenced command value system and marks this by simultaneously setting the bit "position command value status" (S-0-0404).

To inform the drive about the switching in real-time, the bit "position command value status" has to be assigned to a real-time control bit.

The bit "position command value status" has to be set by the control unit independent from the operation mode.

The procedure command is completed by the drive as soon as the bits "position feedback value status" and "position command value status" are set to 1. There is no fixed sequence to set the bits.

For more details see clause Homing.

Name		Displacement to the referenced system procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0173 Marker position A

- **Description**

When the drive recognizes the reference marker pulse of position feedback 1/2 during homing, it stores the instantaneous unreferenced position feedback value 1/2 in the parameter marker position A.

There are groups of two reference marker pulses with a distance coded feedback system.

When the drive recognizes the first reference marker pulse of distance coded position feedback 1/2 during homing, it stores the instantaneous unreferenced position feedback value 1/2 in the parameter marker position A.

Name		Marker position A
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0174 Marker position B

- **Description**

The marker position B is used additionally for distance coded feedback to be able to calculate the absolute position referred to the zero point of the feedback system.

There are groups of two reference marker pulses with a distance coded feedback system.

When the drive recognizes the second reference marker pulse of distance coded position feedback 1/2 during homing, it stores the instantaneous unreferenced position feedback value 1/2 in the parameter marker position B.

Name		Marker position B
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0175 Displacement parameter 1

- **Description**

When the procedure command "calculate displacement" (S-0-0171) or “Set absolute position” (S-0-0447) is active, the drive calculates the difference between the old position feedback value and the new position feedback value. The drive stores the difference as the "displacement Parameter 1" if motor feedback is selected.

For more details see clause Homing.

Name		Displacement parameter 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0176 Displacement parameter 2

- **Description**

When the procedure command "calculate displacement" (S-0-0171) or “Set absolute position” (S-0-0447) is active, the drive calculates the difference between the old position feedback value and the new position feedback value. The drive stores the difference as the "displacement parameter 2" if external feedback is selected.

For more details see clause 1.7.

Name		Displacement parameter 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0177 Absolute distance 1

- **Description**

This parameter describes the distance between the machine zero point and the zero point of an absolute feedback system on the motor.

Name		Absolute distance 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	CP4
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0178 Absolute distance 2

- **Description**

This parameter describes the distance between the machine zero point and the zero point of an absolute feedback system on the machine (external feedback system).

Name		Absolute distance 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	CP4
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0179 Probe status

- **Description**

If the drive with single measuring latches one or several measurement values while the procedure command "probing cycle” (S-0-0170) is activated, it simultaneously sets the assigned bit in the probe status. For every further measuring the control unit shall reset the probe enable (S-0-0405, S-0-0406) and set the probe enable again to 1.

With continuous measuring the drive sets the corresponding bits at the first measuring in the probe status. Therefore the control unit shall evaluate the parameters probe value latched (S-0-0409 to S-0-0412) or difference value latched (S-0-0522, S-0-0523) which the drive increments with every measuring.

If probe 1 enable (S-0-0405) is reset by the control unit, the drive resets the corresponding bits of probe 1.

If probe 2 enable (S-0-0406) is reset by the control unit, the drive resets the corresponding bits of probe 2.

The drive resets all bits of the probe status when the control unit cancels the "probing cycle procedure command" (S-0-0170).

The bits 0 to 3, 6 and 7 are processed identical by the drive.

Structure of probe status:

Bit No.	Value	Description	Comments
15-8		reserved	
7		Extended difference measuring 2	
	0	difference value 2 not latched	
	1	difference value 2 latched	see S-0-0523
6		Extended difference measuring 1	
	0	difference value 1 not latched	
	1	difference value 1 latched	see S-0-0522
5		Status marker losses 2	
	0	marker losses 2 not exceeded	
	1	marker losses 2 exceeded	S-0-0517 > S-0-0519
4		Status marker losses 1	
	0	marker losses 1 not excceded	
	1	marker losses 1 exceeded	S-0-0516 > S-0-0518
3		Probe 2 negative latched	see S-0-0412
	0	not latched	
	1	latched	
2		Probe 2 positive latched	see S-0-0411
	0	not latched	
	1	latched	
1		Probe 1 negative latched	see S-0-0410
	0	not latched	
	1	latched	
0		Probe 1 positive latched	see S-0-0409

	0	not latched	
	1	latched	

Name		Probe status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0180 Spindle relative offset

- **Description**

The parameter is enabled only in connection with the position spindle procedure command (see S-0-0152). The spindle relative offset is added to the absolute position value while being processed.

This parameter is used to drive the spindle of a certain number of revolutions.

Name		Spindle relative offset
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0181 Manufacturer class 2 diagnostic

- **Description**

The drive manufacturer can define additional shut-down warnings in manufacturer class 2 diagnostics. If a warning is set in the manufacturer class 2 diagnostic, the manufacturer-specific warning bit in class 2 diagnostic (S-0-0012) is set as well.

- SERCOS II only: When the manufacturer class 2 diagnostic is read via the service channel, the manufacturer specific warning bit in class 2 diagnostic is reset to 0, but the change bit for C2D (bit 12) in the drive status is not changed.
- SERCOS III only: If the warning doesn't exist, the warning is set to 0 by the drive.

Structure of manufacturer C2D:			
Bit No.	**Value**	**Description**	**Comments**
15-0		C2D	
	0	no warning	all bits are set to 0
	1	warning occurred	one or more bits are set to 1

Name		Manufacturer class 2 diagnostic
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0183 Synchronization velocity window

- **Description**

If during velocity synchronous operation the difference between the synchronous velocity command value of the lead spindle and the velocity feedback value of the synchronous spindle falls within the synchronization window, the drive sets the synchronization operation status (see S-0-0308). If necessary, the synchronization operation status can be assigned to a real-time bit.

Name		Synchronization velocity window
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0184 Synchronization velocity error limit

- **Description**

If during velocity synchronous operation the difference between the synchronous velocity command value of the lead spindle and the velocity feedback value of the synchronous spindle becomes greater than the synchronization velocity error limit value, the drive sets the synchronization error status (see S-0-0309).

If necessary, the synchronization error status can be assigned to a real-time bit.

Name		Synchronization velocity error limit
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0187 IDN-list of configurable data as producer

- **Description**

This IDN list contains all IDNs of operation data (feedback values) which can be processed by the sub-device cyclically in the AT.

If S-0-1060 exists you will find more detailed information of the connection related parameter.

Name		IDN-list of configurable data as producer
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0188 IDN-list of configurable data as consumer

- **Description**

This IDN list contains all IDNs of operation data (command values) which can be processed by the sub-device cyclically.

If S-0-1060 exists you will find more detailed information of the connection related parameter.

Name		IDN-list of configurable data as consumer
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0189 Following distance

- **Description**

The drive uses the operation data of this IDN to store the distance between current position command value and the appropriate position feedback value 1 or 2. Calculation of the following distance:

following distance = position command value – position feedback value 1 or 2

Name		Following distance
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	always
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0190 Drive controlled gear engaging procedure command

- **Description**

When the drive controlled gear engaging procedure command is activated, the drive ignores the cyclic command values and turns in the average engaging speed (S-0-0214). With this procedure command, the "gear engaging function" is activated in the drive using the parameters – engaging dither amplitude (S-0-0213) – average engaging speed (S-0-0214) – engaging dither period (S-0-0215) in order to improve the gear shifting.

When the relevant actual values fulfil the conditions of the gear engaging function, the drive signals the procedure command as performed correctly.

When the master cancels the procedure command, the drive turns off the gear engaging function and the cyclic command values are valid in the drive again.

For more details see 1.16.

Name		Drive controlled gear engaging procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0191 Cancel reference point procedure command

- **Description**

When the master sets and enables the procedure command "cancel reference point" the drive resets the bit "position feedback value status" (S-0-0403).

The procedure command is completed successfully by the drive as soon as the bit "position feedback value status" is reset to 0.

Name		Cancel reference point procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0192 IDN-list of all backup operation data

- **Description**

The IDN-list stores IDNs of all device parameter that have to be loaded in the device in order to guarantee correct operation. The master uses this list to generate a backup copy of the device parameters (e.g., on a floppy disk).

Name		IDN-list of all backup operation data
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0193 Positioning jerk

- **Description**

"Positioning jerk” is the maximum rate of change of acceleration in the operation modes "Interpolation" and "Positioning". Programming a value of zero will cause jerk limiting to be deactivated.

Name		Positioning jerk
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0194 Acceleration command value

- **Description**

The acceleration command value can be transferred as cyclic data or on demand by the control unit via the service channel.

Name		Acceleration command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0160

S-0-0195 Acceleration feedback value 2

- **Description**

The acceleration feedback value 2 is the velocity change referred to the external encoder. Velocity increase is described as positive acceleration. Velocity decrease is described as negative acceleration.

Name		Acceleration feedback value 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0196 Motor rated current

- **Description**

The motor rated current is the current at which the motor produces the rated torque/force according to the motor spec sheet. For all asynchronous motors, this parameter is used as a reference for the scaling of all torque/force data.

Name		Motor rated current
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0197 Set coordinate system procedure command

- **Description**

After activation of the “Set coordinate system procedure command”, the drive ignores the position command value and instead transfers the programmed “Initial coordinate value” (S-

0-0198) into the drive-internal position command. Additionally, the drive re-calculates all absolute values (feedbacks, position limits, etc.), relating them to the “Initial coordinate value”.

The position feedback value status (S-0-0403) and position command value status (S-0-0404) are not affected by this procedure command.

This procedure command is successfully completed by the drive when all necessary calculations are completed, the “Current coordinate offset” (S-0-0283) is calculated, and the drive has based its coordinate system on the “Initial coordinate value” (S-0-0198).

Before the control clears the command, it must also adjust its coordinate system to the same value the drive used. After clearing of the procedure command, the drive will once again act upon the position command.

The procedure command will terminate with a fault, when the drive detects an error during the command specific calculations.

Name		Set coordinate system procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0198 Initial coordinate value

- **Description**

The drives coordinate system will be set to the value programmed as the initial coordinate value during the Set coordinate system procedure command (S-0-0197)

Name		Initial coordinate value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0199 Shift coordinate system procedure command

- **Description**

After activation of the “Shift coordinate system procedure command”, the drive ignores the position command value and instead adds the programmed “Coordinate offset value” (S-0-0275) to the drive-internal position command. Additionally, the drive re-calculates all absolute values (feedbacks, position limits, etc.), relating them to the “Coordinate offset value”.

The position feedback value status (S-0-0403) and position command value status (S-0-0404) are not affected by this procedure command.

This procedure command is successfully completed by the drive when all necessary calculations are completed, the “Current coordinate offset” (S-0-0283) is calculated, and the drive has adjusted its coordinate system based upon the “Coordinate offset value”.

Before the control clears the procedure command, it must also adjust its coordinate system to the same value the drive used. After clearing of the procedure command, the drive will again act upon the position command.

The procedure command will terminate with a fault, when the drive detects an error during the command specific calculations.

Name		Shift coordinate system procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0200 Amplifier warning temperature

- **Description**

When the amplifier temperature exceeds the amplifier warning temperature value, the drive sets

- the warning bit for amplifier over temperature in C2D (S-0-0012),
- the amplifier overtemperature warning (S-0-0311) and
- the diagnostic number (S-0-0390) to 0xC00E2050.

Name		Amplifier warning temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal

	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	always
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0202 Cooling error warning temperature

- **Description**

When an error occurs in the cooling system (e.g., the temperature inside the circuitry housing exceeds the cooling error warning temperature value), the drive

- sets the warning bit for cooling error in C2D (S-0-0012 bit 3),and
- sets the cooling error warning (S-0-0313 bit 0).

Name		Cooling error warning temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	always
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0203 Amplifier shut-down temperature

- **Description**

When the amplifier temperature exceeds the amplifier shut-down temperature value, the drive sets the bit for amplifier over temperature shut-down in C1D (S-0-0011).

Name		Amplifier shut-down temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	never
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0204 Motor shut-down temperature

- **Description**

When the motor temperature exceeds the motor shut-down temperature value, the drive sets the bit for motor over temperature shut-down in C1D (S-0-0011).

Name		Motor shut-down temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	never
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0205 Cooling error shut-down temperature

- **Description**

When an error occurs in the cooling system (e.g., the temperature inside the circuitry housing exceeds the cooling error shut-down temperature value), the drive sets the bit for cooling error shut-down in C1D (S-0-0011).

Name		Cooling error shut-down temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	never
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0206 Drive on delay time

- **Description**

After torque is activated (drive status, bit 14 is set) "drive on delay time" is started. The drive follows the command values after the "drive on delay time" has elapsed.

Name		Drive on delay time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal

	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0207 Drive off delay time

- **Description**

After "drive off" (drive control, bit 15) is reset and nmin (e.g. S-0-0124) is reached, the drive off delay time is started and the locking of the brake is initiated. The torque remains activated in the drive until this drive off delay time is elapsed.

Example: Used as break delay time (clamping or release).

Name		Drive off delay time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0208 Temperature data scaling type

- **Description**

This scaling type parameter determines whether temperature is used in units of degree C or F. Temperature scaling is 0,1 degree C or 0,1 F. The data length of temperature data is fixed to two bytes.

Table 16: Structure of temperature data scaling type

Bit No.	Value	Description	Comments
15-1		reserved	
0		Temperature data scaling type	
	0	entry in 0,1 degree C	
	1	entry in 0,1 degree F	

Name	Temperature data scaling type

Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0209 Lower adaptation limit

- **Description**

Below the lower adaptation limit, the proportional gain adaptation (S-0-0211) and the integral action time adaptation (S-0-0212) are enabled. Above the lower adaptation limit, the adaptation of proportional gain and the integral action time change linearly from the lower level to the values at the upper adaptation limit, as defined for the velocity loop proportional gain (S-0-0100) and the velocity loop integral action time (S-0-0101).

Name		Lower adaptation limit
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0210 Upper adaptation limit

- **Description**

Above the upper adaptation limit, the velocity loop proportional gain (S-0-0100) and the velocity loop integral action time (see S-0-0101) are enabled. Below the upper adaptation limit, the proportional gain and the integral action time velocity loops change linearly from the higher level to the values at the lower adaptation limit as defined for the proportional gain adaptation (S-0-0211) and the integral action time adaptation (S-0-0212).

Name		Upper adaptation limit
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never

	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0211 Adaptation proportional gain

- **Description**

Adaptation proportional gain determines the percentage value below the lower adaptation limit, dependent on the velocity loop proportional gain (S-0-0100). Above the upper adaptation limit, adaptation proportional gain is not enabled. Between the upper and lower adaptation limits, the velocity loop proportional gain changes linearly, dependent on the adaptation proportional gain and the actual velocity (example: see clause 1.6.2.1.

Name		Adaptation proportional gain
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1%

S-0-0212 Adaptation integral action time

- **Description**

Adaptation integral action time determines the percentage value below the lower adaptation limit, dependent on the velocity loop integral action time (S-0-0101). Above the upper adaptation limit, adaptation integral action time is not enabled. Between the lower and upper adaptation limits, the velocity loop integral action time changes linearly, dependent on the adaptation integral action time and the actual velocity (example: see clause 1.6.2.1).

Name		Adaptation integral action time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1%

S-0-0213 Engaging dither amplitude

- **Description**

The engaging dither amplitude defines the maximum speed of the drive in both directions during the drive controlled gear engaging procedure command. Data reference is the motor shaft.

Name		Engaging dither amplitude
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0214 Average engaging speed

- **Description**

During the drive controlled gear engaging procedure command, the drive adds the programmed average engaging speed to the engaging dither amplitude. Data reference is the motor shaft.

Name		Average engaging speed
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0215 Engaging dither period

- **Description**

During the drive controlled gear engaging procedure command, the drive oscillates at its programmed engaging dither amplitude, average engaging speed, and engaging dither period.

Name		Engaging dither period
Attribute	Length (Octet)	2

	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0216 Switch parameter set procedure command

- **Description**

This procedure command allows the system to switch parameter sets and/or gear ratio. The drive switches to the parameter set which is programmed in the parameter set preselection (S-0-0217). If a gear ratio has also been programmed in the gear ratio preselection (S-0-0218), the gear ratio is switched as well.

Name		Switch parameter set procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0217 Parameter set preselection

- **Description**

The parameter set of the drive is selected by means of the parameter set preselection. The switch parameter set procedure command (see S-0-0216) is used to switch parameter sets. If the drive has no switchable parameter sets, it will only accept parameter set 0. Therefore, parameter set 0 must be available in every drive and will be activated during initialization.

Structure of parameter set preselection:			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2–0		parameter set preselection	
	000	parameter set 0	

	001	parameter set 1	
	010	parameter set 2	
	011	parameter set 3	
	100	parameter set 4	
	101	parameter set 5	
	110	parameter set 6	
	111	parameter set 7	

Name		Parameter set preselection
Attribute	Length (Octet)	2
	Data Type and Display Format	hexa decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0218 Gear ratio preselection

- **Description**

The gear ratio preselection selects the gear ratio of the drive. The switch parameter set procedure command (see S-0-0216) is used to switch the gear ratio. If the drive switches to another gear ratio, additional parameters (see S-0-0217, S-0-0219) can be activated.

Structure of the gear ratio preselection:

Bit No.	Value	Description	Comments
15-3		reserved	
2–0		gear ratio preselection	
	000	gear ratio 0	
	001	gear ratio 1	
	010	gear ratio 2	
	011	gear ratio 3	
	100	gear ratio 4	
	101	gear ratio 5	
	110	gear ratio 6	
	111	gear ratio 7	

Name		Gear ratio preselection
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0219 IDN–list of parameter set

- **Description**

IDN-list of S-0-0219 contains all IDNs of parameter set 0 which are also supported in the parameter sets 1 7.

IDN-list S-X-0219 (X =1..7) contains all the IDNs of operation data which are changed automatically when parameter set X is activated.

An IDN-list S-X-0219 (X = 1..7) can contain only parameters of parameter set 0.

Name		IDN-list of parameter set
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0220 Minimum spindle speed

- **Description**

When the speed falls below minimum spindle speed, the drive sets the status nfeedback ≥ minimum spindle speed (S-0-0339). After that the drive can shift to another gear (see also S-0-0216).

Name		Minimum spindle speed
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0221 Maximum spindle speed

- **Description**

The maximum spindle speed indicates the limiting speed for the actual gear. When the speed exceeds the maximum spindle speed, the drive sets the status nfeedback ≥ maximum spindle speed (S-0-0340).

Name		Maximum spindle speed
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0222 Spindle positioning speed

- **Description**

When the position spindle procedure command (see S-0-0152) is received, the drive accelerates or decelerates to the spindle positioning speed, depending upon the current speed.

Name		Spindle positioning speed
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	4
	Write Protection	never
	Conversion Factor	1
Unit		min^{-1}
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		$10^{-4}min^{-1}$

S-0-0223 Drive controlled synchronous operation procedure command

- **Description**

When the master activates the drive controlled synchronous operation procedure command, the synchronous spindle is synchronized on the lead spindle as programmed in the synchronous operation parameter (S-0-0225).

The synchronous operation is cancelled by disabling the procedure command. The command is executed correctly when synchronized operation status (S-0-0308) has been reached. If the synchronous spindle generates an error of C1D, the procedure command results in an error in the procedure command acknowledgment. If the lead spindle generates errors of C1D, synchronized operation continues.

The master transfers the drive controlled synchronous operation procedure command only to the synchronous spindle. Synchronization between lead and synchronous spindle must be maintained during speed changes.

Name		Drive controlled synchronous operation procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0224 Lead spindle address

- **Description**

This parameter contains the drive address of the lead spindle responsible for synchronous operation. Command values are taken from this address during the synchronous operation command.

Name		Lead spindle address
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		maximum slave address
Scaling / Resolution of LSB		1

S-0-0225 Synchronous operation parameter

- **Description**

This parameter contains the control commands for the synchronous operation function.

Structure of the synchronous operation parameter			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2		speed ratio	
	0	changing speed ratio	from lead spindle revolutions (S-0-0226) to synchronous spindle revolutions (S-0-0227).
	1	changing of the speed ratio is complete	the synchronous spindle accepts the ratio.
1–0		synchronous mode	
	00	velocity synchronous mode	For the synchronization only the parameters synchronization velocity window S-0-0183 and synchronization velocity error limit S-0-0184 are necessary.
	01	reserved	
	10	relative angle synchronous mode	**without** regard to a reference point and synchronous position offset S-0-0230.
	11	absolute angle-synchronous mode	**with** regard to a reference point and position-dependent parameter of the synchronous spindle:

Name		Synchronous operation parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0226 Lead spindle revolutions

- **Description**

The speed ratio between the lead spindle and the synchronous spindle is calculated using S-0-0226 and S-0-0227.

$$\text{Speed ratio} = \frac{\text{leadspindlerevolutions}}{\text{synchronousspindlerevolutions}}$$

Lead spindle revolutions shall be entered as integers.

Name		Lead spindle revolutions
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		revolution
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		revolutions

S-0-0227 Synchronous spindle revolutions

- **Description**

Refer to lead spindle revolutions (S-0-0226) for speed ratio.

Synchronous spindle revolutions shall be entered as integers.

Name		Synchronous spindle revolutions
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		revolution
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		revolution

S-0-0228 Synchronization position window

- **Description**

If during synchronization the difference between the synchronous position command value of the lead spindle and the position feedback value of the synchronous spindle falls within the synchronization window, the drive sets the status for synchronization (see S-0-0308). If necessary, the synchronization operation status can be assigned to a real-time bit.

Name		Synchronization position window
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0229 Synchronization position error limit

- **Description**

If during synchronization the difference between the synchronous position command value of the lead spindle and the position feedback value of the synchronous spindle becomes greater than the synchronization error limit value, the drive sets the synchronization error status (see S-0-0309). If necessary, the synchronization error status can be assigned to a real-time bit.

Name		Synchronization position error limit
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0230 Synchronous position offset

- **Description**

This parameter describes the offset angle between the reference points for lead and synchronous spindle.

Name		Synchronous position offset
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never

	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0254 Actual parameter set

- **Description**

This parameter stores the current active parameter set in the drive. If a parameter set is to be switched, it is important that the next consecutive parameter set is updated and correct, and preselected in the parameter set preselection (S-0-0217). Parameter set 0 must be active in every drive before and during initialization.

Structure of the actual parameter set:			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2–0		actual parameter set	
	000	parameter set 0 active	
	001	parameter set 1 active	
	010	parameter set 2 active	
	011	parameter set 3 active	
	100	parameter set 4 active	
	101	parameter set 5 active	
	110	parameter set 6 active	
	111	parameter set 7 active	

Name		Actual parameter set
Attribute	Length (Octet)	2
	Data Type and Display Format	hexa decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0255 Actual gear ratio

- **Description**

This parameter stores the current active gear ratio in the drive. If the gear ratio is to be switched, it is important that the next consecutive gear ratio is updated and correct and it must be preselected in the gear ratio preselection (S-0-0218).

Structure of the actual gear ratio:

Bit No.	Value	Description	Comments
15-3		reserved	
2–0		actual gear ratio	
	000	gear ratio 0 active	
	001	gear ratio 1 active	
	010	gear ratio 2 active	
	011	gear ratio 3 active	
	100	gear ratio 4 active	
	101	gear ratio 5 active	
	110	gear ratio 6 active	
	111	gear ratio 7 active	

Name		Actual gear ratio
Attribute	Length (Octet)	2
	Data Type and Display Format	hexa decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0256 Multiplication factor 1

- **Description**

“Multiplication Factor 1” defines the factor that the motor feedback signal is multiplied by in the drive (see S-0-0116). Multiplication factor 1 can also be determined by the drive, taking the “Maximum travel range” (S-0-0278) into account. The drive-internal resolution for the motor encoder is determined from the resolution of feedback 1 (S-0-0116) and multiplication factor 1.

Name		Multiplication factor 1
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0257 Multiplication factor 2

- **Description**

"Multiplication Factor 2" defines the factor that the external feedback signal is multiplied by the drive (see S-0-0117). Multiplication factor 2 can also be determined by the drive, taking the "Maximum travel range" (S-0-0278) into account. The drive-internal resolution for the external encoder is determined from the resolution of feedback 2 (S-0-0117) and multiplication factor 2.

Name		Multiplication factor 2
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0258 Target position

- **Description**

In the operation mode "Interpolation", the control sends the "Target position" as a command value to the drive. The drive travels to the target position, taking into account the positioning velocity (S-0-0259), positioning acceleration (S-0-0260), and positioning jerk (S-0-0193).

Name		Target position
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0259 Positioning velocity

- **Description**

The “positioning velocity” is used in the operation modes “Interpolation” and "Positioning" as the velocity to travel to the “Active target position” (S-0-0430).

Name		Positioning velocity
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0260 Positioning acceleration

- **Description**

The “positioning acceleration” is used in the operation modes “Interpolation” and "Positioning" as the rate to accelerate to and decelerate from the positioning velocity (S-0-0259).

If the drive supports the positioning deceleration (S-0-0359), then a separate deceleration can be adjusted.

Name		Positioning acceleration
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0261 Coarse position window

- **Description**

When the difference between the accumulated position command value and the position feedback value is within the range of the “coarse position window”, then the drive sets the status “In coarse position” (S-0-0341). If needed, the status 'in coarse position' is assigned to a real-time bit.

Name		Coarse position window
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0262 Load defaults procedure command

- **Description**

Load defaults procedure command

When the master sets and enables the load defaults procedure command, the default parameter values will be activated. The scope and contents of the default parameter values (for example limit values, velocity loop settings, etc.) are determined by the device supplier. The default parameter values are not optimized for the respective application. Therefore the default parameter values allow a problem free inter operation between the sub-device and its connected components.

Name		Load defaults procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0263 Load working memory procedure command

- **Description**

When the master sets and enables the Load working memory procedure command, all data necessary for operation (see S-0-0192) will be loaded from the device’s nonvolatile memory into its “active memory”. After power on, the device automatically transfers the data from non-volatile memory into the active memory.

NOTE! This procedure command will cause active parameters to be overwritten.

Name		Load working memory procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0264 Backup working memory procedure command

- **Description**

Backup working memory procedure command

When the master sets and enables the Backup working memory procedure command, all data necessary for operation (see S-0-0192) will be loaded from the device's "active memory" into its non-volatile memory.

NOTE! This procedure command will cause previously saved parameters to be overwritten.

Name		Backup working memory procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0265 Language selection

- **Description**

This parameter can be used to select one of the languages available in the device (see S-0-0266). By changing the language selection, text from the device such as

- Name (element 2)
- Unit (element 4) and
- all parameter with data type and display format = text (e.g. S-0-0095)

will be displayed in the selected language.

Language Codes			
Bit No.	**Value**	**Description**	**Comments**
15-5		reserved	
4-0		language selection	
	00000	German	
	00001	English	
	00010	French	
	00011	Spanish	
	00100	Italian	
	00101	Portuguese	
	00110	Polish	
	00111	Hungarian	
	01000	Russian	
	01001	Swedish	
	01010	Danish	
	01011	Norwegian	
	01100-01111	reserved	

Name		Language selection
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		1
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0266 List of available languages

- **Description**

This list contains codes for all languages currently available in the device for language selection (See S-0-0265). This list is required if the device cannot manage or save all languages in its memory simultaneously.

Name		List of available languages
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal

		(Language Codes see S-0-0265)
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0267 Password

- **Description**

SERCOS interface supports password that can be used to write protect selected parameters stored in the device. S-0-0267 is used to modify passwords, as well as enable and disable write protection.

A password shall be between 3 and 10 octets long (3 - 10 UTF8 characters except space, UTF8 code 0x20). Acceptable characters include the 26 character alphabet, both upper and lower case (A-Z, a-z). Character recognition is case sensitive. The numeric characters 0-9 are also acceptable. The password is stored in non-volatile memory.

If a control unit or operator interface attempts to modify a IDN listed in the IDN list of password protected data (S-0-0279), the device will generate the following error message in the Service Channel:

Error Code: 0x7009 Operation data is password write-protected.

Reading the current password with activated write protection:

If the password is read (S-0-0267) over the service channel, the device will not send the password. A string of 10 characters (UTF8 code 0x2A = *) will be sent instead (**********).

Modifying the password:

To modify the password, the currently active password, the new password, and a second verification of the new password are transmitted via the service channel to the device. A space (UTF8 code 0x20) character is used to delimit the passwords. The new parameter and the verification copy must match for the device to accept the change. Modifying the password will cause write-protection to be activated.

Activating password write protection:

Password write-protection will be activated:

1. by switching on/off the power of the device
2. by attempting to send S-0-0267 to the device with anything other than the current password.

Activating password write protection does not change bits 28, 29 or 30 in the attribute, and affects all parameters specified in the IDN list of password protected data (S-0-0279).

Deactivating password write-protection:

Sending S-0-0267 to the device with the currently active password will deactivate write-protection. Parameters affected by password write protection may then be altered.

With any write access which does not change the password

- the password write protection also is not changed and
- the device returns the error message Invalid data (Error code 0x7008) via the service channel.

In the case of an unknown password, a supplier designed master password shall be available to deactivate password write protection. This password shall always be available as a current password, in addition to the user specified password. The device supplier shall provide this or another password in documentation shipped with the device, so that the user is able to set up their own password.

Name		Password
Attribute	Length (Octet)	1, variable min = 3, max = 10
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0268 Angular setting

- **Description**

This parameter describes an absolute angular setting related to the reference point of the synchronous spindle. This parameter allows the synchronous spindle to be rotated a specific amount in relation to the lead spindle. Situations such as arrangements where the tool change spindle orientation is different than the spindle transfer position can make use of this parameter.

Name		Angular setting
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076

	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0269 Storage mode

- **Description**

The storage mode parameter setting determines whether data received over the service channel is stored temporarily (e.g., in RAM), or remanent (e.g., EEPROM). Which parameter are affected by the storage mode setting are to be defined by the device supplier in the device documentation.

Table 17: Structure of Storage mode

Bit No.	Value	Description	Comments
15-1		reserved	
0		storage mode	
	0	Data stored remanent	
	1	Data stored not remanent	

Name		Storage mode
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0270 IDN list of selected backup operation data

- **Description**

This IDN list is used to define a subset of a sub-device's entire parameters that should be stored into the non-volatile memory of the sub-device. The selectively backup working memory procedure command (S-0-0293), will only store the operation data of this IDN list into the non-volatile memory.

Name		IDN list of selected backup operation data
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable

	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0272 Velocity window percentage

- **Description**

The velocity window percentage refers to a percentage of the "Velocity command value" (S-0-0036). See S-0-0330 for additional information. If the velocity feedback value (S-0-0040) is found to be within a window of the velocity command defined by this percentage, the drive will set the status 'n feedback = n command' (S-0-0330).

Name		Velocity window percentage
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	2
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,01%

S-0-0273 Maximum drive off delay time

- **Description**

After "drive off" (drive control, bit 15) is reset, the „maximum drive off delay time" is started. After the „maximum drive off delay time" is elapsed, the locking of the brake is initiated and the torque is disabled.

Name		Maximum drive off delay time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	0,1ms

S-0-0275 Coordinate offset value

- **Description**

Using the Shift coordinate system procedure command (S-0-0199), the coordinate system of the drive is shifted by coordinate offset value

Name		Coordinate offset value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0276 Return to Modulo range procedure command

- **Description**

Activating the command “Return to Modulo range procedure command” causes positive and negative commanded and actual positions, as well as angular values, to be recalculated, based upon the Modulo value (S-0-0103), into a positive command and actual positions.

While the command is active, the drive will ignore cyclic command values.

The bits “Position feedback value status” (S-0-0403) and “Position command value status” (S-0-0404) are not affected by this command.

This command is successfully completed by the drive when all necessary calculations are completed.

Before the control clears the command, it must also adjust it’s position command to the value the drive calculates After clearing of the command, the drive will again act upon the command for the selected operation mode.

The command will terminate with a fault, when the drive detects an error during the command specific calculations.

This command is only effective in the drive, when the parameter “Position data scaling type” (S-0-0076) is programmed for absolute format.

Name		Return to Modulo range procedure command
Attribute	Length (Octet)	2

	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0277 Position feedback 1 type

- **Description**

The position feedback 1 type parameter refers only to a motor feedback. This parameter is programmed to define the corresponding conditions which apply to the motor feedback (see also S-0-0115).

Structure of Position Feedback 1 and 2			
Bit No.	**Value**	**Description**	**Comments**
15-10		reserved	
9		cyclic marker evaluation	
	0	not active	
	1	active	for incremental measuring system (no distance coded) with a cyclic marker pulse use the function "cyclic marker pulse detection".
8		reserved	
7,6		Type of measuring system and evaluation	
	x0	relative	(incremental) measuring system, no absolute evaluation
	01	absolute measuring system	Absolute evaluation of the measuring system. The initialization of the position is checked by the drive. Set absolute position S-0-0447 has to be done for homing.
	11	incremental use of a absolute measuring system	No monitoring for the initialization of the position. Set absolute position S-0-0447 is not available. For homing the axis use S-0-0148 or S-0-0146
5		Structure of distance coded feedback	
	0	counting positive with positive direction	

	1	counting negative with positive direction	
4		Marker pulse quantity	
	0	only one reference marker pulse	
	1	multiple cyclic reference marker pulses	
3		Direction polarity	
	0	not inverted	
	1	inverted	
2		Feedback resolution	Feedback 1 see S-0-0116 Feedback 2 see S-0-0117
	0	resolution is metric	
	1	resolution is inches	
1		Distance coded feedback	
	0	no distance coded reference marks	
	1	distance coded reference marks	see S-0-0165, S-0-0166
0		Feedback type	
	0	rotational feedback	
	1	linear feedback	

Name		Position feedback 1 type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0278 Maximum travel range

- **Description**

Via this parameter the valid range of position data is defined. For Modulo axes, at the minimum the Modulo value (S-0-0103) must be programmed. Using the gear ratio, the feed constant, and the motor feedback resolution, the drive calculates Multiplication factor 1 (S-0-0256). Using the gear ratio, the feed constant, and the external feedback resolution, the drive calculates Multiplication factor 2 (S-0-0257).

Name		Maximum travel range
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0279 IDN list of password protected data

- **Description**

The user can enter into this IDN list the IDN numbers whose operation data should be write-protected via the Password (S-0-0267).

Name		IDN list of password protected data
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		1
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0280 Underflow threshold

- **Description**

The underflow and overflow threshold parameters are only required when the operation modes

- Position mode,
- Drive controlled interpolation,
- Positioning or
- Block modes

are used with an endless turning axis.

When the underflow or overflow values are reached or exceeded, the drive conducts an automatic correction calculation on both the actual and commanded position systems.

Underflow and Overflow thresholds are activated in the Position polarity parameter (S-0-0055) when the Position data scaling type parameter (S-0-0076) is set for absolute format.

The difference between “old” and “new” position commands may not be more than one-half the difference between the underflow and overflow thresholds.

- In case of SERCOS II $\frac{(S-0-0281)-(S-0-0280)}{2}$ = Maximum position command difference

- In case of SERCOS III $\frac{(S-0-0281)-(S-0-0280)}{2(tPcyc)}$ = Maximum position command difference

Name		Underflow threshold
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0281 Overflow threshold

- **Description**

The underflow and overflow threshold parameters are only required when the operation modes

- Position mode,
- Drive controlled interpolation,
- Positioning or
- Block modes

are used with an endless turning axis.

When the underflow or overflow values are reached or exceeded, the drive conducts an automatic correction calculation on both the actual and commanded position systems.

Underflow and Overflow thresholds are activated in the Position polarity parameter (S-0-0055) when the Position data scaling type parameter (S-0-0076) is set for absolute format.

The difference between “old” and “new” position commands may not be more than one-half the difference between the underflow and overflow thresholds.

- In case of SERCOS II $\frac{(S-0-0281)-(S-0-0280)}{2}$ = Maximum position command difference

- In case of SERCOS III $\frac{(S-0-0281)-(S-0-0280)}{2(tPcyc)}$ = Maximum position command difference

Name		Overflow threshold
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0282 Positioning command value

- **Description**

The positioning command value is sent to the drive as a relative (incremental) or absolute position command when in "Positioning" operation mode. Toggling bit 0 of “Positioning control word” (S-0-0346) will cause the drive to add or copy the positioning command value to the “Active target position” (S-0-0430), and move to that new position taking into account the positioning velocity (S-0-0259), positioning acceleration (S-0-0260), and positioning jerk (S-0-0193).

Name		Positioning command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter

	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0283 Current coordinate offset

- **Description**

Via the "Set coordinate system procedure command" (S-0-0197) and the "Shift coordinate system procedure command" (S-0-0199), the drive calculates the offset from the current system to the new coordinate system (referenced to the machine reference point). The control is then always able to reference the absolute position to the machine reference point, even when the coordinate system has been shifted several times.

Name		Current coordinate offset
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0284 Secondary operation mode 4

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 4
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0285 Secondary operation mode 5

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 5
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0286 Secondary operation mode 6

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 6
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0287 Secondary operation mode 7

- **Description**

The drive modes of operation defined by this parameter become active when the operation mode is selected via bits 10, 9 and 8 in the drive control. The activated operation mode is indicated by bits 10, 9 and 8 of the drive status.

A drive shall only support bit 10 in the drive control and drive status if it supports secondary operation modes 4 to 7.

Name		Secondary operation mode 7
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0292 List of supported operation modes

- **Description**

In this list, the drive enters the codes (see Table 18) for all operation modes it supports. Hereby not only SERCOS specified operation modes supported, but also supplier designated coding of operation modes. Using this list, the control as well as the operator interface can determine what operation modes the drive supports.

Name		List of supported operation modes
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hex (codes)
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

18: Structure of drive operation 1= manufacturer operation mode 0= SERCOS operation mode	reserved	0= without axis control word 1= with axis control word (S-0-0520)	0= without transition support 1= with transition support	Expanded operation modes	0= with following error 1= without following error	Basic operation modes	Operation modes
15	14 - 10	9	8	7 - 4	3	2 1 0	Bit number
1	xx xxx	x	x	xxxx	x	xxx	Operation mode (bits 14-0) defined by manufacturer
0	000 00	0	0	0000	0	000	No operation mode defined
0	000 00	0	0	0000	0	001	Torque/Force control
0	000 00	0	0	0000	0	010	Velocity control
0	000 00	0	0/1	0000	0/1	011	Position control using position feedback value 1
0	000 00	0	0/1	0000	0/1	100	Position control using position feedback value 2
0	000 00	0	0/1	0000	0/1	101	Position control using position feedback values 1 and 2
0	000 00	1	0	0000	0	101	Position control using axis control word
0	000 00	0	0	0000	0	110	Pressure control
0	000 00	0	0	0000	0	111	Operation mode without control loops
0	000 00	0	0/1	0001	0/1	011	Interpolation using position feedback value 1
0	000 00	0	0/1	0001	0/1	100	Interpolation using position feedback value 2
0	000 00	0	0/1	0001	0/1	101	Interpolation using position feedback

							values 1 and 2
0	000 00	1	0	0001	0	101	Interpolation using axis control word
0	000 00	0	0/1	0010	0/1	011	Positioning using position feedback value 1
0	000 00	0	0/1	0010	0/1	100	Positioning using position feedback value 2
0	000 00	0	0/1	0010	0/1	101	Positioning using position feedback values 1 and 2
0	000 00	1	0	0010	0	101	Positioning using axis control word
0	000 00	0	0/1	0011	0/1	011	Block mode using position feedback value 1
0	000 00	0	0/1	0011	0/1	100	Block mode using position feedback value 2
0	000 00	0	0/1	0011	0/1	101	Block mode using position feedback value 1 and 2
0	000 00	1	0	0011	0	101	Block mode using axis control word
0	000 00	0	0	0100	0	010	Synchronous Operation with velocity control
0	000 00	0	0/1	0100	0/1	011	Synchronous Operation with position control using position feedback value 1
0	000 00	0	0/1	0100	0/1	100	Synchronous Operation with position control using position feedback value 2
0	000 00	0	0/1	0100	0/1	101	Synchronous Operation with position control using position feedback values 1 and 2
0	000 00	1	0	0100	0	101	Synchronous Operation with position control using axis control word

S-0-0293 Selectively backup working memory procedure command

- **Description**

When the master sets and enables the selectively backup working memory procedure command, all parameter programmed in the IDN list of selected backup operation data (S-0-0270) will be loaded from the device's "active memory" and stored into its non-volatile memory.

NOTE! This procedure command will cause previously saved parameters to be overwritten.

Name		Selectively backup working memory procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0294 Divider modulo value

- **Description**

do not use this IDN, it will be deleted (TWG Drive)
If the physical modulo value isn't programmed in the modulo value (S-0-0103), a factor can be entered in this parameter to adapt the modulo value on the physical modulo value.

If the modulo value corresponds to the physical modulo value, then the factor must be programmed on the value 1.

Name		Divider modulo value
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0295 Drive enable delay time

- **Description**

When "drive enable" is set (drive control, bit 14) the "drive enable delay time" is started. Motor current (torque) will first be activated after this time delay. The enable delay is required at use of a contactor in the motor cable.

Name		Drive enable delay time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1ms

S-0-0296 Velocity feed forward gain

- **Description**

Velocity feed forward is effective in the operation mode “Position control without following error (lag-less)”, and serves to reduce the velocity-dependent following distance. The control unit has to check the data length before access the data.

Name		Velocity feed forward gain
Attribute	Length (Octet)	SERCOS III: 4 SERCOS II: 2 or 4 (manufacturer specific)
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0297 Homing distance

- **Description**

The homing distance is necessary to also use the "drive controlled homing procedure command" (S-0-0148) (e.g. distance coded measurement systems) on mechanical coupled axes (e.g. gantry axes). The procedure command must be synchronized via the drive start / stop bit.

The function is selected via bit 8 in the homing parameter (S-0-0147).

Remark:

On distance coded measurement systems the homing distance must be chosen long enough, that the drive can surely recognise two distance coded reference marks (S-0-0165, S-0-0166).

After activation of "drive controlled homing procedure command ", the drive travels the homing distance. At the first reference mark the actual feedback value is saved into the marker position A (S-0-0173), at the second reference mark the actual feedback value is saved into the marker position B (S-0-0174).

Name		Homing distance
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0298 Suggest home switch distance

- **Description**

There is an optimum distance between the marker pulse of a feedback and the home switch. To assist setup personnel in adjusting this during the first commissioning, this parameter displays the distance between the home switch and the ideal point. The home switch must be either mechanically or electronically shifted by this value. For electronic shifting, this value is entered into the parameter "home switch offset 1 or 2" (S-0-0299 or S-0-0358).

Name		Suggest home switch distance
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0299 Home switch offset 1

- **Description**

With digital drives, it is no longer technically possible to mechanically rotate the feedback system on the motor in order to shift the relationship between the marker pulse and home switch. With milled switch dog rails neither can the home switch dog be shifted.

Based on these conditions, the home switch offset 1 parameter is programmed during startup so that the home position is unambiguous. The required offset between the home switch and marker pulse is provided by the drive with the aid of the programmed home switch offset 1.

The "suggest home switch distance" (S-0-0298) can be transferred into this parameter.

Name		Home switch offset 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0308 Synchronization operation status

- **Description**

This parameter defines an IDN for the synchronization operation status. Therefore, the synchronization operation status can be assigned to a real-time status bit (see S-0-0305).

Case 1, Angle synchronous mode: The synchronization operation status is set when the difference between the synchronous position command value of the lead spindle and the position feedback value of the synchronous spindle is within the value range of the synchronization position window (S-0-0228).

Case 2, Velocity synchronous mode: The synchronization operation status is also set when the difference between the synchronous velocity command value of the lead spindle and the velocity feedback value of the synchronous spindle is within the value range of the synchronization velocity window (S-0-0183).

Bit 0 is defined for operation data only.

Structure of the synchronization operation status			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		Synchronization operation status	

	0	not synchronized	
	1	synchronized	

Name		Synchronization operation status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0309 Synchronization error status

- **Description**

This parameter defines an IDN for the synchronization error status. Therefore, the synchronization error status can be assigned to a real-time status bit (see S-0-0305).

Case 1, Angle synchronous mode: The synchronization error status is set when the difference between the synchronous position command value of the lead spindle and the position feedback value of the synchronous spindle falls outside the value range of the the synchronization position error limit value (S-0-0229).

Case 2, Velocity synchronous mode: The synchronization error status is also set when the difference between the synchronous velocity command value of the lead spindle and the velocity feedback value of the synchronous spindle falls outside the value range of the the synchronization velocity error limit value (S-0-0184).

Bit 0 is defined for operation data only.

Structure of the synchronization error status

Bit No.	Value	Description	Comments
15-1		reserved	
0		synchronization error status	
	0	not exceeded	
	1	exceeded	

Name		Synchronization error status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0

	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0310 Overload warning

- **Description**

This parameter is used to define an IDN for the overload warning. This allows the overload warning to be assigned, for instance, to a real-time status bit (see S-0-0305). The overload warning is set appropriately according to the overload limit value (see S-0-0114).

Bit 0 is defined for operation data only.

Structure of overload warning			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		overload warning	
	0	no overload warning	
	1	overload warning	

Name		Overload warning
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0311 Amplifier overtemperature warning

- **Description**

This parameter is used to define an IDN for the amplifier overtemperature warning. This allows the amplifier overtemperature warning to be assigned, for instance, to a real-time status bit (see S-0-0305). The amplifier overtemperature warning is defined as a C2D bit (S-0-0012) and is set appropriately according to the amplifier temperature warning (see S-0-0200).

Bit 0 is defined for operation data only.

Structure of amplifier overtemperature warning			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		Amplifier overtemperature warning	
	0	no amplifier overtemperature warning	
	1	amplifier overtemperature warning	

Name		Amplifier overtemperature warning
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0312 Motor overtemperature warning

- **Description**

This parameter is used to define an IDN for the motor over temperature warning. This allows the motor over temperature warning to be assigned, for instance, to a real-time status bit (see S-0-0305). The motor over temperature warning is defined as a C2D bit (S-0-0012) and is set appropriately according to the motor temperature warning (see S-0-0201).

Bit 0 is defined for operation data only.

Structure of motor overtemperature warning			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		Motor overtemperature warning	
	0	no motor overtemperature warning	
	1	motor overtemperature warning	

Name		Motor overtemperature warning
Attribute	Length (Octet)	2
	Data Type and Display Format	binary

	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0313 Cooling error warning

- **Description**

This parameter is used to define an IDN for the cooling error warning. This allows the cooling error warning to be assigned, for instance, to a real-time status bit. The cooling error warning is defined as a C2D bit (S-0-0012) and is set appropriately according to the cooling error warning temperature (see S-0-0202).

Bit 0 is defined for operation data only.

Cooling error warning			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
15-1		cooling error warning	
	0	no cooling error warning	
	1	cooling error warning	

Name		Cooling error warning
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0323 Target position outside of travel range

- **Description**

This parameter is used to define an IDN for the target position outside of travel range warning. This allows the warning to be assigned, for instance, to a real-time status bit (for example S-0-0305). The Warning is set appropriately when the active target position (S-0-0430) is outside the position limit values (positive or negative, S-0-0049, S-0-0050).

- The target position outside of travel range warning is defined in (S-0-0437 bit 14), and
- sets the diagnostic number (S-0-0390) to 0xC00E2053.

NOTE: If the actual position exceeds a position limit value, the bit over travel limit is exceeded in C1D is set.

Bit 0 is defined for operation data only.

Structure of target position outside of travel range			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		target position outside of travel range	
	0	target position within position limit values	
	1	target position outside of position limit values	

Name		Target position outside of travel range
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0326 Parameter checksum

- **Description**

After being switched on, the control unit is able to find out by comparing, whether the parameters or firmware have changed in the device. The device calculates the parameter checksum, when the IDN is read via the service channel.

For the calculation of the checksum, the parameters which are listed in S-0-0327 are used. If S-0-0327 is not supported by the device, the IDN list S-0-0192 is taken for the checksum calculation.

Name		Parameter checksum
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal

	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0327 IDN list of checksum parameter

- **Description**

This IDN-list contains all ident numbers for the calculation of the parameter checksum. See S-0-0326.

Name		IDN list of checksum parameter
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0328 Bit number allocation list for signal status word

- **Description**

In this configuration list the bit numbers of the operation data are programmed, which are copied into the signal status word (S-0-0144). The sequence of the bit numbers in the configuration list sets the numerical order in the signal status word. The first bit number in the configuration list sets bit 0, the last bit number sets bit 15 into the signal status word. Maximum 16 bit number can be taken into this list. (see also S-0-0026).

Name		Bit number allocation list for signal status word
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal (bit number)
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	63
Scaling / Resolution of LSB	1

S-0-0329 Bit number allocation list for signal control word

- **Description**

In this configuration list the bit numbers of the operation data are programmed, which are contained in the signal control word (S-0-0145). The sequence of the bit numbers in the configuration list sets the numerical order in the signal control word. The first bit number in the configuration list sets bit 0, the last bit number sets bit 15 in the signal control word. Maximum 16 bit number can be taken into this list. (see also S-0-0027).

Name		Bit number allocation list for signal control word
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal (bit number)
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		63
Scaling / Resolution of LSB		1

S-0-0330 Status 'nfeedback = ncommand'

- **Description**

This parameter is used to define an IDN for the status 'n feedback = n command' This allows the status 'n feedback = n command' to be assigned to a real-time status bit (see S-0-0305). The status 'n feedback = n command' is set when the velocity feedback value (S-0-0040) lies within the calculated command value for the velocity window (S-0-0157 and / or S-0-0272) which is based upon the velocity command values (see S-0-0036, S-0-0037, etc.).

Bit 0 is defined for operation data only.

Structure of status 'nfeedback = ncommand'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	

0		status	
	0	nfeedback ≠ ncommand	
	1	nfeedback = ncommand	$\|nfeedback - ncommand\| \leq \|ncommand\|$ * S-0-0272 + S-0-0157

Name		Status 'nfeedback = ncommand'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0331 Status 'nfeedback = 0'

- **Description**

This parameter is used to define an IDN for the status 'nfeedback = 0'. This allows the status 'nfeedback = 0' to be assigned to a real-time status bit (see S-0-0305).

The status 'nfeedback = 0' is set when the velocity feedback value (S-0-0040) is within the standstill window (see S-0-0124).

Bit 0 is defined for operation data only.

Structure of status 'nfeedback = 0'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	nfeedback = outside standstill window	
	1	nfeedback = within standstill window	

Name		Status 'nfeedback = 0'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0332 Status 'nfeedback less then nx'

- **Description**

This parameter is used to define an IDN for the status 'nfeedback < nx'. This allows the status 'nfeedback < nx' to be assigned to a real-time status bit (see S-0-0305). The status 'nfeedback < nx' is set when the velocity feedback value (see S-0-0040) is smaller than the velocity threshold nx (see S-0-0125).

Bit 0 is defined for operation data only.

Structure of status nfeedback < nx

Bit No.	Value	Description	Comments
15-1		reserved	
0		status	
	0	$\|nfeedback\| \geq \|nx\|$	
	1	$\|nfeedback\| < \|nx\|$	

Name		Status 'nfeedback < nx'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0333 Status 'T higher than Tx'

- **Description**

This parameter is used to define an IDN for the status 'T ≥ Tx'. This allows the status 'T ≥ Tx' to be assigned to a real-time status bit (see S-0-0305). The status 'T ≥ Tx' is set when the torque (force) feedback value (see S-0-0084) is greater than the torque (force) threshold value Tx (see S-0-0126).

Bit 0 is defined for operation data only.

Structure of Status 'T ≥ Tx'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	$\lvert T\rvert < \lvert Tx\rvert$	T = S-0-0084, Tx = S-0-0126
	1	$\lvert T\rvert \geq \lvert Tx\rvert$	

Name		Status 'T ≥ Tx'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0334 Status 'T greater than Tlimit '

- **Description**

This parameter is used to define an IDN for the status 'T ≥ Tlimit.' This allows the status 'T ≥ Tlimit' to be assigned to a real-time status bit (see S-0-0305). The status 'T ≥ Tlimit' is set when the torque feedback value (see S-0-0084) exceeds the programmed torque limits (see S-0-0082, S-0-0083, S-0-0092).

Bit 0 is defined for operation data only.

Structure of status 'T ≥ Tlimit'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	$\lvert T\rvert < \lvert Tlimit\rvert$	
	1	$\lvert T\rvert \geq \lvert Tlimit\rvert$	

Name		Status 'T ≥ Tlimit '
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0335 Status 'ncommand greater than nlimit'

- **Description**

This parameter is used to define an IDN for the status ncommand > nlimit'.' This allows the status ncommand > nlimit' to be assigned to a real-time status bit (see S-0-0305). The status ncommand > nlimit' is set when the sum of velocity command values (see S-0-0036, S-0-0037, etc.) is greater than the velocity limit value (see S-0-0038, S-0-0039, S-0-0091).

Bit 0 is defined for operation data only.

Structure of status ncommand > nlimit

Bit No.	Value	Description	Comments
15-1		reserved	
0		status	
	0	$\lvert ncommand \rvert \leq \lvert nlimit \rvert$	
	1	$\lvert ncommand \rvert > \lvert nlimit \rvert$	

Name		Status ncommand > nlimit
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0336 Status 'In position'

- **Description**

This parameter is used to define an IDN for the status 'in position' to be assigned to a real-time status bit (see S-0-0305). The status 'in position' is set when the position feedback value falls within the position window (see S-0-0057) relative to the position command value (see S-0-0047).

Bit 0 is defined for operation data only.

Structure of status 'in position'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		In position	
	0	outside of position window	
	1	within position window	

Name		Status In position
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0337 Status 'P greater Px'

- **Description**

This parameter is used to define an IDN for the status 'P ≥ Px.' This allows the status 'P ≥ Px' to be assigned to a real-time status bit (see S-0-0305). The status 'P ≥ Px' is set when the actual supplied power exceeds the power threshold Px (see S-0-0158).

Bit 0 is defined for operation data only.

Structure of status 'P ≥ Px'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	$\|P\| < \|Px\|$	
	1	$\|P\| \geq \|Px\|$	

Name		Status 'P ≥ Px'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0338 Status “Position feedback = active target position“

- **Description**

This parameter is used to define an IDN for the status „position feedback = active target position“. This allows the status „position feedback = active target position“ to be assigned to a real-time status bit (see S-0-0305).

The status „position feedback = active target position“ is set when position feedback value falls within the position window (see S-0-0057) relative to the active target position (see S-0-0430).

Bit 0 is defined for operation data only.

Structure of Postion feedback = active target position

Bit No.	Value	Description	Comments
15-1		reserved	
0		status	
	0	Position feedback outside position window	
	1	Position feedback within position window	(S-0-0430 – S-0-0386) < S-0-0057

Name		Status “Position feedback = active target position"
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0339 Status 'nfeedback less than minimum spindle speed'

- **Description**

This parameter is used to define an IDN for the status 'nfeedback ≤ minimum spindle speed'. This allows the status 'nfeedback ≤ minimum spindle speed' to be assigned to a real-time status bit (see S-0-0305).

The status 'nfeedback ≤ minimum spindle speed' is set when the velocity feedback value (S-0-0040) is lower than or equal to the programmed minimum spindle speed (S-0-0220).

Bit 0 is defined for operation data only.

Structure of 'nfeedback ≤ minimum spindle speed'

Bit No.	Value	Description	Comments
15-1		reserved	
0		status	
	0	$\|nfeedback\| >$ minimum spindle speed	
	1	$\|nfeedback\|$ ≤ minimum spindle speed	

Name		Status 'nfeedback ≤ minimum spindle speed'
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0340 Status 'nfeedback exceeds maximum spindle speed’

- **Description**

This parameter is used to define an IDN for the status 'nfeedback ≥ maximum spindle speed'. This allows the status 'nfeedback ≥ maximum spindle speed' to be assigned to a real–time status bit (see S-0-0305).

The status 'nfeedback ≥ maximum spindle speed' set when the velocity feedback value (S-0-0040) is greater than or equal to the programmed maximum spindle speed (S-0-0221).

Bit 0 is defined for operation data only.

Structure of 'nfeedback ≥ maximum spindle speed'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	$\lvert nfeedback \rvert <$ maximum spindle speed	
	1	$\lvert nfeedback \rvert \geq$ maximum spindle speed	

Name		Status 'nfeedback exceeds maximum spindle speed’
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0341 Status “In Coarse position”

- **Description**

This parameter is used to define an IDN for the ‘in coarse position’ status. This allows the status to be assigned, for instance, to a real-time status bit (for example S-0-0305). The status 'in coarse position' is set when the position feedback value falls within the coarse position window (see S-0-0261) relative to the position command value (see S-0-0047).

Bit 0 is defined for operation data only.

Structure of status 'in position'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	

	0	outside of coarse position window	
	1	within coarse position window	

Name		Status “In Coarse position”
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0342 Status “Target position attained”

- **Description**

This parameter is used to define an IDN for the 'target position attained' status. This allows the status to be assigned, for instance, to a real-time status bit (for example S-0-0305). The status 'target position attained' is set when the position command of the drive interpolator (S-0-0047) is equal to the Active target position (S-0-0430).

Bit 0 is defined for operation data only.

Structure of status 'target position attained'

Bit No.	Value	Description	Comments
15-1		reserved	
0		status	
	0	target position not reached	
	1	target position reached	position command = active target position

Name		Status “Target position attained”
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0343 Status “Interpolator halted”

- **Description**

This parameter is used to define an IDN for the 'interpolator halted' status. This allows the status to be assigned, for instance, to a real-time status bit (for example S-0-0305). The status 'interpolator halted' is set when the drive interpolator (IPO) has not yet attained the Active target position (S-0-0430), but the position command is not changing. The control can stop the interpolator with bit 13 in the drive control.

Bit 0 is defined for operation data only.

Structure of status 'interpolator halted'			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		status	
	0	interpolator not halted	
	1	interpolator halted	target position not attained, position command not changing

Name		Status “Interpolator halted”
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0346 Positioning control

- **Description**

This parameter is used in the operation mode Positioning only to implement positioning functions. The structure is shown in Table 19.

Table 19: Structure of Positioning control

Structure of Positioning control			
Bit	**Value**	**Description**	**Comments**

No.			
15-8		reserved	
7-5		**Block transitions**	
	xx0	**Active target position remains stored:**	the following functions for the block transitions are defined, (bit 6 and 7 are active).
	000	**Block transition with halt:**	Drive always positions on active target position, before positioning to next active target position (Target position - Actual value < Positioning window)
	010	**Block transition without halt (Mode 1):**	The target position of the start block is passed with the velocity of the start block. When the target position of the start block is reached the acceleration of the following block is activated.
	100	**Block transition without halt (Mode 2):**	The target position of the start block is passed with the velocity of the following block. Required accelerations or decelerations are already activated during start block. The target position of the following block is activated only, when reaching the target position of the start block.
	xx1	**Active target position is overwritten:**	On acceptance the drive overwrites immediately the active target position and positions to the new active target position. The drive does not move to the previous target position. Bit 6 and 7 are inactive.
4		Positioning command value is refered to	Reference of relative positioning command value (bit 3 is 1). This bit controls the behave of the positioning command value.
	0	Active target position	the positioning command value is added to the active target position.
	1	Current position feedback value	the positioning command value is added to the current position feedback value.
3		Positioning command value (relative or absolute)	
	0	absolute	Positioning command value is an absolute position value
	1	relative	Positioning command value is a relative position value
2-1		Positioning modes	
	00	**Positioning**	the positioning starts after takeover of positioning command value (bit 0). The positioning is interrupted by changing the Bit 1 and 2 from 00 to xx. The positioning is restarted with Bit 0. A still available remains distance is deleted (see bit 4).

	01	**Jogging +**	Continuous movement in positive direction with the current positioning parameters (S-0-0259, S-0-0260, S-0-0359).
	10	**Jogging -**	Continuous movement in negative direction with the current positioning parameters (S-0-0259, S-0-0260, S-0-0359).
	11	**Positioning Halt**	Stop with positioning deceleration (S-0-0359)
0			Takeover of positioning command value
	0/1	toggle	Takeover of positioning command value with every changing edge. The bit shall be inverted by the control unit to take on the next positioning command value.

Name		Positioning control
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0347 Velocity error

- **Description**

The current difference between the commanded velocity and actual velocity is placed in this parameter.

Name		Velocity error
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0348 Acceleration feed forward gain

- **Description**

Acceleration feed forward is effective in the operation mode “Position control without following distance (lag-less)”, and serves to reduce acceleration / deceleration-dependent following distance. The control unit has to check the data length before access the data.

Name		Acceleration feed forward gain
Attribute	Length (Octet)	SERCOS III: 4 SERCOS II: 2 or 4 (manufacturer specific)
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	manufacturer specific
	Write Protection	never
	Conversion Factor	manufacturer specific
Unit		manufacturer specific
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		manufacturer specific

S-0-0349 Bipolar jerk limit

- **Description**

"Bipolar jerk" limits the maximum rate of change of acceleration. Programming a value of zero will cause jerk limiting to be deactivated.

Name		Bipolar jerk limit
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0356 Distance home switch - marker puls

- **Description**

For the relative position of the home switch to the marker pulse of the encoder exists an optimal distance. In order to make the commissioning easy for the user, the distance from the home switch to the marker pulse is measured by the drive and shown in this parameter. Then the user has to calculate a value following the indications of the drive manufacturer and set this value in the home switch offset 1 / 2 (S-0-0299, S-0-0358), or shift mechanically the home switch corresponding to the calculated value .

Name		Distance home switch - marker puls
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never

	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0357 Marker pulse distance

- **Description**

In order to calculate in the drive the optimal distance between marker pulse and home switch, the distance between two marker pulses is set in this parameter.

Name		Marker pulse distance
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0358 Home switch offset 2

- **Description**

With digital drives, it is no longer technically possible to mechanically shift the external feedback system in order to shift the relationship between the marker pulse and home switch. With milled switch dog rails neither can the home switch dog be shifted.

Based on these conditions, the home switch offset 2 parameter is programmed during startup so that the home position is unambiguous. The required offset between the home switch and marker pulse is provided by the drive with the aid of the programmed home switch offset 2.

The "suggest home switch distance" (S-0-0298) can be transferred into this parameter.

Name		Home switch offset 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	see S-0-0076

S-0-0359 Positioning deceleration

- **Description**

With the "positioning deceleration" the positioning velocity (S-0-0259) is reduced during stop function in the operation modes "Interpolation" and "Positioning".

Name		Positioning deceleration
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0360 MDT data container A1

- **Description**

For the standard data container function in the MDT,

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 and A9 (S-0-0360 and S-0-0457). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0361 and S-0-0459) as well as one configuration list (S-0-0370) for all MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's

data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The MDT data container A1 (S-0-0360) and A9 (S-0-0457) are used also in the extended data container function.

Name		MDT data container A1
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0361 MDT data container B1

- **Description**

For the standard data container function in the MDT,

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 and A9 (S-0-0360 and S-0-0457). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0361 and S-0-0459) as well as one configuration list (S-0-0370) for all MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The MDT data container A1 (S-0-0360) and A9 (S-0-0457) are used also in the extended data container function.

Name		MDT data container B1
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0362 MDT data container A list index

- **Description**

If in the MDT data container an IDN with a variable length (list parameter) is configured,

- the corresponding list element of this list parameter will be addressed via the list index.
- The master writes the addressed list element into the MDT data container.

The list index of the MDT data container consists of a 16 bit address. Via list index 65535 the MDT data container can be defined not valid by the master.

The list index of MDT data container can be configured in the cyclic data of the MDT. Thereby, a switching of the list elements in the MDT data container during the next communication cycle is possible.

If the list index is situated outside of the list parameter, the data in the corresponding MDT data containers will be ignored by the slave. In this case the slave shall set the list index (acknowledgment) on value 65535 and optionally the pointer of MDT data container on value 255.

The list index of MDT data container can also be configured into the cyclic data of the ATs. In this way an acknowledgment of the MDT data container is possible. The drive reads the list index of MDT data container from the MDT and acknowledges it in the AT.

Bit No.	Value	Description	Comments
15-0		MDT data container A list index structure:	
	0-65534	index of the configured list parameter	
	65535	MDT data container A not valid	Error

Name	MDT data container A list index

Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0363 MDT data container B list index

- **Description**

If in the MDT data container an IDN with a variable length (list parameter) is configured,

- the corresponding list element of this list parameter will be addressed via the list index.
- The master writes the addressed list element into the MDT data container.

The list index of the MDT data container consists of a 16 bit address. Via list index 65535 the MDT data container can be defined not valid by the master.

The list index of MDT data container can be configured in the cyclic data of the MDT. Thereby, a switching of the list elements in the MDT data container during the next communication cycle is possible.

If the list index is situated outside of the list parameter, the data in the corresponding MDT data containers will be ignored by the slave. In this case the slave shall set the list index (acknowledgment) on value 65535 and optionally the pointer of MDT data container on value 255.

The list index of MDT data container can also be configured into the cyclic data of the ATs. In this way an acknowledgment of the MDT data container is possible. The drive reads the list index of MDT data container from the MDT and acknowledges it in the AT.

Bit No.	Value	Description	Comments
15-0		MDT data container B list index structure:	
	0-65534	index of the configured list parameter	
	65535	MDT data container B not valid	Error

Name		MDT data container B list index
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0364 AT data container A1

- **Description**

For the standard data container function in the AT

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 and A9 (S-0-0364 and S-0-0487). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0365 and S-0-0489) as well as one configuration list (S-0-0371) for all AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The AT data container A1 (S-0-0364) and A9 (S-0-0487) are used also in the extended data container function.

Name		AT data container A1
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		1
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-0365 AT data container B1

- **Description**

For the standard data container function in the AT

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 and A9 (S-0-0364 and S-0-0487). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0365 and S-0-0489) as well as one configuration list (S-0-0371) for all AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The AT data container A1 (S-0-0364) and A9 (S-0-0487) are used also in the extended data container function.

Name		AT data container B1
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		1
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0366 AT data container A list index

- **Description**

If in the AT data container an IDN with a variable length (list parameter) is configured,

- the corresponding list element of this list parameter will be addressed via the list index.
- The slave writes the addressed list element into the AT data container.

The list index of the AT data container consists of a 16 bit address. Via list index 65535 the AT data container can be defined not valid by the slave.

The list index of AT data container can be configured in the cyclic data of the MDT. Thereby, a switching of the list elements in the AT data container during the next communication cycle is possible.

If the list index is situated outside of the list parameter, the data in the corresponding AT data containers will be ignored by the master. In this case the slave shall set the list index (acknowledgment) on value 65535 and optionally the pointer of AT data container on value 255.

The list index of AT data container can also be configured into the cyclic data of the ATs. In this way an acknowledgment of the AT data container is possible. The slave reads the list index of AT data container from the MDT and acknowledges it in the AT.

Bit No.	Value	Description	Comments
15-0		AT data container A list index structure:	
	0-65534	index of the configured list parameter	
	65535	AT data container A not valid	Error

Name		AT data container A list index
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0367 AT data container B list index

- **Description**

If in the AT data container an IDN with a variable length (list parameter) is configured,

- the corresponding list element of this list parameter will be addressed via the list index.

- The slave writes the addressed list element into the AT data container.

The list index of the AT data container consists of a 16 bit address. Via list index 65535 the AT data container can be defined not valid by the slave.

The list index of AT data container can be configured in the cyclic data of the MDT. Thereby, a switching of the list elements in the AT data container during the next communication cycle is possible.

If the list index is situated outside of the list parameter, the data in the corresponding AT data containers will be ignored by the master. In this case the slave shall set the list index (acknowledgment) on value 65535 and optionally the pointer of AT data container on value 255.

The list index of AT data container can also be configured into the cyclic data of the ATs. In this way an acknowledgment of the AT data container is possible. The slave reads the list index of AT data container from the MDT and acknowledges it in the AT.

Bit No.	Value	Description	Comments
15-0		AT data container B list index structure:	
	0-65534	index of the configured list parameter	
	65535	AT data container B not valid	Error

Name		AT data container B list index
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0368 Data container A pointer

- **Description**

The data container pointers contain an 8-bit pointer, that defines which operation data should be placed in the MDT and AT data container. The data container pointer is the offset, within the corresponding data container configuration list

- for standard data container S-0-0370 and S-0-0371,
- additionally for extended data container S-0-0490 to S-0-0498 and S-0-0500 to S-0-0508.

Herewith inside the configuration list an IDN is addressed for the MDT data container or AT data container.

The master writes the addressed operation data into the MDT data container. The slave writes the addressed operation data into the AT data container.

The IDN "Data container A pointer and B pointer" (S-0-0368, S-0-0369) can be configured in the cyclic data of the MDT. Thereby, a switching of the operation data in the data containers during the next communication cycle is possible.

The IDN "Data container A pointer and B pointer" (S-0-0368, S-0-0369) can also be configured in the cyclic data of the AT. In this case the addressing (acknowledgment) according to the contents of the data container will be transmitted. The slave generates the acknowledgment by copying the pointer of MDT to the pointer of AT.

If the pointer of the data container is situated outside of the configuration list for the MDT or AT data container or the data is longer than the data container, the contents of the data container are not valid. The slave sets the pointer (acknowledgment) in the AT on 255. The data in MDT or AT data container will be ignored by the slave or master.

Data container A pointer structure			
Bit No.	**Value**	**Description**	**Comments**
15-8		address for AT data container	
	0-254	address for AT data container A	
	255	AT data container A not valid	Error
7-0		address for MDT data container	
	0-254	address for MDT data container A	
	255	MDT data container not valid	Error

Name		Data container A pointer
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0369 Data container B pointer

- **Description**

The data container pointers contain an 8-bit pointer, that defines which operation data should be placed in the MDT and AT data container. The data container pointer is the offset, within the corresponding data container configuration list

- for standard data container S-0-0370 and S-0-0371,

- additionally for extended data container S-0-0490 to S-0-0498 and S-0-0500 to S-0-0508.

Herewith inside the configuration list an IDN is addressed for the MDT data container or AT data container.

The master writes the addressed operation data into the MDT data container. The slave writes the addressed operation data into the AT data container.

The IDN "Data container A pointer and B pointer" (S-0-0368, S-0-0369) can be configured in the cyclic data of the MDT. Thereby, a switching of the operation data in the data containers during the next communication cycle is possible.

The IDN "Data container A pointer and B pointer" (S-0-0368, S-0-0369) can also be configured in the cyclic data of the AT. In this case the addressing (acknowledgment) according to the contents of the data container will be transmitted. The slave generates the acknowledgment by copying the pointer of MDT to the pointer of AT.

If the pointer of the data container is situated outside of the configuration list for the MDT or AT data container or the data is longer than the data container, the contents of the data container are not valid. The slave sets the pointer (acknowledgment) in the AT on 255. The data in MDT or AT data container will be ignored by the slave or master.

Data container A pointer structure

Bit No.	Value	Description	Comments
15-8		address for AT data container	
	0-254	address for AT data container B	
	255	AT data container B not valid	Error
7-0		address for MDT data container	
	0-254	address for MDT data container B	
	255	MDT data container not valid	Error

Name		Data container B pointer
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0370 MDT data container A/B configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A/B configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0371 AT data container A/B configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A/B configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0372 Drive Halt acceleration bipolar

- **Description**

This parameter is only activated if no internal drive interpolator is active and “Drive Halt” (drive control, bit13) is changed by the control unit.

Name		Drive Halt acceleration bipolar
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0377 Velocity feedback monitoring window

- **Description**

With help of the "Velocity feedback monitoring window" the maximum velocity deviation between velocity feedback value 1 (S-0-0040) and velocity feedback value 2 (S-0-0156) can be monitored. If the velocity deviation exceeds the monitoring window, the drive will generate either a warning or an error.

- Warning: in class 2 diagnostics (S-0-0012)
- Error: C1D bit in the drive status.

With this function a lack in the mechanics can be monitored. With the value "0" the monitoring is switched off.

Name		Velocity feedback monitoring window
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0378 Absolute encoder range 1

- **Description**

If the motor encoder is an absolute measuring system, then the absolute range with consideration of the mechanics and programming will be displayed or entered in this parameter.

Name		Absolute encoder range 1
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0379 Absolute encoder range 2

- **Description**

If the external encoder is an absolute measuring system, then the absolute range with consideration of the mechanics and programming will be displayed or entered in this parameter.

Name		Absolute encoder range 2
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0380 DC bus voltage

- **Description**

The drive’s DC (intermediate) bus voltage value is placed in this parameter.

Name		DC bus voltage
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		V
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1V

S-0-0381 DC bus current

- **Description**

The drive places the measured (actual) DC (intermediate) bus current in this parameter.

Name		DC bus current
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0382 DC bus power

- **Description**

The drive places the measured (actual) DC (intermediate) bus power in this parameter.

Name		DC bus power
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		W
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1W

S-0-0383 Motor temperature

- **Description**

The drive places the measured (actual) motor temperature in this parameter

Name		Motor temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	never
	Conversion Factor	see S-0-0208

Unit	see S-0-0208
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0208

S-0-0384 Amplifier temperature

- **Description**

The drive places the measured (actual) amplifier temperature (output stage) in this parameter.

Name		Amplifier temperature
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0208
	Write Protection	never
	Conversion Factor	see S-0-0208
Unit		see S-0-0208
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0208

S-0-0385 Active power

- **Description**

The drive places the calculated active power of the motor in this parameter.

Name		Active power
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		W
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1W

S-0-0386 Active position feedback value

- **Description**

The "Active position feedback value" is necessary, when the drive changes in the operation modes between position feedback value 1 and 2 automatically. The position feedback value which is activated by the operation mode is written in this IDN. To use this function, the control unit has to configure S-0-0386 in the AT.

Name		Active position feedback value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0387 Power overload

- **Description**

The drive places the calculated power overload in this parameter.

Name		Power overload
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	2
	Write Protection	never
	Conversion Factor	1
Unit		%
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,01%

S-0-0388 Braking current limit

- **Description**

With this parameter, the user can reduce the braking current limit during deceleration phase (active braking).

Name		Braking current limit
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0389 Effective current

- **Description**

The drive places the measured (actual) or calculated effective current (Root Mean Square) in this parameter.

Name		Effective current
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		A
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001A

S-0-0390 Diagnostic number

- **Description**

For every diagnostic, the sub-device stores a specific code in this IDN. Using this, the operator interface has the ability to display diagnostic message text in languages which are not supported by the sub-device.

The structure of the diagnostic number is shown in the table.

Bit No.	Value	Description	Comments
31-30		interpretation of bits 29-0	The bits 31-30 defines the interpretation of groups source type, class and status codes.
	00	manufacturer specific status codes	bits 29-24 type and class defined by SERCOS bits 15-0 status codes defined by manufacturer
	01	fully manufacturer specific	bits 29-0 are defined by manufacturer
	10	reserved	
	11	Standard	bits 29-0 are defined by SERCOS
29-24		source type	**responsible for codes defined by SERCOS**
	0x00	FSP Drive	TWG Drive
	0x01	FSP IO	TWG IO
	0x02	GDP	TWG Communication
	0x03	SCP	TWG Communication

	0x04	FSP Safety	TWG Safety
	0x05..0x3E	reserved	
	0x3F	unknown	
23-20		reserved	
19-16		class	
	0x00-0x08	are reserved	
	0x09	info	
	0x0A	operational state	
	0x0B	reserved	
	0x0C	procedure command specific state	
	0x0D	reserved	
	0x0E	warnings	
	0x0F	faults	
15-00		Status Code	

Name		Diagnostic number
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0391 Position feedback monitoring window

- **Description**

Using the position feedback monitoring window, the maximum permissible position error between feedback position 1 and feedback position 2 can be defined. If the position difference between these two feedback devices exceeds the limit in this parameter, the drive generates an error

- for excessive feedback position difference in C1D (S-0-0011), and
- sets the diagnostic number (S-0-0390) to 0xC00F2037.

Entering the value ‘0’ disables the monitoring.

Name	Position feedback monitoring window

Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0392 Velocity feedback filter

- **Description**

The time constant for the velocity feedback filter is entered into this parameter. If the filter time constant is 0, the filter is inactive.

Name		Velocity feedback filter
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1µs

S-0-0393 Command value mode

- **Description**

When the Modulo function is active, the interpretation of position commands is dependent upon the command value mode setting. In operation modes "Interpolation" and "Positioning" active only.

Structure of command value mode			
Bit No.	**Value**	**Description**	**Comments**
15-2		reserved	
1-0		Direction with modulo function	
	00	clockwise, positive direction	
	01	counter-clockwise, negative direction	

	10	shortest path	
	11	reserved	

Name		Command value mode
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0398 IDN list of configurable real-time/status bits

- **Description**

This IDN list contains ident numbers, whose bits are configurable as real-time bits or signal status bits in a producer connection.

Name		IDN list of configurable real-time/status bits
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0399 IDN list of configurable real-time/control bits

- **Description**

This IDN list contains ident numbers, whose bits are configurable as real-time bits or control bits in a consumer connection.

Name		IDN list of configurable real-time/control bits
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always

	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0400 Home switch

- **Description**

This parameter is used to assign an IDN to the home switch (external signal). This allows the home switch to be allocated to a real-time status bit (see S-0-0305). If the procedure command "control unit controlled homing" (S-0-0146) is active, the "home switch" is only valid if the signal "homing enable" (S-0-0407) is set.

Bit 0 is defined for operation data only.

Structure of home switch

Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		home switch	
	0	inactive switch	
	1	active switch	

Name		Home switch
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0401 Probe 1

- **Description**

This parameter is used to assign an IDN to probe 1 (external signal). This allows probe 1 to be assigned to a real-time status bit (see S-0-0305). Additional parameters are S-0-0130 and S-0-0131.

The signal "probe 1" is checked and updated by the drive only if the procedure command "probing cycle" (S-0-0170) is active and the signal "probe 1 enable" (S-0-0405) is set.

Bit 0 is defined for operation data only.

Structure of probe 1			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		probe 1	
	0	inactive probe	
	1	active probe	

Name		Probe 1
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0402 Probe 2

- **Description**

This parameter is used to assign an IDN to probe 2 (external signal). This allows probe 2 to be assigned to a real-time status bit (see S-0-0305). Additional parameters are S-0-0132 and S-0-0133.

The signal "probe 2" is checked and updated by the drive only if the procedure command "probing Cycle" (S-0-0170) is active and the signal "probe 2 enable" (S-0-0406) is set.

Bit 0 is defined for operation data only.

Structure of probe 2			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		probe 2	
	0	inactive probe	

	1	active probe	

Name		Probe 2
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0403 Position feedback value status

- **Description**

When the drive switches the position feedback values to the coordinates referred to the machine zero point the drive sets bit 0 of this parameter in order to inform the control unit that all actual position values are based on the zero point of the machine. Bit 0 is reset when either the procedure command "displacement to the referenced system" (S-0-0172) or the procedure command "drive controlled homing procedure" (S-0-0148) or the procedure command "cancel reference point" (S-0-0191) or “set absolute position procedure command” (S-0-0447) is started or when the drive loses its reference to the zero point of the machine. The position feedback value status can be assigned to a real-time bit.

SERCOS II only: Bit 0 is mapped to CD3 (S-0-0013) bit 14 for an immediate real time reference in the control unit.

Structure of position feedback value status			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2		Status of position feedback value 2	
	0	not referenced to machine zero point	
	1	referenced to machine zero point	
1		Status of position feedback value 1	
	0	not referenced to machine zero point	
	1	referenced to machine zero point	
0		Status of activated position feedback value	
	0	not referenced to machine zero point	
	1	referenced to machine zero point	

Name		Position feedback value status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0404 Position command value status

- **Description**

When the position command values are switched to the coordinates referred to the machine zero point the control unit sets bit 0 of this parameter. This signals to the drive that all position command values are based on the zero point of the machine. Simultaneously the control unit transfers the new position command values in the cyclic data. Bit 0 is reset when the procedure command "displacement to the referenced system" (S-0-0172) is started. The position command value status can be assigned to a real-time control bit and therefore it can be permanently signaled to the drive in the control word (see S-0-0301).

Bit 0 is defined for operation data only.

Structure of position command value status			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		position command value status	
	0	position command value not referenced to machine zero point	
	1	position command value referenced to machine zero point	

Name		Position command value status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0405 Probe 1 enable

- **Description**

This parameter is used to assign an IDN to probe 1 enable. This allows the status "probe 1 enable" to be assigned to a real-time control bit (see S-0-0301).

Probe 1 enable is checked by the drive only as long as the procedure command "probing cycle" (S-0-0170) is active. For a new probing cycle with probe 1 the control unit has to handle the probe 1 enable according to clause 1.8.

Bit 0 is defined for operation data only.

Structure of probe 1 enable

Bit No.	Value	Description	Comments
15-1		reserved	
0		Probe 1 enable	
	0	probe 1 not enabled	
	1	probe 1 enabled	

Name		Probe 1 enable
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0406 Probe 2 enable

- **Description**

This parameter is used to assign an IDN to probe 2 enable. This allows the status "probe 2 enable" to be assigned to a real-time control bit (see S-0-0301). Probe 2 enable is checked by the drive only as long as the procedure command "probing cycle" (S-0-0170) is active. For a new probing cycle with probe 2 the control unit has to handle the probe 2 enable according to clause 1.8.

Bit 0 is defined for operation data only.

Structure of probe 2 enable			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		probe 2 enable	
	0	probe 2 not enabled	
	1	probe 2 enabled	

Name		Probe 2 enable
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0407 Homing enable

- **Description**

This parameter is used to assign an IDN to the status homing enable. This allows the status "homing enable" to be assigned to a real-time control bit (see S-0-0301). The drive interprets the homing enable only while the procedure command "control unit controlled homing" (S-0-0146) is active.

Bit 0 is defined for operation data only.

Structure of homing enable			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		homing enable	
	0	homing not enabled	
	1	homing enabled	

Name		Homing enable
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0408 Reference marker pulse registered

- **Description**

This parameter is used to assign an IDN to reference marker pulse registered. This allows the status "reference marker pulse registered" to be assigned to a real-time status bit (see S-0-0305). The drive sets this bit to 1 if the procedure command "control unit controlled homing” (S-0-0146) is active, if the homing is enabled (S-0-0407) and the marker pulse of the feedback (external signal) is registered. Simultaneously the drive stores the not referenced position feedback value in the related marker position (S-0-0173 or S-0-0174). The drive resets this bit to 0 when the control unit activates the procedure command “control unit controlled homing” (S-0-0146). While the procedure command (S-0-0146) is active, the ”reference marker pulse registered” is valid. The procedure command ”drive controlled homing” (S-0-0148) has no effect on this bit (S-0-0408).

Bit 0 is defined for operation data only.

Structure of reference marker pulse registered			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		reference marker pulse registered	
	0	reference marker pulse not registered	
	1	reference marker pulse registered	

Name		Reference marker pulse registered
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0409 Probe 1 positive latched

- **Description**

- With single measuring bit 0 of status "probe 1 positive latched" is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 1 enable" (S-0-0405) is set to 1 and the positive edge of "probe 1" (S-0-0401) is registered. Simultaneously the drive stores the measuring data allocated in S-0-0426 in the "probe 1 positive edge" (S-0-0130). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 1 enable is reset to 0.

For more details see clause 1.8.

Structure of probe 1 positive latched			
Bit No.	**Value**	**Description**	**Comments**
		Single measuring	
15-1		reserved	
0		probe 1 positive	bit 0 is valid only
	0	probe 1 positive not latched	
	1	probe 1 positive latched	
		Continuous measuring	
15-0		counter	every measuring is counted

Name		Probe 1 positive latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0410 Probe 1 negative latched

- **Description**

- With single measuring bit 0 of status "probe 1 negative latched" is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 1 enable" (S-0-0405) is set to 1 and the negative edge of "probe 1" (S-0-0401) is registered. Simultaneously the drive stores the measuring data allocated in S-0-0426 in the "probe 1 negative edge" (S-0-0131). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 1 enable is reset to 0.

For more details see clause 1.8.

Structure of probe 1 negative latched			
Bit No.	**Value**	**Description**	**Comments**
		Single measuring	
15-1		reserved	
0		probe 1 negative	bit 0 is valid only
	0	probe 1 negative not latched	
	1	probe 1 negative latched	
		Continuous measuring	
15-0		counter	every measuring is counted

Name		Probe 1 negative latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		1
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0411 Probe 2 positive latched

- **Description**

- With single measuring bit 0 of status "probe 2 positive latched" is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 2 enable" (S-0-0406) is set to 1 and the positive edge of "probe 2" (S-0-0402) is registered. Simultaneously the drive stores the measuring data allocated in S-0-0427 in the "probe 2 positive edge" (S-0-0132). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 2 enable is reset to 0.

For more details see clause 1.8.

Structure of probe 2 positive latched			
Bit No.	**Value**	**Description**	**Comments**
		Single measuring	
15-1		reserved	
0		probe 2 positive	bit 0 is valid only
	0	probe 2 positive not latched	
	1	probe 2 positive latched	
		Continuous measuring	
15-0		counter	every measuring is counted

Name		Probe 2 positive latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0412 Probe 2 negative latched

- **Description**

- With single measuring bit 0 of status "probe 2 negative latched" is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 2 enable" (S-0-0406) is set to 1 and the negative edge of "probe 2" (S-0-0402) is registered. Simultaneously the drive stores the measuring data allocated in S-0-0427 in the "probe 2 negative edge" (S-0-0133). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 2 enable is reset to 0.

For more details see clause 1.8.

Structure of probe 2 negative latched			
Bit No.	**Value**	**Description**	**Comments**
		Single measuring	
15-1		reserved	

0		probe 2 negative	bit 0 is valid only
	0	probe 2 negative not latched	
	1	probe 2 negative latched	
		Continuous measuring	
15-0		counter	every measuring is counted

Name		Probe 2 negative latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0417 Positioning velocity threshold in modulo mode

- **Description**

In modulo mode and shortest path programmed in command value mode S-0-0393 the last direction of rotation is kept if the value of the current velocity feedback value is greater than the positioning velocity threshold in modulo mode.

The „positioning velocity threshold in modulo mode“ shall be programmed greater then the level of the noise signal of the velocity feedback value.

If the value 0 is programmed, then this parameter is not active and the movement direction programmed in the command value mode is valid.

Name		Positioning velocity threshold in modulo mode
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	never
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0418 Target position window in modulo mode

- **Description**

With modulo mode and the programmed positive or negative movement direction in the command value mode, the direction shortest path is activated if the active target position and in addition the stopping distance lie within the target position window in modulo mode.

The shortest path in modulo mode will proceed if the following condition is valid:
Distance in this parameter > position feedback value - target position + stopping distance

If the value 0 is programmed, then this parameter is not active and the movement direction programmed in the command value mode is valid.

Name		Target position window in modulo mode
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0419 Positioning acknowledge

- **Description**

The drive acknowleges the take-over of the positioning command value, by copying bit 0 of S-0-0346 to bit 0 of S-0-0419. This bit is set to 0, if the operation mode is activated or the control unit sets in the positioning control word (S-0-0346) bit 0 to 0.

Bit 0 is defined for operation data only.

Structure of positioning acknowledge

Bit No.	Value	Description	Comments
15-1		reserved	
0		Takeover of positioning command value	
	toggle	the drive has taken on the positioning command value related on positioning control word, bit 0.	

Name		Positioning acknowledge
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never

	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0420 Activate parametrization level procedure command (PL)

- **Description**

Activating this procedure command causes all monitoring systems to shut down associated with the connected hardware systems. While the procedure command is active, the sub-device does not report a C1D error (S-0-0011). Corresponding status bits are reset by the sub-device.

During parameterization level the control unit can change all parameters except the communication parameters and timing parameters.

The parameterization level is automatically deleted by the sub-device in CP0, or with the exit parameterization level procedure command (S-0-0422).

The procedure command is positively acknowledged in CP2, CP3 or CP4 if

- the sub-device is disabled (status word, bit 14=0 or bit 15=0) and
- the monitoring system mentioned above is turned off and
- the Device Status (S-Dev) bit parametrization level active is set to 1.

Otherwise the sub-device generates a negative acknowledge or an error message (error code of 0x7012) over the service channel.

Name		Activate parametrization level procedure command (PL)
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0422 Exit parameterization level procedure command

- **Description**

When the Exit parameterization level procedure command is activated, all parameters are checked and all monitoring systems are turned on again. Necessary references shall be recovered by the control unit (e.g. homing etc...).

Activating this procedure command results in leaving the parameterization level.

The procedure command is positively acknowledged

- if the corresponding parameters are checked without faults and
- the monitoring systems are switched on again and
- the device status parametrization level is set to 0 (see S-0-0420).

The procedure command is also positively acknowledged if the sub-device is already in operation level.

The procedure command is negatively acknowledged if a fault has appeared during the checks. The IDNs which have caused a fault are stored in the IDN list (S-0-0423). The sub-device remains in parameterization level.

Name		Exit parameterization level procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0423 IDN-list of invalid data for parameterization level

- **Description**

IDNs which are considered invalid by the sub-device prior to exit parameterization level (see S-0-0422) are stored in this IDN-list.

Case 1: procedure command S-0-0422 is performed correctly; the IDN-list (S-0-0423) contains no IDNs.

Case 2: procedure command S-0-0422 results in an error; the IDN-list (S-0-0423) contains all IDNs of invalid parameter.

Name		IDN-list of invalid data for parameterization level
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable

	Data Type and Display Format	
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0426 Measuring data allocation 1

- **Description**

The IDN list of configurable measuring data (S-0-0428) contains all IDNs which ones the drive as measuring data supports. To assign a measuring data for probe 1 (S-0-0401) an IDN from the IDN list shall be programmed in the measuring data allocation 1.

Name		Measuring data allocation 1
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0427 Measuring data allocation 2

- **Description**

The IDN list of configurable measuring data (S-0-0428) contains all IDNs which ones the drive as measuring data supports. To assign a measuring data for probe 2 (S-0-0402) an IDN from the IDN list shall be programmed in the measuring data allocation 2.

Name		Measuring data allocation 2
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0428 IDN list of configurable measuring data

- **Description**

This IDN list contains all IDNs which can be processed by the drive as measuring data for probe 1 and probe 2.

Name		IDN list of configurable measuring data
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0429 Emergency stop deceleration

- **Description**

When the control unit activates „Drive OFF“ (drive control, bit 15 = 0, bit 14 = 1) the drive decelerates as best as possible limited by the „emergency stop deceleration“, followed by disabling of the torque at nmin.

Name		Emergency stop deceleration
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0430 Active target position

- **Description**

At the operation mode interpolation, the drive writes the target position (S-0-0258) into the active target position.

At the operation mode positioning, the drive writes the positioning command value (S-0-0282) either into the target position (absolute position data) or the positioning command value is added (relative position data) to the active target position. See control word positioning (S-0-0346).

Active target position is only an intermediate memory in the drive, therefore cannot be configured as cyclic data and is readable only over the service channel.

Name		Active target position
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0431 Spindle positioning acceleration bipolar

- **Description**

When the position spindle procedure command (see S-0-0152) is received, the drive accelerates or decelerates to the spindle positioning speed (S-0-0222), depending on the spindle positioning acceleration bipolar.

Name		Spindle positioning acceleration bipolar
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0160
	Write Protection	never
	Conversion Factor	see S-0-0160
Unit		see S-0-0160
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0160

S-0-0437 Positioning status

- **Description**

In this parameter all status messages of the operation modes "interpolation" and "positioning" are combined.

Bit 1 is supported in operation mode interpolation only.

Bits 2 and 12 are supported in operation mode positioning only.

Structure of Positioning status

Bit No.	Value	Description	Comments
15		reserved	
14		Warning target position outside of travel range	see S-0-0323
	0	target position within position limit values	
	1	target position outside of position limit values	
13		Warning positioning velocity > n Limit:	see S-0-0315
	0	positioning velocity ≤ n Limit	
	1	positioning velocity > n Limit	
12		Jog mode	Positioning only
	0	jog mode is inactive	
	1	jog mode is active	S-0-0346, bit 2 and 1 are set to “01” or “10”
11-7		reserved	
6		drive decelerates	
	0	drive doesn’t decelerate	
	1	drive decelerates	
5		drive accelerates	
	0	drive doesn’t accelerate	
	1	drive accelerates	
4		constant interpolation velocity	
	0	no constant velocity	
	1	constant velocity	
3		Status 'interpolator halted'	see S-0-0343, the control unit can stop the drive interpolator with bit 13 of drive control.
	0	interpolator not halted	

	1	interpolator halted	target position not attained
2		Status Postion feedback = active target position	Positioning only, see S-0-0338, (S-0-0430 – S-0-0386) < S-0-0057
	0	Position feedback outside position window	
	1	Position feedback within position window	
1		Status Postion feedback = target position	Interpolation only, (S-0-0258 – S-0-0386) < S-0-0057
	0	Position feedback outside position window	
	1	Position feedback within position window	
0		Status 'target position attained'	see S-0-0342, drive internal position command = active target position
	0	target position not reached	
	1	target position reached	

Name		Positioning status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0440 Marker position feedback 1

- **Description**

The position of the marker of feedback 1 is stored when the function cylic marker detection is activ. The position transferred from the drive to the control unit for supervision incremental feedback with cyclic marker pulses.

Name		Marker position feedback 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	always
	Conversion Factor	see S-0-0076
Unit		see S-0-0076

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	see S-0-0076

S-0-0441 Marker position feedback 2

- **Description**

The position of the marker of feedback 2 is stored when the function cyclic marker detection is active (S-0-0115 Bit 9). The position transferred from the drive to the control unit for supervision incremental feedback with cyclic marker pulses.

Name		Marker position feedback 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	always
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0442 Counter marker position feedback 1

- **Description**

The position of the marker of feedback 1 is stored when the function cyclic marker detection is active. For every correct detection of an marker the counter in incremented. The value is transferred from the drive to the control unit for supervision incremental feedback with cyclic marker pulses.

Name		Counter marker position feedback 1
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0443 Counter marker position feedback 2

- **Description**

The position of the marker of feedback 2 is stored when the function cyclic marker detection is active. For every correct detection of an marker the counter in incremented. The value is transferred from the drive to the control unit for supervision incremental feedback with cyclic marker pulses.

Name		Counter marker position feedback 2
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0444 IDN-list of configurable data in the AT data container

- **Description**

This list consists the IDNs of operation data which can be processed by the sub-device in the AT data containers. This IDN list is used in the standard and extended data container.

Name		IDN-list of configurable data in the AT data container
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0445 IDN-list of configurable data in the MDT data container

- **Description**

This list consists the IDNs of operation data which can be processed by the sub-device in the MDT data containers. This IDN list is used in the standard and extended data container.

Name		IDN-list of configurable data in the MDT data container
Attribute	Length (Octet)	SERCOS III: 4,variable

		SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0446 Ramp reference velocity

- **Description**

With acceleration scaling “ramp time” the acceleration is programmed as time. The parameter “ramp reference velocity” serves as a reference value for the acceleration calculation (see Scaling of operation data).

acceleration parameter [time] = ramp reference velocity / acceleration

Name		Ramp reference velocity
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0044
	Write Protection	OL
	Conversion Factor	see S-0-0044
Unit		see S-0-0044
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0044

S-0-0447 Set absolute position procedure command

- **Description**

After activation of the “Set absolute position procedure command” the drive

- ignores the command value from the control unit,
- evaluates the Set absolute position control (S-0-0448),
- calculates the difference between the position feedback value 1 or 2 (S-0-0051 or S-0-0053) and the Reference Distance 1 or 2 (S-0-0052 or S-0-0054),
- stores the calculated difference in the Displacement Parameter 1 or 2 (S-0-0175 or S-0-0176).

The positive acknowledgment is generated if

- the displacement value is transferred in the position feedback value (the drive checks whether it can immediately transfer the displacement value or not),
- the position feedback value status (S-0-0403) reflects that homing has been completed once.
- **SERCOS II only:** the position feedback value status is mapped to CD3 (S-0-0013).

How the absolute reference is set and maintained within the drive is a function of the drive.

The procedure command will terminate with a fault, when the drive detects an error during the procedure command specific calculations.

For more details see clause 1.7.3.

Name		Set absolute position procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0448 Set absolute position control word

- **Description**

The control word specifies which encoder is referenced by the set absolute position procedure command (S-0-0447). Encoder 1, encoder 2 or both.

A drive may or may not support setting the absolute position while it is enabled. The control unit can verify if the drive supports set absolute position with or without the drive enabled. In the case that bit 2 is set to “1” by the control unit, and the drive responses with an error, it does not support set absolute position while enabled.

Table 20: Structure of set absolute position control

Bit No.	Value	Description	Comments
15-3		reserved	
2		absolute homing mode in the drive	
	0	the control unit requests the drive to support absolute homing “without drive enabled”	
	1	the control unit requests the drive to support absolute homing “with drive enabled”	

1		encoder 2 (external encoder)	
	0	not selected	
	1	selected for absolute homing	
0		encoder 1 (motor encoder)	
	0	not selected	
	1	selected for absolute homing	

Name		Set absolute position control
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0450 MDT data container A2

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's

data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A2
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0451 MDT data container A3

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A3
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0452 MDT data container A4

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A4
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0453 MDT data container A5

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A5
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0454 MDT data container A6

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and

- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A6
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0455 MDT data container A7

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A7
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0456 MDT data container A8

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):

- The configured operation data is represented in the data container in hexadecimal, without the units.
  - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
  - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A8
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0457 MDT data container A9

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):

  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A9
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0458 MDT data container A10

- **Description**

For the extended data container function in the MDT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0360, S-0-0450 to S-0-0458), as well as a configuration list (S-0-0370, S-0-0490 to S-0-0498) for each of the MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		MDT data container A10
Attribute	Length (Octet)	8

	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0459 MDT data container B2

- **Description**

For the standard data container function in the MDT,

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the MDT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 and A9 (S-0-0360 and S-0-0457). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0361 and S-0-0459) as well as one configuration list (S-0-0370) for all MDT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the MDT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The MDT data container A1 (S-0-0360) and A9 (S-0-0457) are used also in the extended data container function.

Name		MDT data container B2
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0

	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0460 Position switch point off 1

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0461 Position switch point off 2

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0462 Position switch point off 3

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 3
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0463 Position switch point off 4

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 4
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0464 Position switch point off 5

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 5
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0465 Position switch point off 6

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 6
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0466 Position switch point off 7

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 7
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter

	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0467 Position switch point off 8

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 8
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0468 Position switch point off 9

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 9
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0469 Position switch point off 10

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 10
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0470 Position switch point off 11

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 11
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0471 Position switch point off 12

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 12
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0472 Position switch point off 13

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 13
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0473 Position switch point off 14

- **Description**

The position switches consist of a “position switch point off” and a position switch flag (S-0-0059). If the position feedback value is less than the “position switch point off”, the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the “position switch point off”, the appropriate position switch flag is set to 0.

Name		Position switch point off 14
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter

	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0474 Position switch point off 15

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 15
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0475 Position switch point off 16

- **Description**

The position switches consist of a "position switch point off" and a position switch flag (S-0-0059). If the position feedback value is less than the "position switch point off", the appropriate position switch flag is set to 1. If the position feedback value is equal to or greater than the "position switch point off", the appropriate position switch flag is set to 0.

Name		Position switch point off 16
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0476 Position switch control

- **Description**

In this parameter the position switch flags 1 to 16 (S-0-0059) are activated. The drive shall process only the position switch flags, which are activated. The bits 16-31 define whether it is position switches or cam switches. At a cam switch the position switch on (S-0-0060 to S-0-0075) and the position switch off (S-0-0460 to S-0-0475) are necessary.

Structure of position switch control

Bit No.	Value	Description	Comments
31		Cam switch 16	S-0-0075 and S-0-0475
	0	cam switch disabled	only S-0-0075 is active
	1	cam switch enabled	S-0-0075 and S-0-0475 are active
30 to 17		Cam switch 15 to 2	
16		Cam switch 1	S-0-0060 and S-0-0460
	0	cam switch disabled	only S-0-0060 is active
	1	cam switch enabled	S-0-0060 and S-0-0460 are active
15		Position switch flag 16	S-0-0075, S-0-0475
	0	disabled	
	1	enabled	
14 to 1		Position switch flag 15 to 2	
0		Position switch flag 1	S-0-0060, S-0-0460
	0	disabled	
	1	enabled	

Name		Position switch control
Attribute	Length (Octet)	4
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0477 Position switch hysteresis

- **Description**

The hysteresis has an effect on all position switches (S-0-0060 to S-0-0075 and S-0-0460 to S-0-0475) and take into account if the flags (S-0-0059) are switched on or off.

Name		Position switch hysteresis
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0478 Limit switch status

- **Description**

This parameter shows the input level of the limit switches and is used for homing and monitoring.

Structure of limit switch status

Bit No.	Value	Description	Comments
15-2		reserved	
1		Limit switch negative direction	
	0	low level (no voltage e.g. 0V)	
	1	high level (voltage e.g. +24V)	
0		Limit switch positive direction	
	0	low level (no voltage e.g. 0V)	
	1	high level (voltage e.g. +24V)	

Name		Limit switch status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0480 AT data container A2

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A2
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0481 AT data container A3

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and

- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A3
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0482 AT data container A4

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A4
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0483 AT data container A5

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):

    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A5
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0484 AT data container A6

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):

- The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A6
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0485 AT data container A7

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A7
Attribute	Length (Octet)	4

	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0486 AT data container A8

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A8
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0487 AT data container A9

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A9
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0488 AT data container A10

- **Description**

For the extended data container function in the AT,

- eight data containers (4 octets long, A1 to A8) and
- two data containers (8 octets long, A9 and A10)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer (S-0-0368) is required for the containers A1 to A10 (S-0-0364, S-0-0480 to S-0-0488), as well as a configuration list (S-0-0371, S-0-0500 to S-0-0508) for each of the AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
    - The configured operation data is represented in the data container in hexadecimal, without the units.
        - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
        - Units: Not present
- Maximum required data block (access via service channel):
    - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Name		AT data container A10
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0489 AT data container B2

- **Description**

For the standard data container function in the AT

- two data containers (4 octets long, A1 and B1) and
- two data containers (8 octets long, A9 and B2)

are defined, serving as placeholders in the AT. The contents of the data containers can be dynamically changed by the master as necessary. Additionally, the data container A pointer

(S-0-0368) is required for the containers A1 and A9 (S-0-0364 and S-0-0487). The data container B pointer (S-0-0369) is required for the containers B1 and B2 (S-0-0365 and S-0-0489) as well as one configuration list (S-0-0371) for all AT container. If the configured operation data is only 2 or 4 octets long, it is placed in the lower part of the AT data container. The higher part is not used.

In configuring data container operation data, the slave can select between a minimum requirement and maximum requirement.

- Minimum required data block (access via service channel):
  - The configured operation data is represented in the data container in hexadecimal, without the units.
    - Attribute: Data type and display format' are set hexadecimal (bits 22-20 = 011)
    - Units: Not present
- Maximum required data block (access via service channel):
  - The configured operation data is represented in the data container not with the data block of the data container itself, rather the configured operation data's data block. In this case the operation data will be displayed with the IDN of the data container exactly the same as it would be with its own IDN.

Note: The AT data container A1 (S-0-0364) and A9 (S-0-0487) are used also in the extended data container function.

Name		AT data container B2
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0490 MDT data container A2 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A2 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable

		SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0491 MDT data container A3 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A3 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0492 MDT data container A4 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name	MDT data container A4 configuration list

Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0493 MDT data container A5 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A5 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0494 MDT data container A6 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A6 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0495 MDT data container A7 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A7 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0496 MDT data container A8 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A8 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0497 MDT data container A9 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A9 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0498 MDT data container A10 configuration list

- **Description**

The Master enters into the MDT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding MDT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the master to the slave.

The IDNs in this IDN list may be taken from S-0-0445 or from S-0-0188, if S-0-0445 does not exist.

Name		MDT data container A10 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0500 AT data container A2 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A2 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0501 AT data container A3 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A3 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0502 AT data container A4 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A4 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0503 AT data container A5 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A5 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0504 AT data container A6 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A6 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0505 AT data container A7 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A7 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0506 AT data container A8 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container 8 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0507 AT data container A9 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A9 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0508 AT data container A10 configuration list

- **Description**

The Master enters into the AT data container configuration list the IDN numbers for the operation data that are to be sent via the corresponding AT data containers (A and B of standard data container, and A1 to A10 of extended data container) as needed from the slave to the master.

The IDNs in this IDN list may be taken from S-0-0444 or from S-0-0187, if S-0-0444 does not exist.

Name		AT data container A10 configuration list
Attribute	Length (Octet)	SERCOS III: 4,variable SERCOS II: 2,variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0509 Extended probe control

- **Description**

In this parameter the extended measuring functions are activated. Extended functions are:

- probing window,
- marker monitoring and
- extended difference measuring.

This parameter shall be programmed according to the application before the extended measuring function is activated. For more details see clause 1.8.

Structure of extended probe control			
Bit No.	**Value**	**Description**	**Comments**
15-14		reserved	
13-12		Edge selection difference 2	
	00	Difference 2	from negative to negative edge
	01	Difference 2	from positive to negative edge
	10	Difference 2	from negative to positive edge
	11	Difference 2	from positive to positive edge
11-10		Edge selection difference 1	
	00	Difference 1	from negative to negative edge
	01	Difference 1	from positive to negative edge
	10	Difference 1	from negative to positive edge
	11	Difference 1	from positive to positive edge
9		Extended difference measuring 2	
	0	disable difference measuring 2	
	1	enable difference measuring 2	
8		Extended difference measuring 1	
	0	disable difference measuring 1	
	1	enable difference measuring 1	
7-4		reserved	
3		Marker monitoring 2	
	0	disable Marker monitoring 2	
	1	enable Marker monitoring 2	
2		Marker monitoring 1	
	0	disable Marker monitoring 1	
	1	enable Marker monitoring 1	
1		Probing window 2	
	0	disable probing window 2	
	1	enable probing window 2	defined by S-0-0514 and S-0-0515
0		Probing window 1	
	0	disable probing window 1	

	1	enable probing window 1	defined by S-0-0512 and S-0-0513

Name		Extended probe control
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0510 Difference value probe 1

- **Description**

For the difference measuring both edges of probe 1 shall be enabled in the extended probe control (S-0-0509). The drive calculates the difference of two measurement values of probe 1 and stores the result in this parameter.

Name		Difference value probe 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0511 Difference value probe 2

- **Description**

For the difference measuring both edges of probe 2 shall be enabled in the extended probe control (S-0-0509). The drive calculates the difference of two measurement values of probe 2 and stores the result in this parameter.

Name		Difference value probe 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0512 Start position probing window 1

- **Description**

If the probing window 1 is enabled in the extended probe control (S-0-0509), then the value of this parameter defines the start position of the probing window 1.

For more details see clause 1.8.

Name		Start position probing window 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0513 End position probing window 1

- **Description**

If the probing window 1 is enabled in the extended probe control (S-0-0509), then the value of this parameter defines the end position of the probing window 1.

For more details see clause 1.8.

Name		End position probing window 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0514 Start position probing window 2

- **Description**

If the probing window 2 is enabled in the extended probe control (S-0-0509), then the value of this parameter defines the start position of the probing window 2.

For more details see clause 1.8.

Name		Start position probing window 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0515 End position probing window 2

- **Description**

If the probing window 2 is enabled in the extended probe control (S-0-0509), then the value of this parameter defines the end position of the probing window 2.

For more details see clause 1.8.

Name		End position probing window 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0076
	Write Protection	never
	Conversion Factor	see S-0-0076
Unit		see S-0-0076
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0076

S-0-0516 Marker losses probe 1

- **Description**

The marker monitoring 1 is enabled in the extended probe control (S-0-0509). The marker monitoring 1 is executed within the probing window 1 (S-0-0512 and S-0-0513) only. The successive marker losses are added in the marker losses probe 1.

If the marker losses probe 1 exceed the programmed value of the maximum marker losses probe 1 (S-0-0518), then bit 4 is set to 1 in the probe status. If a marker is recognized within the probing window 1 the marker losses probe 1 are set to 0.

For more details see clause 1.8.

Name		Marker losses probe 1
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0517 Marker losses probe 2

- **Description**

The marker monitoring 2 is enabled in the extended probe control (S-0-0509). The marker monitoring 2 is executed within the probing window 2 (S-0-0514 and S-0-0515) only. The successive marker losses are added in the marker losses probe 2.

If the marker losses probe 2 exceed the programmed value of the maximum marker losses probe 2 (S-0-0519), then bit 5 is set to 1 in the probe status. If a marker is recognized within the probing window 2 the marker losses probe 2 are set to 0.

For more details see clause 1.8.

Name		Marker losses probe 2
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0518 Maximum marker losses probe 1

- **Description**

When the programmed value of maximum marker losses of probe 1 is exceeded by the marker losses probe 1 (S-0-0516), then the drive sets the bit status marker losses 1 in the probe status (S-0-0179).

For more details see clause 1.8.

Name		Maximum marker losses probe 1
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0519 Maximum marker losses probe 2

- **Description**

When the programmed value of maximum marker losses of probe 2 is exceeded by the marker losses probe 2 (S-0-0517), then the drive sets the bit status marker losses 2 in the probe status (S-0-0179).

For more details see clause 1.8.

Name		Maximum marker losses probe 2
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0520 Axis control word

- **Description**

The axis control word is active, if bit 9 = 1 and bit 15 = 0 in the active operation mode.

Structure of axis control word			
Bit No.	**Value**	**Description**	**Comments**

15-3		reserved	
2		Position control with or without following distance	
	0	with following distance	
	1	without following distance	
1		Hybrid position control	
	0	Bit 0 defines the activated feedback	
	1	Position control with feedback 1 and 2 (hybrid)	
0		Position control feedback 1 or 2	
	0	with feedback 1 (motor encoder)	
	1	with feedback 2 (external encoder)	

Name		Axis control word
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0521 Axis status word

- **Description**

The axis status word is active, if bit 9 = 1 and bit 15 = 0 in the active operation mode.

Structure of axis status word			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2		Position control with or without following distance	
	0	with following distance	
	1	without following distance	
1		Hybrid position control	
	0	Bit 0 defines the activated feedback	
	1	Position control with feedback 1 and 2 (hybrid)	
0		Position control feedback 1 or 2	

	0	with feedback 1 (motor encoder)	
	1	with feedback 2 (external encoder)	

Name		Axis status word
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0522 Difference value 1 latched

- **Description**

- With single measuring bit 0 of difference value 1 latched is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 1 enable" (S-0-0405) is set to 1 and edges of "probe 1" (S-0-0401) are registered. The drive calculates the selected difference and stores the result in the "difference value probe 1" (S-0-0510). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 1 enable is reset to 0.

For more details see clause 1.8.

Structure of difference value 1 latched

Bit No.	Value	Description	Comments
		Single measuring	
15-1		reserved	
0		difference value 1	bit 0 is valid only
	0	difference value 1 not latched	
	1	difference value 1 latched	
		Continuous measuring	
15-0		counter	every difference measuring is counted

Name		Difference value 1 latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal

	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0523 Difference value 2 latched

- **Description**

- With single measuring bit 0 of difference value 2 latched is valid only and can be assigned to a real-time status bit (see S-0-0305).
- With Continuous measuring this parameter is a 16-bit counter.

This parameter is processed by the drive only if the procedure command "probing cycle" (S-0-0170) is active, the signal "probe 2 enable" (S-0-0405) is set to 1 and edges of "probe 2" (S-0-0401) are registered. The drive calculates the selected difference and stores the result in the "difference value probe 2" (S-0-0511). The drive resets this parameter when the control unit cancels the procedure command "probing cycle" or when probe 2 enable is reset to 0.

For more details see clause 1.8.

Structure of difference value 2 latched

Bit No.	Value	Description	Comments
		Single measuring	
15-1		reserved	
0		difference value 2	bit 0 is valid only
	0	difference value 2 not latched	
	1	difference value 2 latched	
		Continuous measuring	
15-0		counter	every difference measuring is counted

Name		Difference value 2 latched
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0524 Time compensation positive edge, probe 1

- **Description**

The value of this parameter defines the time compensation of the positive edge. Time delays arising in the I/Os, sensors and mechanical imprecision can be compensated.

Name		Time compensation positive edge, probe 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 µs

S-0-0525 Time compensation negative edge, probe 1

- **Description**

The value of this parameter defines the time compensation of the negative edge. Time delays arising in the I/Os, sensors and mechanical imprecision can be compensated.

Name		Time compensation negative edge, probe 1
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 µs

S-0-0526 Time compensation positive edge, probe 2

- **Description**

The value of this parameter defines the time compensation of the positive edge. Time delays arising in the I/Os, sensors and mechanical imprecision can be compensated.

Name	Time compensation positive edge, probe 2

Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 µs

S-0-0527 Time compensation negative edge, probe 2

- **Description**

The value of this parameter defines the time compensation of the negative edge. Time delays arising in the I/Os, sensors and mechanical imprecision can be compensated.

Name		Time compensation negative edge, probe 2
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 µs

S-0-0530 Clamping torque

- **Description**

If the torque feedback value exceeds the clamping torque, the drive recognizes that the positive stop is reached. The clamping torque is used at positive stop functions (e.g. homing S-0-0147, positive stop drive procedure command S-0-0149, etc.).

Name		Clamping torque
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0532 Limit switch control

- **Description**

The configuration of the limit switch is programmable.

Structure of limit switch control			
Bit No.	**Value**	**Description**	**Comments**
15-3		reserved	
2		Error reaction	
	0	activation of limit switch generates an error (C1D)	
	1	activation of limit switch generates a warning (C2D)	
1		Enable of limit switches	
	0	both limit switches are disabled	
	1	both limit switches are enabled	
0		Inversion of limit switches	
	0	both limit switches are not inverted	
	1	both limit switches are inverted	

Name		Limit switch control
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0533 Motor continuous stall torque/force

- **Description**

With the motor continuous stall current (S-0-0111) the motor produces the Motor continuous stall torque/force according to the motor spec sheet. This parameter is not used for asynchronous motors.

Name		Motor continuous stall torque/force
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter

	Places after Decimal Point	3
	Write Protection	OL
	Conversion Factor	1
Unit		N or Nm
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 N or Nm

S-0-0534 Motor peak torque/force

- **Description**

With the motor peak current (S-0-0109)(or pressure) the motor produces the peak torque/force according to the motor spec sheet.

Name		Motor peak torque/force
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0800 Pressure command value

- **Description**

During the operation mode pressure control the pressure command value is transferred from the control unit to the drive.

Name		Pressure command value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806
	Write Protection	never
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0801 Pressure command value additive

- **Description**

This IDN is used if an additional pressure offset is required during pressure control operation mode in the drive. The additive pressure command value is added to the pressure command value S-0-0800.

Name		Pressure command value additive
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806
	Write Protection	never
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0802 Bipolar pressure limit value

- **Description**

The bipolar pressure limit value describes the maximum allowable pressure in both directions. If the pressure limit value is exceeded, the drive responds by setting the status 'p > plimit' (S-0-0830) and sets the status code ?? (S-0-0390).

Name		Bipolar pressure limit value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806
	Write Protection	never
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0803 Pressure feedback value A

- **Description**

The pressure feedback value A is transferred from the drive to the control unit in order to allow the control unit to periodically display the pressure. The pressure feedback value A refers to the piston area A.

Name		Pressure feedback value A
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806

	Write Protection	always
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0804 Pressure feedback value B

- **Description**

The pressure feedback value B is transferred from the drive to the control unit in order to allow the control unit to periodically display the pressure. The pressure feedback value B refers to the piston area B.

Name		Pressure feedback value B
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806
	Write Protection	always
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0805 Pressure polarity parameter

- **Description**

This parameter is used to switch polarities of reported pressure data for specific applications. Polarities are switched outside (i.e., on the input and output) of a closed loop system.

Table 21: Structure of the Pressure polarity parameter

Bit No.	Value	Description	Comments
15-3		reserved	
2		Pressure feedback value	S-0-0809, S-0-0803 and S-0-0804
	0	Non-inverted	
	1	Inverted	
1		Additive pressure command value	S-0-0801
	0	Non-inverted	
	1	Inverted	
0		Pressure command value	S-0-0800

	0	Non-inverted	
	1	Inverted	

Name		Pressure polarity parameter
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0806 Pressure data scaling type

- **Description**

A variety of scaling methods can be selected by means of this scaling type parameter (see Scaling of operation data).

Table 22: Structure of pressure data scaling type

Bit No.	Value	Description	Comments
15-5		reserved	
4		Units for pressure	
	0	bar	
	1	psi (pound force square inch)	optional
3			
	0	preferred scaling	
	1	parameter scaling	uses S-0-0807 and S-0-0808
2–0		Scaling method	
	000	reserved	
	001	pressure scaling	

Name		Pressure data scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0807 Pressure data scaling factor

- **Description**

This parameter defines the scaling factor for all pressure data in a sub-device (see Scaling of operation data).

Name		Pressure data scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0808 Pressure data scaling exponent

- **Description**

This parameter defines the scaling exponent for all pressure data in a sub-device (see Scaling of operation data).

Table 23: Structure of the scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Pressure data scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-0809 Pressure feedback value

- **Description**

The pressure feedback value is transferred from the drive to the control unit in order to allow the control unit to periodically display the pressure. The pressure feedback value is calculated by pressure feedback value A and B.

Name		Pressure feedback value
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0806
	Write Protection	always
	Conversion Factor	see S-0-0806
Unit		see S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0806

S-0-0810 Pressure ramp

- **Description**

This parameter contains the pressure difference per time of the pressure command value.

Name		Pressure ramp
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0814
	Write Protection	never
	Conversion Factor	see S-0-0814
Unit		see S-0-0814
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0814

S-0-0813 System pressure

- **Description**

The system pressure is the current pressure of the hydraulic aggregate.

Name		System pressure
Attribute	Length (Octet)	4
	Data Type and Display Format	signed decimal

	Function	Parameter
	Places after Decimal Point	S-0-0806
	Write Protection	never
	Conversion Factor	S-0-0806
Unit		S-0-0806
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		S-0-0806

S-0-0814 Pressure change per time scaling type

- **Description**

A variety of scaling methods can be selected by means of this scaling type parameter (see Scaling of operation data).

Table 24: Structure of pressure change per time scaling type

Bit No.	Value	Description	Comments
15-6		reserved	
5		Time units	
	0	Seconds [s]	
	1	Minutes [min]	optional
4		Pressure units	
	0	bar [bar]	1 bar equals 14,504 psi
	1	psi [psi]	1 psi equals 0,068948 bar
3			
	0	preferred scaling	
	1	parameter scaling	uses S-0-0815 and S-0-0816
2–0		Scaling method	
	000	reserved	
	001	pressure change per time scaling	Default [bar/s]
	010-111	reserved	

Name		Pressure change per time scaling type
Attribute	Length (Octet)	2
	Data Type and Display Format	Binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a

Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-0815 Pressure change per time scaling factor

- **Description**

This parameter defines the scaling factor for all pressure change per time data in a drive (see Scaling of operation data).

Name		Pressure change per time scaling factor
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-0816 Pressure change per time scaling exponent

- **Description**

This parameter defines the scaling exponent for all pressure change per time data in a drive (see Scaling of operation data).

Table 25: Structure of this scaling exponent

Bit No.	Value	Description	Comments
15		Sign of the exponent	
	0	positive	
	1	negative	
14-0		Exponent	

Name		Pressure change per time scaling exponent
Attribute	Length (Octet)	2
	Data Type and Display Format	signed decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-0822 Torque/force ramp reference

- **Description**

This parameter contains the torque/force difference which can be reached within the torque/force ramp time (S-0-0823).

Name		Torque/force ramp reference
Attribute	Length (Octet)	see S-0-0086
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	see S-0-0086
	Write Protection	never
	Conversion Factor	see S-0-0086
Unit		see S-0-0086
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		see S-0-0086

S-0-0823 Torque/force ramp time

- **Description**

This is the reference time for the torque/force ramp.

Name		Torque/force ramp time
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	1
	Write Protection	never
	Conversion Factor	1
Unit		ms
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,1 ms

S-0-1000 SCP Type & Version

- **Description**

The SCP Type & Version contains a list of the SERCOS Communication capabilities of this SERCOS slave and the dedicated version.

The bits 15 ... 8 indicate the SCP class which is available in the SERCOS slave, bits 7 ... 0 indicate the version of this SCP class.

Table 26: SCP Type & Version

Bit No.	Value	Description	Comments
15-8		SCP class ID	
	0x01	SCP_FixCFG	
	0x02	SCP_VarCFG	
	0x03	SCP_Sync	
	0x04	SCP_WD	
	0x05	SCP_Diag	
	0x06	SCP_RTB	
	0x07	SCP_HP	
	0x08	SCP_SMP	
	0x09	SCP_MuX	
	0x0A	SCP_NRT	
	0x0B	SCP_SIG	
7-0		Version	
	0x00	reserved	
	0x01	1.st version	

Name		SCP Type & Version
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1002 Communication Cycle time (tScyc)

- **Description**

The interfaces communication cycle time (t_{Scyc}) defines the intervals during which the real-time data are transferred. The interface cycle times are defined as 31,25µs, 62,5µs, 125µs, 250µs, 500µs, 750 µs, 1ms, 1,25 ms .... up to 65ms in steps of 250µs. In CP2, t_{Scyc} needs to be transferred from the master to the slave and has to be activated in CP3 and CP4. Min/max values are mandatory.

Name	Communication cycle time (tScyc)

Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		31 250 (31,250 µs)
Maximum Value		65 000 000 (65000,000 µs)
Scaling / Resolution of LSB		0,001 µs

S-0-1003 Allowed MST losses in CP3/CP4

- **Description**

This parameter defines the maximum number of successive communication cycles in which a slave may not receive the MST in CP3 and CP4. The slave shall switch from CP3 or CP4 back to NRT mode if it did not receive its MST within more than this user-defined quantity. After half the amount of cycles specified in this parameter, the slave shall set the communication warning bit (bit 15) in its device status and set the error code in S-0-0390.

Whereas this parameter shall be set to a value that is large enough (compare to S-0-1050.x.11), so that the master can keep communicating with undisturbed sub-devices for safely stopping the other parts of the machine.

Name		Allowed MST losses in CP3/CP4
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1005 Minimum feedback processing time (t5)

- **Description**

t_5 determines the slave-specific maximum time duration that it needs between latching its producer data (e.g. feedback values) and making them ready for transmission within ATs.

Name		Minimum feedback processing time (t5)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3

	Write Protection	always
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001µs

S-0-1006 AT0 transmission starting time (t1)

- **Description**

The AT0 transmission starting time (t1) determines the nominal time interval between the end of MST and begin of AT0. The master sends its AT0 based on the MST in CP3 and CP4. This parameter shall be transferred by the master to the slave during CP2.

Name		AT0 transmission starting time (t1)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		0
Maximum Value		t_{Scyc}
Scaling / Resolution of LSB		0,001µs

S-0-1007 Feedback acquisition capture point (T4)

- **Description**

The time t_4 determines the time duration between the end of the MST and the feedback acquisition capture point. The master shall set the acquisition capture point of the feedback values T_4 related to the synchronization reference time(TSref). The master sets the time t_4 to be less to the SERCOS cycle time.

At the time T_4 the command values shall be activated in the sub-device.

The sub-device shall enable the acquisition capture point of the feedback during CP3.

Name		Feedback acquisition capture point (T4)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs

Minimum Value	0
Maximum Value	$< t_{Scyc}$
Scaling / Resolution of LSB	0,001µs

S-0-1008 Command value valid time (t3)

- **Description**

The command value valid time indicates the time after which the slave can access the new values from the MDT related to the synchronization time. Thus the master can preset the same command value valid time for all coordinated applications.

Name		Command value valid time (t3)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		0
Maximum Value		t_{Scyc}
Scaling / Resolution of LSB		0,001µs

S-0-1009 Device Control (C-Dev) offset in MDT

- **Description**

The device control offset in the MDT defines MDT number and the position within this MDT for the device control. This parameter shall be transferred by the master to each slave during CP2. It shall become active during CP3 in the master and slave. This offset shall start after the SERCOS III header within the MDT.

Structure of C-Dev offset in MDT

Bit No.	Value	Description	Comments
15-14		reserved	
13-12		MDT number	
	00	MDT0	
	01	MDT1	
	10	MDT2	
	11	MDT3	
11-0		Offset in MDT	in octets
	0...1492	C-Dev offset in MDT	

Name	Device Control (C-Dev) offset in MDT

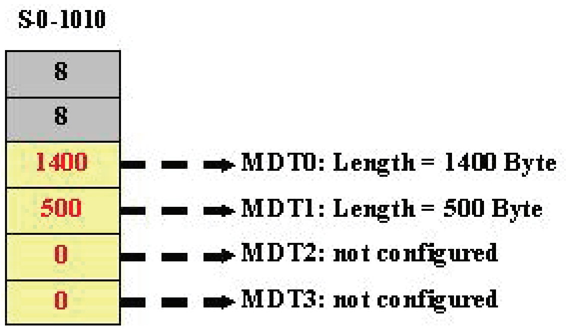
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1010 Lengths of MDTs

- **Description**

The lengths of the MDTs shall be expressed in octets. The parameter shall contain the lengths of the four possible master data telegrams. The lengths are necessary for the initialization of the SERCOS III hardware. Always all four lengths have to be specified. Not configured MDTs shall be marked with length = 0 and the master shall not transmit these MDTs. Each slave shall be informed by the master during CP2 of the lengths of all configured MDTs. It shall become active in the master and slave during CP3. The length includes all data between the end of MDT header to the begin of FCS. The number of list elements of this parameter is fixed to four.

Example Lengths of MDTs



Name		Lengths of MDTs
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		1494
Scaling / Resolution of LSB		1

S-0-1011 Device Status (S-Dev) offset in AT

- **Description**

The device status offset in AT defines the position of device status field of the slave in one of the ATs, expressed as a OCTET position. The offset shall start after the SERCOS III header within the AT. This parameter shall be transferred by the master to each slave during CP2. It shall become active during CP3 in the master and slave.

Structure of Device Status (S-Dev) offset in AT			
Bit No.	**Value**	**Description**	**Comments**
15-14		reserved	
13-12		AT number	
	00	AT0	
	01	AT1	
	10	AT2	
	11	AT3	
11-0		S-Dev offset in AT	in octets
	0...1492	S-Dev offset in AT	

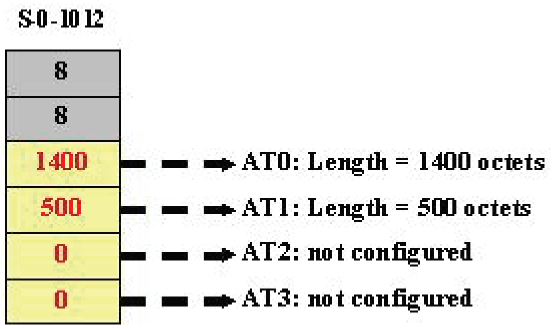
Name		Device Status (S-Dev) offset in AT
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1012 Lengths of ATs

- **Description**

The lengths of the ATs shall be expressed in octets. The parameter shall contain the lengths of the four possible ATs. The lengths are necessary for the initialization of the SERCOS III hardware. Always all four lengths have to be specified. Not configured ATs shall be marked with length = 0 and the master shall not transmit these ATs. Each slave shall be informed by the master during CP2 of the lengths of all configured ATs. It shall become active in the master and slave during CP3. The length includes all data between the end of AT header to the begin of FCS. The number of list elements of this parameter is fixed to four.

Example Lengths of ATs



Name		Lengths of ATs
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		1494
Scaling / Resolution of LSB		1

S-0-1013 SVC offset in MDT

- **Description**

The SVC offset in the MDT defines the position of the service channel for the slave. This offset shall start after the SERCOS III header within the related MDT. Every slave shall be informed by the master during CP2 of the offset of the service channel in the MDT. This parameter shall become active during CP3 in the master and slave.

Structure of SVC offset in MDT			
Bit No.	**Value**	**Description**	**Comments**
15-14		reserved	
	0	No others values allowed	
13-12		MDT number	
	00	MDT0	
	01	MDT1	
	10	MDT2	
	11	reserved	
11-0		MDT SVC-Offset	in octets
	0...1484	MDT SVC-Offset	

Name		SVC offset in MDT
Attribute	Length (Octet)	2

	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1014 SVC offset in AT

- **Description**

The SVC offset in AT defines the position of a service channel for the slave. The offset shall start after the SERCOS III header within the related AT. Every slave shall be informed by the master during CP2 of the offset of the service channel in the AT. This parameter shall become active during CP3 in the master and slave.

Structure of SVC offset in AT			
Bit No.	**Value**	**Description**	**Comments**
15-14		reserved	
	0	No others values allowed	
13-12		AT number	
	00	AT0	
	01	AT1	
	10	AT2	
	11	reserved	
11-0		AT SVC-Offset	in octets
	0...1484	AT SVC-Offset	

Name		SVC offset in AT
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1015 Ring delay

- **Description**

The master determines the entire ring delay and assigns it to the slaves. The slaves need the ring delay to determine their synchronization. More details see synchronization.

Name		Ring delay
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	never
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		1048,575
Scaling / Resolution of LSB		0,001µs

S-0-1016 Slave delay (P/S)

- **Description**

After the master has assign the ring delay (S-0-1015) to the slaves, the slaves determine SYNCCNT-P/S, when the procedure command S-0-1024 is executed. If the topology is ring, both list elements shall be filled. In case of line topology the corresponding list element holds the value, the other list element is set to zero.

- List element 0 is SYNCCNT-P
- List element 1 is SYNCCNT-S

Name		Slave delay (P/S)
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	always
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001µs

S-0-1017 NRT transmission time

- **Description**

The NRT transmission time includes two list elements(t_6 & t_7).

- The first list element t_6 (begin of the NRT channel) defines the time for the slave to switch the topology from RT communication to NRT communication. If no NRT channel is required, the master shall set t_6 to 0. If NRT channel is required master shall set t_6 as specified in clause medium access.

- The second list element t_7 (end of NRT channel) defines the time for the slave to switch from any topology to the current commanded topology of RT communication . The master shall set t_7 as specified in clause medium access even if the NRT channel is not used.

If the length of the NRT channel is less than 125µs, S-0-1027.0.1 shall be adjusted accordingly.

Name		NRT transmission time
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		0
Maximum Value		t_{Scyc}
Scaling / Resolution of LSB		0,001µs

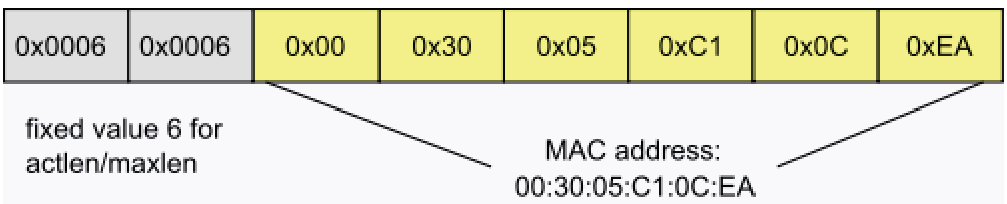
S-0-1019 MAC Address

- **Description**

The slave inserts its MAC Address in this parameter.

Name		MAC Address
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

Strucure of MAC Address



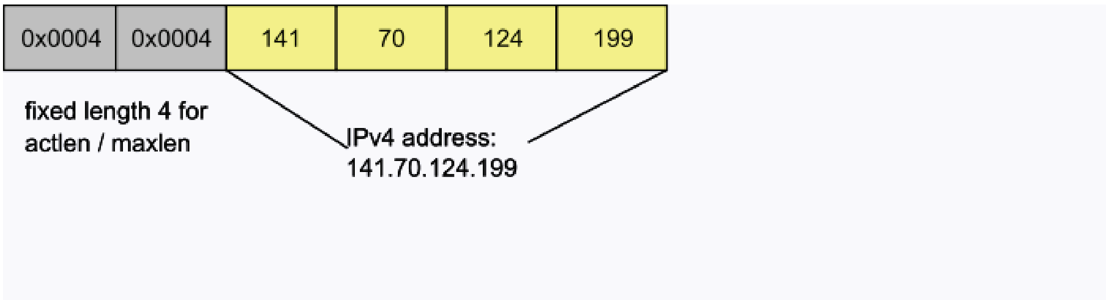
S-0-1020 IP address

Description

This IDN contains the IP address of the slave's SERCOS III interface. The master may change the IP address by writing this IDN.

Name		IP Address
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

Structure of IP Address



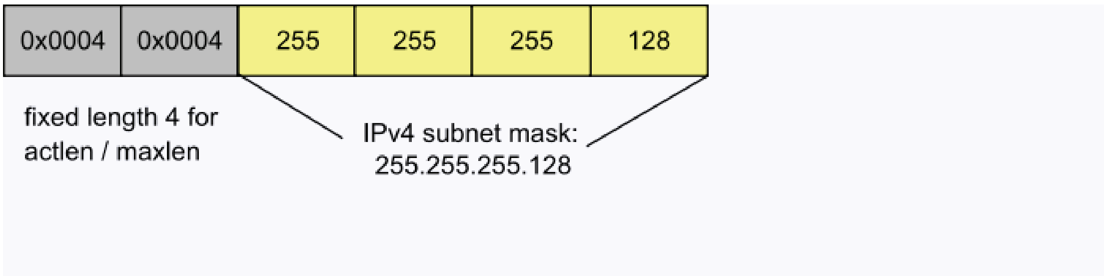
S-0-1021 Subnet Mask

- **Description**

This IDN contains the subnet mask. The master may change the subnet mask for IP communication inside the NRT channel.

Name		Subnet mask
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

Structure of Subnet mask



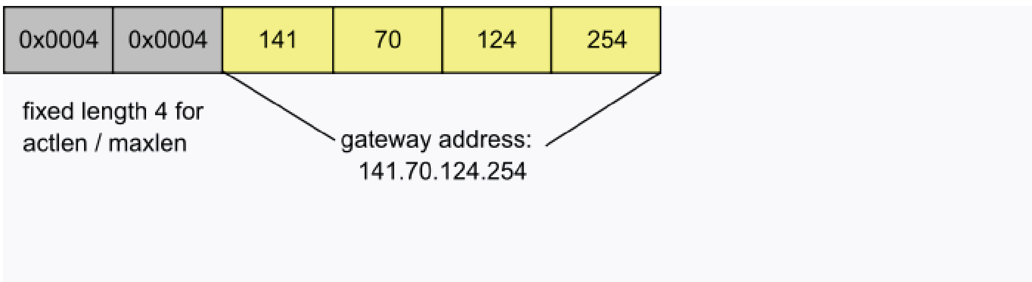
S-0-1022 Gateway address

Description

This IDN contains the gateway address. The master may change the gateway address for IP communication inside the NRT channel.

Name		Gateway address
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

Structure of Gateway address



S-0-1023 SYNC jitter

- **Description**

The master shall program the maximum synchronization jitter. The jitter is used in the slave to determine the width of the MST receive window. The MST receive window is 2x synchronization jitter.

Name	SYNC jitter

Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		0,001 µs

S-0-1024 SYNC delay measuring procedure command

- **Description**

After activation of this procedure command the slave shall determine the slave delay (P/S) (S-0-1016) depending on the ring delay (S-0-1015).

- In CP2 the positive acknowledgment is generated if the slave was able to determine valid slave delay (P/S). In this case the slave synchronizes in CP3 automatically.
- In CP3/4 the positive acknowledgment is generated if the slave was able to determine valid slave delay (P/S) and the slave has synchronized again.

Otherwise the slave will generate a negative acknowledgment and diagnostic codes in the diagnostic number (S-0-0390).

The master has to activate this procedure command

- and wait until it is finished in CP2 before activation of S-0-0127,
- in CP3 and CP4 after the ring recovery

in every slave which has to be synchronized.

Name		SYNC delay measuring procedure command
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	procedure command
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1026 Version of communication hardware

- **Description**

This parameter includes the SERCOS III-specific communication hardware identification.

e.g.

- FPGA version and revision
- netX version and revision

Example: SERCON100 Vxx.rr

Name		Version of communication hardware
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	Text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1027 MTU

- These IDNs are related to the MTU:

- S-0-1027.0.1 Requested MTU
- S-0-1027.0.2 Effective MTU

S-0-1027.0.1 Requested MTU

Description

The requested MTU defines the maximum number of octets, that may be sent within the NRT channel by higher layers. This IDN only defines the target value for the MTU and is used to calculate the effective MTU which can be read from S-0-1027.0.2. The effective value may be different from the target value, if the target value does not lie in between the limits of the current phase.

The effective MTU has to be set at the interface immediately and updated when this IDN is written or a phase change has ocurred.

E.g. if this IDN is set to 80, the effective MTU during NRT, CP0, CP1, CP2 and HP0 will be 576.

Note:

- This parameter shall be used to calculate the time of the $\text{event}_{\text{LastTransmit}}$ for the FPGA.
- NetX does not need the $\text{event}_{\text{LastTransmit}}$.

Calculation of the effective MTU:

$t_{NRT} = (t_7 - t_6) > 6,72us$ (see S-0-1017)

$$MTU(t_{NRT}) = min\{1500; \quad \frac{t_{NRT}}{s} * 12.498.750 - 38\}$$

Communication phase(cp)	upper limit MTU	lower limit MTU
NRT	1500	576
CP0	1500	576
CP1	1500	576
CP2	1500	576
CP3	$MTU(t_{NRT})$	46
CP4	$MTU(t_{NRT})$	46
HP0	1500	576
HP1	$MTU(t_{NRT})$	46
HP2	$MTU(t_{NRT})$	46

$$MTU_{interim} = min\{upper - limit(cp); \quad MTU_{target}\}$$

$$MTU_{effective} = max\{lower - limit(cp); \quad MTU_{interim}\}$$

Default value for this IDN shall be 576.

Name		Requested MTU
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		46
Maximum Value		1500
Scaling / Resolution of LSB		1

S-0-1027.0.2 Effective MTU

- **Description**

This IDN is the current MTU. The current MTU is calculated using S-0-1017 and S-0-1027.0.1. For further information see S-0-1027.0.1.

Name		Effective MTU
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		46
Maximum Value		1500
Scaling / Resolution of LSB		1

S-0-1028 Error counter MST-P/S

- **Description**

The MST error counter is incremented if a valid MST is neither received on port 1 nor on port 2 in communication phases 3 and 4.

The MST error counter shall stop counting as soon as it reaches 65 535. It means that if the counter has a value of 65 535, there may have been more than 65 535 invalid MST's (e.g., noisy transmission over a long period of time).

The error counter MST-P/S is reset with S-0-0127.

Name		Error counter MST-P/S
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1031 Test pin assignment Port 1 & Port 2

- **Description**

This parameter is used to assign communication related hardware signals to the test pins TS1 and TS2. The assigned signal is mapped to the test pins TS1 or TS2.

Structure of test pin assignment Port 1 & Port 2			
Bit No.	**Value**	**Description**	**Comments**
15-12		reserved	

11-8		the selected signal is mapped to the test pin TS2	selectable signals are shown in the following table.
7-4		reserved	
3-0		the selected signal is mapped to the test pin TS1	selectable signals are shown in the following table.

Selectable output signals

Value	Signal Slave	Description
0000	Port 1 MST	MST pulse from the Rx MAC of port 1 (40 ns duration)
0001	Port 2 MST	MST pulse from the Rx MAC of port 2 (40 ns duration)
0010	TMST	TMST signal after MST generator (TSref)
0011	CON_CLK	CON Clock from the TCNT timer
0100	DIV_CLK	DIV Clock from the DIV Clock unit (only if present)
0101	TCNT Reload	Overflow of the TCNT timer
0110	Port 1 TCNT Reload	Overflow of the Port 1 timer
0111	Port 2 TCNT Reload	Overflow of the Port 2 timer
1000	Port 1 IP Open	Port 1 IP window
1001	Port 1 IP Open Write	Port 1 IP transmit window
1010	Port 2 IP Open	Port 2 IP window
1011	Port 2 IP Open Write	Port 2 IP transmit window
1100	Port 1 MST Window Open	Port 1 MST window
1101	Port 2 MST Window Open	Port 2 MST window
1110	Port 1 Rx Frame	Reception of a frame on port 1
1111	Port 2 Rx Frame	Reception of a frame on port 2

Name		Test pin assignment Port 1 & Port 2
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1035 Error counter Port1 and Port2

- **Description**

This counter counts all Ethernet errors such as

- received Ethernet frames with defective frame check sequence or RxER indications,
- received Ethernet frames with a defective frame. Defective frames are wrong aligned frames.

The low word represents port 1, the high word represents port 2.

The counter shall start in CP0 and will be incremented maximal one time per SERCOS cycle. The counter is writable, so a human machine interface may reset this counter. The maximum value is 0xFFFF. The counters are not buffered, so they are 0x0000 on power-on.

Name		Error counter Port 1 & Port 2
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1040 SERCOS address

- **Description**

This parameter contains the SERCOS address which is assigned to the SERCOS sub-device.

- If the sub-device has no address switch the sub-device shall store the known address in this parameter.
- If the sub-device contains an address switch so this operational data is not changeable.

The default value shall be 0.

Name		SERCOS address
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		511
Scaling / Resolution of LSB		1

S-0-1041 AT Command value valid time (t9)

- **Description**

After the AT Command value valid time (t_9) the slave can access the new command values from the AT. Thus the master can preset the same AT Command value valid time (t_9) for all coordinated applications.

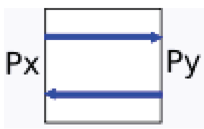
Name		AT Command value valid time (t9)
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		0
Maximum Value		t_{Scyc}
Scaling / Resolution of LSB		0,001 µs

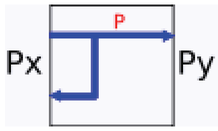
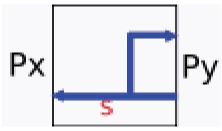
S-0-1044 Device Control (C-Dev)

- **Description**

The device control contains the control information (e.g. topology control, etc.) which are set by the master and evaluated by the slave. The device control is not part of a connection.

Device control field (C-Dev):

Bit No.	Value	Description	Comments
15		Identification	
	0	No Identification request	
	1	Identification request	Slave shows the condition of this bit at the SERCOS III LED or at the display. This function is used for the remote address allocation or for configuration errors between master and slave.
14		Topology HS	Initial value is 0 in every CP
	toggle		Master toggles every time it requires a topology change
13-12		Topology control	Master selects the new topology
	00	Fast-Forward on both ports	

	01	Loopback with Forward of P-Telegrams	
	10	Loopback with Forward of S-Telegrams	
	11	reserved	
11		Status physical topology	used in the NRT channel only. If the slave detects a toggle, then it has to drop Source address table
	0	physical ring is broken	
	1	physical ring is closed	
10		reserved for SERCOS III time	
9-0		reserved	

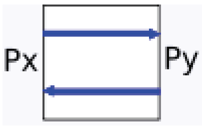
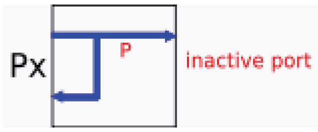
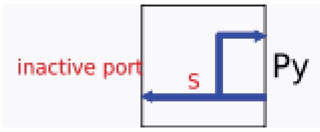
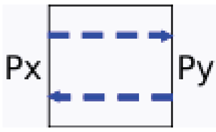
Name		Device Control
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1045 Device Status (S-Dev)

- **Description**

The device status contains the status information (e.g. topology status, etc.) which are set by the slave and evaluated by the master. The device status is not part of a connection.

Device status field (S-Dev):			
Bit No.	**Value**	**Description**	**Comments**
15		Communication warning interface	
	0	No warning	
	1	Communication warning	E.g. Number of permitted MST losses has

		occurred	exceeded the half value of S-0-1003
14		Topology HS	Initial value is 0 in every CP
	toggle		Slave toggles, if the request of the master has been processed
13-12		Topology status	shows the current topology (Px or Py can be either P1 or P2)
	00	Fast-Forward on both ports	
	01	Loopback with Forward of P-Telegrams	
	10	Loopback with Forward of S-Telegrams	
	11	NRT mode (store and forward)	
11-10		Status of inactive port	these bits are don't care if the topology bits (13-12) are 00 or 11
	00	no link on inactive port	no device is connected
	01	link on inactive port	device is connected
	10	P-Telegram on inactive port	SERCOS III device is connected
	11	S-Telegram on inactive port	SERCOS III device is connected
9		Error connection	
	0	Error-free connection	
	1	Error in the connection occurred	Consumer recognized an error in a connection
8		Slave valid	See Switching of communication phases (CPS)
	0	Slave not valid	is set to 0 when entering CP0 or if CPS = 1
	1	Slave valid	is set to 1 when new CP is prepared and CPS = 0
7		Error of device	inclusive sub-device and resource errors
	0	no error	
	1	error	detailed information is shown in S-0-0390
6		Warning of device	inclusive sub-device and resource warnings
	0	no Warning	

	1	warning	detailed information is shown see S-0-0390
5		Procedure command change bit	
	0	No change in procedure command acknowledgement	
	1	Changing procedure command acknowledgment	procedure command is positive or negative acknowledged
4		Parameterization level	PL
	0	PL is not active	is set to 0 with procedure command S-0-0422
	1	PL is active	is set to 1 with procedure command S-0-0420
3		reserved for SERCOS time	
2		reserved for RT-Plug	
1-0		reserved	

Name		Device Status
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1046 List of SERCOS addresses in device

- **Description**

The device stores the SERCOS addresses of the slaves that participate in the communication. If only one slave exists in the device, this IDN may be absent.

Name		List of SERCOS addresses in device
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050 Connections

- These are the parameter which describes the configuration of SERCOS III connections

- S-0-1050.x.01 Connection setup
- S-0-1050.x.02 Connection Number
- S-0-1050.x.03 Telegram Assignment
- S-0-1050.x.04 Max. Length Of Connection
- S-0-1050.x.05 Current length of connection
- S-0-1050.x.06 Configuration List
- S-0-1050.x.07 Assigned connection capability
- S-0-1050.x.08 Connection Control (C-Con)
- S-0-1050.x.10 Producer Cycle Time
- S-0-1050.x.11 Allowed Data Losses
- S-0-1050.x.12 Error Counter Data Losses
- S-0-1050.x.20 IDN Allocation of real-time bit
- S-0-1050.x.21 Bit allocation of real-time bit

S-0-1050.x.01 Connection setup

- **Description**

This parameter configures connections.

Bit No.	Value	Description	Comments
15		usage of configuration	
	0	not used	slave shall not use this connection
	1	used	slave shall use this connection
14		function within connection	
	0	consumer	
	1	producer	
13-12		source of connection configuration	
	00	master	configured by master
	01	reserved	
	10	external	not configured by master
	11	reserved	
11-6		reserved	
5-4		type of configuration	
	00	variable configuration of IDNs with S-0-1050.x.6	
	01	configuration with connection length, see S-0-1050.x.5. S-0-1050.x.6 will not be considered	in case of FSP I/O: The connection assignment is defined as follows: (C-CON - IO Control - S-0-1500.x.5) (C-CON - IO Status - S-0-1500.x.9)

	10	standard telegram see S-0-0015	used by FSP-Drive only
	11	reserved	
3		mechanism of producing	
	0	producer cycle synchronous	
	1	asynchronous	
2		reserved	
1-0		mechanism of monitoring	for consumer
	00	producer cycle synchronous operation	
	01	asynchronous operation with watchdog	time defined by S-0-1050.x.10
	10	asynchronous operation without watchdog	
	11	reserved	

Name		Connection setup
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.02 Connection Number

- **Description**

The connection number is used to identify a connection. The producer and all consumers of the same connection shall have the same connection number.

Name		Connection Number
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		65535

Scaling / Resolution of LSB	1

S-0-1050.x.03 Telegram Assignment

- **Description**

The telegram assignment defines the

- telegram type (MDT or AT), the
- telegram number and the
- offset of connection control

for this connection. The telegram offset points to the Connection control (C-Con) of this connection. This offset shall start with 0x8 octets for the initial data after the SERCOS III header within the MDT or AT.

Structure of telegram assignment			
Bit No.	**Value**	**Description**	**Comments**
15-14		reserved	
13-12		Telegram number	
	00	MDT0 / AT0	
	01	MDT1 / AT1	
	10	MDT2 / AT2	
	11	MDT3 / AT3	
11		Telegram type	
	0	AT	
	1	MDT	
10-0		Offset of connection control	in octets
	0...1492	Offset in MDT or AT	

Name		Telegram assignment
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		1492
Scaling / Resolution of LSB		1

S-0-1050.x.04 Max. Length Of Connection

- **Description**

This parameter defines the maximum length of this connection. The 2 octets for the connection control (C-Con) are part of this length. If the slave shows a length of n octets this length contains 2 octets C-Con and n-2 octets data.

Name		Max. Length of Connection
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		2
Maximum Value		product specific or S-0-1060.x.04
Scaling / Resolution of LSB		1

S-0-1050.x.05 Current length of connection

- **Description**

This parameter defines the current length of this connection. The 2 octets for the connection control (C-Con) are part of this length. If the slave shows a length of n octets this length contains 2 octets C-Con and n-2 octets data. This parameter shall be updated by the slave if configuration parameter are changed.

Name		Current length of connection
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.06 Configuration List

- **Description**

If the connection data is configured with IDNs (type of connection, bit 5-4 = 00, in S-0-1050.x.1) this parameter contains the list of ident numbers (4 octets) within this connection. The sequence of the IDNs in this parameter and the sequence of the corresponding operation data in the connection is identical.

Name	Configuration List

Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.08 Connection Control (C-Con)

- **Description**

This parameter contains the image of the control word C-Con of this connection.

Table 27: Connection control (C-Con)

Structure of connection control (C-Con)			
Bit No.	**Value**	**Description**	**Comments**
15-10		reserved	
9		reserved	for Real-time bit 4
8		reserved	for Real-time bit 3
7		Real-time bit 2	
6		Real-time bit 1	
5-4		reserved	
3		Producer synchronization	
	0	not synchronized	producer is not synchronous with the synchronization reference time (TSref)
	1	synchronized	producer is synchronous with the synchronization reference time (TSref)
2		Data field delay	the consumer shall prefer taking the data of the port at which this bit has the value 0.
	0	no delay	data are transmitted without delay in the same SERCOS cycle.
	1	delay	master has copied the data and therefore the data are transmitted with additional delay of **one SERCOS cycle**.
1		New data	new producer data
	toggle		detailed description of new data bit:

			• initial value of this bit is 0 in CP4 • when changes occur, this bit has to be toggled • every toggle of this bit announces new data in the connection and then the data are exchanged between connection and application. This implies, that in the consumer when the data of a certain producer cycle has not been received, the prospected value for this bit has also to be toggled for the next communication cycle. • at a cycle synchronous connection this bit shall always be toggled in the related SERCOS cycle. • at an asynchronous connection this bit triggers a watchdog, i.e. after toggle of this bit the monitoring is always started again with the monitoring time (t_{Pcyc}). • this bit shall be toggled once per monitoring time (t_{Pcyc}), after producer ready (bit 0) is set to 1.
0		Producer ready	
	0	not valid	the producer does not generate any data in this connection yet
	1	valid	the producer generates data in this connection. The consumer can process the connection data if the producer has toggled the New data (bit 1). The producer ready bit shall be evaluated in CP4 only.

Name		Connection Control (C-Con)
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.10 Producer Cycle Time

- **Description**

The Producer cycle time should be an integer multiple of the communication cycle time:

- $t_{Pcyc} = t_{Scyc} \cdot n \quad \forall\, n \in \mathbb{N} > 0$.

Minimum and maximum input values are mandatory.

Name		Producer cycle time
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal

	Function	Parameter
	Places after Decimal Point	3
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		µs
Minimum Value		31 250 (31,250 µs)
Maximum Value		product specific
Scaling / Resolution of LSB		0,001µs

S-0-1050.x.11 Allowed Data Losses

- **Description**

This parameter indicates the maximum amount of consecutive lost producer data, before a connection is broken. If this connection is broken the consumer shall not process data anymore and sets the connection error in the device status. The default value = 1

Name		Allowed Data Losses
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4
	Conversion Factor	1
Unit		n/a
Minimum Value		1
Maximum Value		65535
Scaling / Resolution of LSB		1

S-0-1050.x.12 Error Counter Data Losses

- **Description**

This parameter counts the amount of lost producer data. This counter will be reset with the positive edge of the designated Producer ready in the Connection control. This counter does not have any overrun and ends with 65535.

Name		Error Counter Data Losses
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.20 IDN Allocation of real-time bit

- **Description**

In order to assign signals to the real-time bits (see S-0-0398/S-0-0399), the IDN of the signal is written to this parameter. After the allocation of the IDN and the bit number (see S-0-1050.x.21), the assigned signal is copied in the corresponding real-time bit. If the S-0-1050.x.21 isn't supported by the slave, the bit 0 of the IDN is configured automatically. This parameter contains maximum 2 list elements.

- List element 0 corresponds to real-time bit 1: IDN of assigned signal
- List element 1 corresponds to real-time bit 2: IDN of assigned signal

See S-0-1050.x.21.

Name		IDN Allocation of real-time bit
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1050.x.21 Bit allocation of real-time bit

- **Description**

This parameter contains the bit number of the operation data assigned in the S-0-1050.x.20. The signal assigned by the IDN in S-0-1050.x.20 and the bit number in this parameter is copied into the corresponding real-time bit. This parameter contains a maximum of 2 list elements.

- List element 0 corresponds to real-time bit 1: bit number of assigned signal
- List element 1 corresponds to real-time bit 2: bit number of assigned signal

see S-0-1050.x.20

Name		Bit allocation of real-time bit
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a

Minimum Value	0
Maximum Value	63
Scaling / Resolution of LSB	1

S-0-1051 Image of connection setups

- **Description**

This IDN shows the actual state of all the connections of the slave, corresponding to S-0-1050.x.1.

The quantity of list elements shows the maximum number of connections of this slave.

Name		Image of connection setups
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1100 SMP Diagnostics

- The structure elements of S-0-1100 provide diagnostic information about the SMP stack.

- S-0-1100.0.01 Diagnostic counter sent SMP fragments
- S-0-1100.0.02 Diagnostic counter received SMP fragments
- S-0-1100.0.03 Diagnostic counter dropped SMP fragments

S-0-1100.0.01 Diagnostic counter sent SMP fragments

- **Description**

This parameter displays the number of SMP fragments transmitted via the SMP stack since system startup. This counter has an auto-rollover at 2^{32}-1 to 0.

Name		Diagnostic counter sent SMP fragments
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-1100.0.02 Diagnostic counter received SMP fragments

- **Description**

This parameter displays the number of SMP fragments received by the SMP stack since system startup. This counter has an auto-rollover at 2^{32}-1 to 0.

Name		Diagnostic counter received SMP fragments
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1100.0.03 Diagnostic counter dropped SMP fragments

- **Description**

This parameter displays the number of SMP fragments received by SMP stack that were dropped because its header did not match the receiver's expectations.

Reasons for this include:

- invalid session ID,
- wrong sequence count,
- incorrect sequence of the FOS/LOS bits.

This counter shows the number of dropped fragments since system startup and has an auto-rollover at 2^{32}-1 to 0.

Name		Diagnostic counter dropped SMP fragments
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-1101 SMP Configuration

- The structure elements of S-0-1101 contain information related to an SMP container. Each instance of S-0-1101 describes one container.

- S-0-1101.x.01 SMP Container Data
- S-0-1101.x.02 List of session identifiers
- S-0-1101.x.03 List of session priorities

S-0-1101.x.01 SMP Container Data

- **Description**

This parameter contains the actual data transmitted via a SMP container.

Name		SMP Container Data
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1101.x.02 List of session identifiers

- **Description**

This parameter contains the list of all session identifiers currently set up for a SMP container. Each list entry corresponds to the list entry with the same index in S-0-1101.x.3 and defines the identifier for this session.

The lists in S-0-1101.x.2 and S-0-1101.x.3 shall have the same current length.

Name		List of session identifiers
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-1101.x.03 List of session priorities

- **Description**

This parameter contains a list of priority values for the sessions in this SMP container. Each list entry corresponds to the list entry with the same index in S-0-1101.x.2 and defines the priority for this session. The highest priority is 0, the lowest priority is 3.

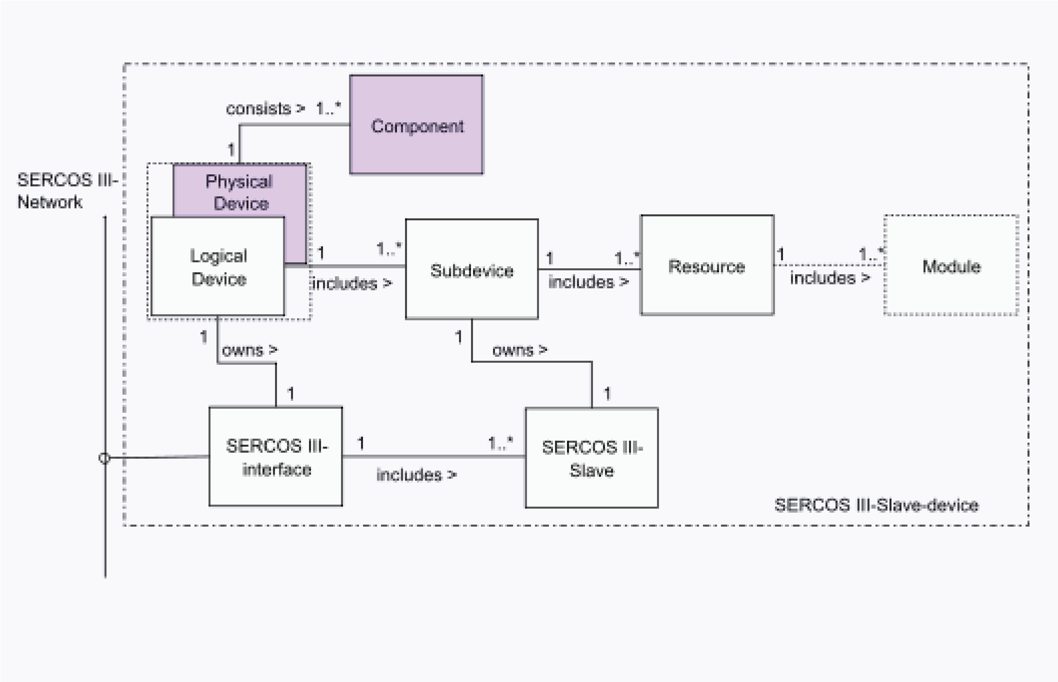
The lists in S-0-1101.x.2 and S-0-1101.x.3 shall have the same current length.

Name		List of session priorities
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		3
Scaling / Resolution of LSB		1

S-0-1300 Electronic Label

- The IDNs S-0-1300.x.x are encompassed by the term "Electronic Label". They are used to identify SERCOS III devices by a master or a configurator device. Because of this the two IDNs S-0-1300.x.3 and S-0-1300.x.5 are mandatory and have to be present in every SERCOS III device. All other IDNs of the list below are optional and can be used by the manufacturer to assign further information to the device.

The structure instance 0 of all "Electronic Label" IDNs assigns the physical device which contains the SERCOS III Interface itself and the other structure instances can be assigned to any component of the device (e.g. modules) according to the manufacturer's requirements.



- S-0-1300.x.01 Component Name
- S-0-1300.x.02 Vendor Name
- S-0-1300.x.03 Vendor Code
- S-0-1300.x.04 Device Name
- S-0-1300.x.05 Vendor Device ID
- S-0-1300.x.06 Connected to sub-device
- S-0-1300.x.07 Function revision
- S-0-1300.x.08 Hardware Revision
- S-0-1300.x.09 Software Revision
- S-0-1300.x.10 Firmware Loader Revision
- S-0-1300.x.11 Order Number
- S-0-1300.x.12 Serial Number
- S-0-1300.x.13 Manfacturing Date
- S-0-1300.x.14 QS Date
- S-0-1300.x.20 Operational Hours
- S-0-1300.x.21 Service Date
- S-0-1300.x.22 Calibration Date
- S-0-1300.x.23 Calibration Due Date

S-0-1300.x.01 Component Name

- **Description**

The content of this IDN is manufacturer specific and contains the component name depending to the device e.g. motor, amplifier, power supply, bus coupler. The contents may be used for display purpose only.

Name		Component Name
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.02 Vendor Name

- **Description**

This IDN contains the vendor name of the device. E.g. andron, AUTOMATA, Bachmann, Baumüller, Bosch Rexroth AG, Phoenix, etc.

Name		Vendor Name
Attribute	Length (Octet)	1, variabel
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.03 Vendor Code

- **Description**

The vendor code is a unique number assigned to each vendor and helps identifying a SERCOS device installed in the SERCOS ring. The vendor shall apply for the vendor-code to SERCOS International.

0x0000	unregistered vendors
<> 0x0	registrated vendors

Name		Vendor Code
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.04 Device Name

- **Description**

The content of this IDN is manufacturer specific and identifies the device name published in vendor's price list. E.g. R-ILB S3 24 DI16. DIO16-2.

Name		Device Name
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.05 Vendor Device ID

- **Description**

The vendor device ID is a unique device ID managed by the vendor and identifies the component number e.g. R911258345.

Name		Vendor Device ID
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.06 Connected to sub-device

- **Description**

If a device holds more then one sub-device and the component is assigned to a specific sub-device the operational data shows the assigned slave number.

Name		Connected to sub-device
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		511
Scaling / Resolution of LSB		1

S-0-1300.x.07 Function revision

- **Description**

The function revision shall be adjusted on the case of functional corrections of this component. The manufacturer of the device guarantees the compatibleness. The function revision shall not to be checked on system startup.

Name		Function revision
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		0
Maximum Value		9999
Scaling / Resolution of LSB		1

S-0-1300.x.08 Hardware Revision

- **Description**

This parameter contains the hardware revision of the device, therefore it can be used for the identification. The hardware revision includes the software revision S-0-1300.x.09 if it is not supported by the device. The hardware revision is specified by the manufacturer (e.g. 103).

Name		Hardware Revision
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.09 Software Revision

- **Description**

This parameter contains the software or firmware version of the device, therefore it can be used for the identification. The software revision is specified by the manufacturer.

Name		Software Revision
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.10 Firmware Loader Revision

- **Description**

This parameter contains the firmware loader or boot loader revision which is implemented in the device. The firmware loader revision is specified by the manufacturer.

Name		Firmware Loader Revision
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.11 Order Number

- **Description**

This parameter contains the order number of the device, therefore it can be used for the identification. The customer needs the order number to order this certain device. The order number is specified by the manufacturer.

Name		Order number
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0

	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.12 Serial Number

- **Description**

This parameter contains the serial number of the device, therefore it can be used for the identification. The serial number is defined and assigned by the manufacturer that uniquely identifies each individual device (e.g. 1234567890).

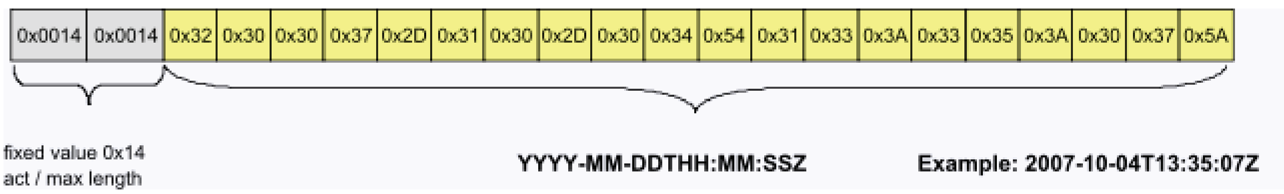
Name		Serial Number
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.13 Manfacturing Date

- **Description**

This IDN contains the manufacturing date and time of the component.

The information is provided by a text string formatted as described within ISO 8601:2006-09 / EN 28601 (extended format):



Date information is separated by hyphen "-" and time information by colon ":". Date and time are divided by a "T" character. The resulting text is terminated with Z (time zone UTC) with a fixed length of 20 characters.

Name	Manufacturing date

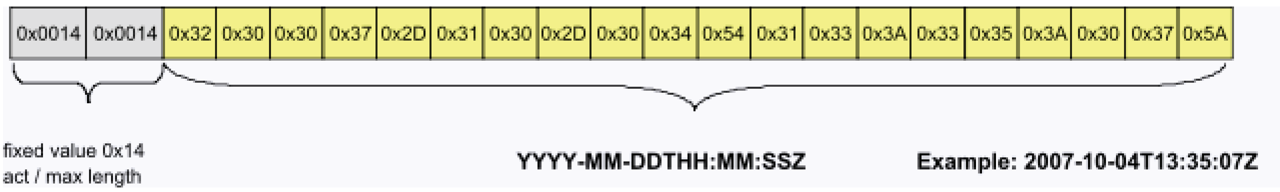
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.14 QS Date

- **Description**

This IDN contains the date and time when the quality assurance test has been performed for this component. Quality assurance is one of the final product test to ensure that the product keeps the desired quality level.

The information is provided by a text string formatted as described within ISO 8601:2006-09 / EN 28601 (extended format):



Date information is separated by hyphen "-" and time information by colon ":". Date and time are divided by a "T" character. The resulting text is terminated with Z (time zone UTC) with a fixed length of 20 characters.

Name		QS Date
Attribute	Length (Octet)	1,variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.20 Operational Hours

- **Description**

This parameter contains the operational hours of the component. The device shall store this data in a retain memory.

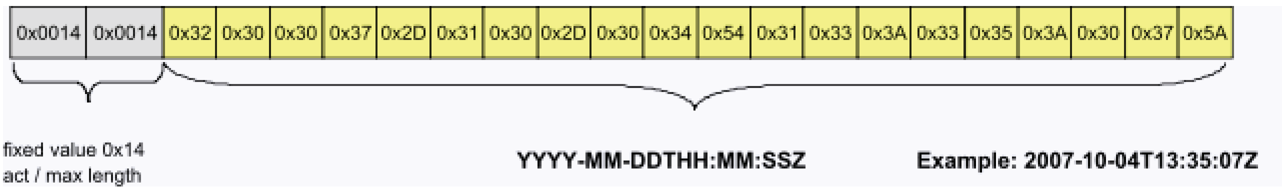
Name		Operational hours
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		h
Minimum Value		0
Maximum Value		n/a
Scaling / Resolution of LSB		1h

S-0-1300.x.21 Service Date

- **Description**

This IDN contains the date and time of the last service maintenance for the device, e.g. firmware update.

The information is provided by a text string formatted as described within ISO 8601:2006-09 / EN 28601 (extended format):



Date information is separated by hyphen "-" and time information by colon ":". Date and time are divided by a "T" character. The resulting text is terminated with Z (time zone UTC) with a fixed length of 20 characters.

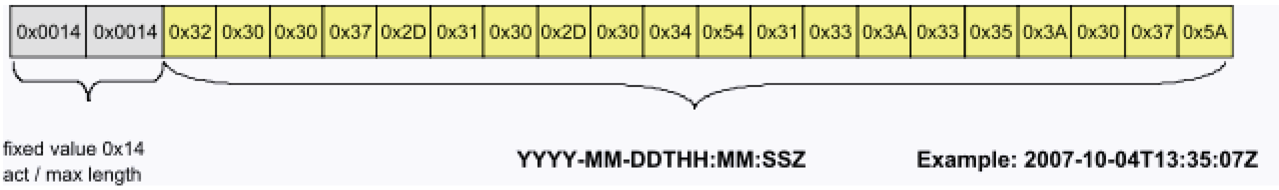
Name		Service Date
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.22 Calibration Date

- **Description**

This IDN contains the date and time of the last calibration of the device. A service engineer shall write the current date and time into this IDN after a calibration service is done. At the same time, the date and time of the next required calibration date shall be written to S-0-1300.x.23.

The information is provided by a text string formatted as described within ISO 8601:2006-09 / EN 28601 (extended format):



Date information is separated by hyphen "-" and time information by colon ":". Date and time are divided by a "T" character. The resulting text is terminated with Z (time zone UTC) with a fixed length of 20 characters.

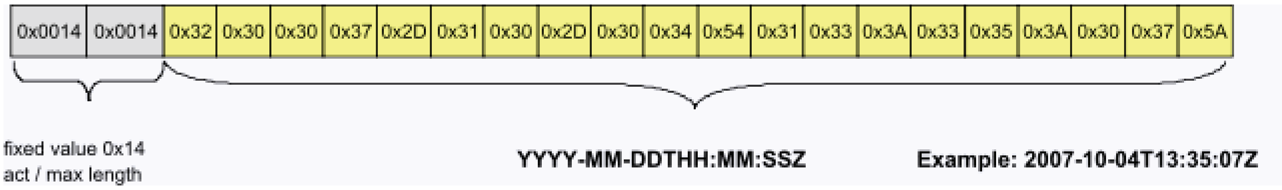
Name		Calibration Date
Attribute	Length (Octet)	1,variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1300.x.23 Calibration Due Date

- **Description**

This IDN contains the date and time of the next calibration service of the device. After a service engineer has performed a calibration, the current date and time shall be written to S-0-1300.x.22 and the date and time of the next required calibration date shall be written to this IDN.

The information is provided by a text string formatted as described within ISO 8601:2006-09 / EN 28601 (extended format):



Date information is separated by hyphen "-" and time information by colon ":". Date and time are divided by a "T" character. The resulting text is terminated with Z (time zone UTC) with a fixed length of 20 characters.

Name		Calibration Due Date
Attribute	Length (Octet)	1,variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1301 List of GDP classes & Version

- **Description**

This parameter contains a list of the generic capabilities and the dedicated versions of the sub-device.

The bits 15 ... 8 indicate the GDP classes which are available in the device. All capabilities have to be flagged in this IDN.

The bits 7 ... 0 indicates the version.

Bit No.	Value	Description	Comments
15-8		ProfileID	
0x01	GDP_Basic		
0x02-0xFF	reserved		
7-0		Version	
	0x00	reserved	
	0x01	1.st version	

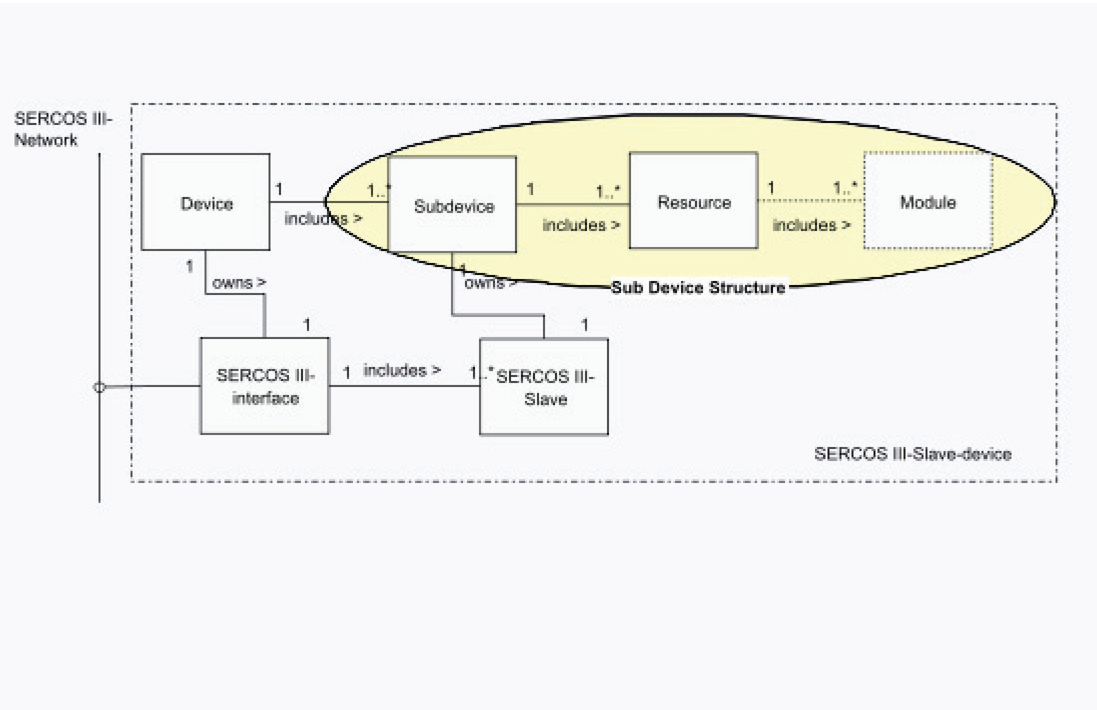
Name	List of GDP classes & Version

Attribute	Length (Octet)	2, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1302 Resource Structures of sub-device

- The resource structures of sub-device is a SERCOS structure which contains resource dependent information. Each instance holds information of one resource. Following structure elements hold this information.

- S-0-1302.x.01 FSP Type & Version
- S-0-1302.x.02 Function groups
- S-0-1302.x.03 Application Type



S-0-1302.x.01 FSP Type & Version

- **Description**

The FSP Type & Version indicates the function specific type and the function dependent version of the resource.

Bit No.	Value	Description	Comments
31		P/S	
	0	SERCOS standard	
	1	manufacturer specific	
30-16		SERCOS standard types	
	0x0000	reserved	
	0x0001	FSP_IO	
	0x0002	FSP_Drive	
	0x0003-0x7FFF	reserved	
15-0		Version	
	0x0000	reserved	
	0x0001	1.st version	

Name		FSP Type & Version
Attribute	Length (Octet)	4
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

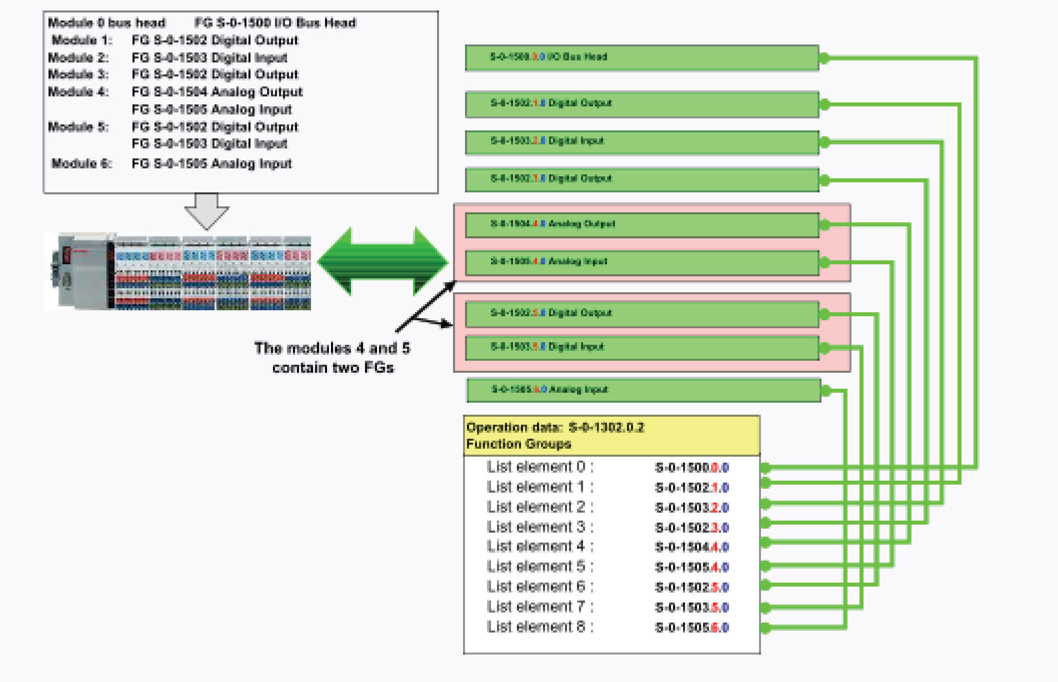
S-0-1302.x.02 Function groups

- **Description**

The operation data of this IDN contains a list of all instanced function groups.

This IDN is only present In case of a modular structured device acoording to e.g. FSP_IO.In case of a resource FSP IO this IDN is a list of IO function groups of FSP-IO. The elements of this list contain IDNs. The structure instance shows the slot number of the module (structure instance = module position), the structure element (SE) is always = 0. If the leftmost component is the bus coupler this list starts with the FG bus coupler (S-0-1500.0.0). Otherwise the list starts with the leftmost IO-module at slot number 1, e.g. S-0-15xx.1.0. If a module contains more than one IO function group every IO function group has to be designated.

The mapping of data into the container InputData S-0-1500.x.9 and container OutputData S-0-1500.x.5 occurs in sequence of the list entries.



Name		Function groups
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1302.x.03 Application Type

- **Description**

The operation data of the application type contains the type of the sub-device application (e.g., main spindle drive, round axis, X axis, etc.) The user can program this parameter if desired.

Name		Application Type
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	text
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a

Scaling / Resolution of LSB	1

S-0-1303 Diagnosis trace

- The diagnosis trace is a SERCOS structure and contains the following parameter.

- S-0-1303.0.01 Diagnosis trace configuration
- S-0-1303.0.02 Diagnosis trace control
- S-0-1303.0.03 Diagnosis trace state
- S-0-1303.0.10 Diagnosis trace buffer no1
- S-0-1303.0.11 Diagnosis trace buffer no2

This collection of IDN's controls every change of the diagnostic number S-0-0390 and stores extended diagnostic informations in a time based order. Since S-0-0390 Diagnostic Number is a detailed information structure according to a single error event, the diagnosis trace can be used to localize an error.

This information shall stored in SERCOS lists starts with S-0-1303.0.10 to S-0-1303.0.127 as a maximum.

These informations are

- mandatory part of diagnosis trace
  - S-0-1303.0.10 contains the contents of S-0-0390
  - S-0-1303.0.11 contains the contents of Timestamp IEC 61588 (PTP format)

  IEC 61588 defines data type and its representation see S-0-1305.0.1

- optional part of diagnosis trace are shown in S-0-1303.0.1 by the device such as Error source or channel or submodule (e.g. function group)

The Diagnosis trace configuration S-0-1303.0.1 adjusts the default configuration diagnostic number & time stamp with additional diagnostic data e.g. see Figure 1 for FSP I/O.

S-0-1303.0.10/m	S-0-1303.0.11/m	S-0-1303.0.12
Diagnostic Number (S-0-0390)	SERCOS current time (S-0-1305.0.1)	IO Diagnosis Message (S-0-1500.0.32)
(1)	(1)	(1)
(2)	(2)	(2)
(n)	(n)	(n)

Figure 1: Example Diagnosis_Trace

S-0-1303.0.01 Diagnosis trace configuration

- **Description**

The IDN S-0-1303.0.1 describes the configured extension of the diagnosis trace listed in S-0-1303.0.12 to S-0-1303.0.127. This IDN shows a list of the additional IDNs which are added to the diagnosis trace in the same sequence (e.g. S-0-1500.x.32 IO Diagnosis Message in case of FSP IO).

The contents of IDN S-0-1303.0.1 are defined by the manufacturer.

Name		Diagnosis trace configuration
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1303.0.02 Diagnosis trace control

- **Description**

This parameter controls the diagnosis trace flow.

Diagnosis trace control			
Bit No.	**Value**	**Description**	**Comments**
15-8		reserved	
7-4		threshold	threshold of Diagnosis class corresponds to Bits 19..16 S-0-0390 only diagnosis with classes equal or higher will be captured
3-1		reserved	

0		recording	
	0	on	capture
	1	off	freeze

Name		Diagnosis trace control
Attribute	Length (Octet)	2
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1303.0.03 Diagnosis trace state

- **Description**

This parameter shows the state of the diagnosis trace flow.

Diagnosis trace state			
Bit No.	**Value**	**Description**	**Comments**
15-1		reserved	
0		recording	
	0	on	diagnosis trace records information
	1	off	diagnosis trace is stopped

Name		Diagnosis trace state
Attribute	Length (Octet)	2 octets
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1303.0.10 Diagnosis trace buffer no1

- **Description**

This parameter contains a row of the diagnosis trace, and contains information organized as a ring buffer or list. The configuration is fixed to diagnostic number S-0-0390 of the related time stamp to the diagnostic number. The configuration of the assembly can be extended by using S-0-1303.0.1.

Name		Diagnosis trace buffer no1
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	Hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		unit of the parameter
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1303.0.11 Diagnosis trace buffer no2

- **Description**

This parameter contains a row of the diagnosis trace, and contains information organized as a ring buffer or list. The configuration is fixed to SERCOS current time S-0-1305.0.1 of the related diagnostic number.

Name		Diagnosis trace buffer no2
Attribute	Length (Octet)	8, variable
	Data Type and Display Format	SERCOS Time
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		unit of parameter
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1305 SERCOS time

- These are the IDNs related to the SERCOS current time

- S-0-1305.0.01 SERCOS current time
- S-0-1305.0.02 SERCOS requested time

S-0-1305.0.01 SERCOS current time

- **Description**

The S-0-1305.0.1 SERCOS current time contains the SERCOS time in IEC 61588 Format. The device shall mark events with this parameter. E.g. SERCOS current time marks the Diagnosis trace with time stamps.

This parameter shall become active within startup of the device with 0 Seconds 0 Nanoseconds (1.1.1970) indicating that SERCOS current time isn't set to an initial value from the master.

Table 3: Structure of SERCOS time

Bit No.	Value	Description	Comments
63-32		Seconds	
31-0		Nanoseconds	

Name		SERCOS current time
Attribute	Length (Octet)	8
	Data Type and Display Format	SERCOS time
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1305.0.02 SERCOS requested time

- **Description**

The S-0-1305.0.2 SERCOS requested time contains the SERCOS time in IEC 61588 Format.

The SERCOS requested time shall be set during the bus master in CP2. Bus master indicates that S-0-1305.0.2 SERCOS requested time shall become active in S-0-1305.0.1 SERCOS current time setting ??? in Device Control.

Structure of SERCOS time			
Bit No.	**Value**	**Description**	**Comments**
63-32		Seconds	
31-0		Nanoseconds	

Name		SERCOS requested time
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always

	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1310 IDN List of operation data changed from default

- **Description**

This IDN list is managed by the device and contains a list of all IDN's which operation data is changed from default.

Name		IDN List of operation data changed from default
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500 IO Bus coupler

- S-0-1500.x.01 IO Control
    - S-0-1500.x.02 IO Status
    - S-0-1500.x.03 Module Type Code
    - S-0-1500.x.04 List of inserted Function groups
    - S-0-1500.x.05 Container OutputData
    - S-0-1500.x.09 Container InputData
    - S-0-1500.x.11 List of replaced function groups
    - S-0-1500.x.19 Parameter Channel Receive
    - S-0-1500.x.20 Parameter Channel Transmit
    - S-0-1500.x.23 Local Bus Cycle Time
    - S-0-1500.x.30 List of monitored functions
    - S-0-1500.x.31 Monitored Diagnosis Classes
    - S-0-1500.x.32 IO Diagnosis Message
- **----- S-0-1500.x.03 Module Type Code -----**
  Description

This operating data contains a manufacturer specific list of the module type codes of all plugged modules (except passive modules, which can't be recognized via the local bus) . Exactly one module type code should exist for every module type of the respective manufacturer. E. g. this can be a unique hexadecimal module identifier or a order number.

The list is arranged in increasing order of the module's slot number, from left to right beginning with "0". Passive modules have no module type code. These module type codes are used for generating the necessary SERCOS III IDNs (in case of a modular I/O) which is being done by assigning the relevant IO Function Groups to the suitable slot-numbers with respect to the plugged IO modules during start-up. Further they are used for checking the configuration in communication phase 2. The module type codes 0xFFFFFFFF.FFFF0000 - 0xFFFFFFFF.FFFFFFFF are reserved.

see Appendix C: Example Module Type Codes for a Modular IO

Name		Module Type Code
Attribute	Length (Octet)	8, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.04 List of inserted Function groups

- **Description**

The operation data of this IDN contains a list of adapted function groups which can be added by a configuration tool. Each function group shall be a member of IO function groups described within the FSP-IO. The format of the IDNs inserted to the list comprises the function group type and the desired SI of the new module. The structure element identifier (SE) is always zero (= 0):

Definition

IDN S-0-15xx.y.0
xx = function group identifier
y = destination SI

The entries of the list must be in a rising order starting with destination SI one. To adapt two (or more) function groups (FG) on one structure instance (SI) they have to be followed one behind the other with the same SI.

These function groups can be of type virtual or passive:

a) Virtual Modules: virtual place holder

- are physically not present
- do not participate in the local-bus communication
- have no own slot but own structure-instance (SI)
- fulfill a placeholder function in the Container InputData or OutputData

- their channel-amount and -width is adjustable by writing to the related
- structure elements
- can provide an optional electronic label instance in S-0-1300 Electronic Label

b) Passive Modules (e.g. power-supply module or a local bus expansion):

- are physically present
- do not participate in the local-bus communication
- have an own slot and structure-instance (SI)
- does not fulfill a placeholder function in the Container InputData or OutputData
- their channel-amount and -width is 0

The bus coupler shall update the IDNs like S-0-0017, S-0-1302.x.02, S-0-0187, S-0-0188, S-0-1300 according to IDN parameter changes. Changes shall take effect within the same SVC write access to S-0-1500.x.04 List of insterted function groups (refer to IO_state_machine for further information).

Since channel amount and width values are unknown for inserted parameters, the following structure elements shall initially be set to zero:

- IO_FG.x.3 Channel Amount PDOUT
- IO_FG.x.4 Channel With PDOUT
- IO_FG.x.7 Channel Amount PDIN
- IO_FG.x.8 Channel Width PDIN
- IO_FG.x.11 Channel Amount DIAGOUT
- IO_FG.x.12 Channel Width DIAGOUT
- IO_FG.x.15 Channel Amount DIAGIN
- IO_FG.x.16 Channel Width DIAGIN

These *new* structure elements are write protected in (CP3/CP4) & OL.

Example I: (adding passive modules S-0-1516)

```
In this example, an external tool has configured two additional passive
modules
S-0-1516 to be placed on slot number 2 and 5. The bus coupler shall read
out the entries and insert the IDNs one after another into the list of
function groups
S-0-1302.x.2 FunctionGroups. At the end of this job, the list of
function groups is updated as shown below.
```

List element	S-0-1302.x.2	S-0-1500.x.04	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1516.2.0	S-0-1500.0.0
2	S-0-1502.1.0	S-0-1516.5.0	S-0-1502.1.0
3	S-0-1503.2.0	--	**S-0-1516.2.0**
4	S-0-1506.3.0	--	S-0-1503.3.0
5	S-0-1505.4.0	--	S-0-1506.4.0
6	S-0-1503.5.0	--	**S-0-1516.5.0**
7	--	--	S-0-1505.6.0
8	--	--	S-0-1503.7.0

Example II: (adding virtual modules)

In this example, an external tool has configured two additional virtual modules (one analog and one digital output) to be placed on slot number 2 and 5. As described in the example above, the bus coupler reads out the entries and insert the IDNs into the list of function groups S-0-1302.x.2 FunctionGroups.

List element	S-0-1302.x.2	S-0-1500.x.04	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1504.2.0	S-0-1500.0.0
2	S-0-1502.1.0	S-0-1502.5.0	S-0-1502.1.0
3	S-0-1503.2.0	--	**S-0-1504.2.0**
4	S-0-1506.3.0	--	S-0-1503.3.0
5	S-0-1505.4.0	--	S-0-1506.4.0
6	S-0-1503.5.0	--	**S-0-1502.5.0**
7	--	--	S-0-1505.6.0
8	--	--	S-0-1503.7.0

After the additional IDN has been added by the bus coupler, the channel amount and width can be configured by the master during configuration part of IO power up sequence in CP2 and PL.

Additionally, Bit 7 in IO_FG.x.2 Configuration of IO FG has to be set by the bus coupler to indicate that this module is not really existing on the local bus. **Note: Adding virtual modules will destroy the physical addressing scheme on the local bus!** This means that the structure instances of physically present modules that follow after virtual modules don't represent any more their physical slot number. So by using virtual modules the physical addressing scheme becomes a logical one. Furthermore the addressing scheme will be logical if there are any passive modules which have **not** been added to the list of inserted function groups. This may be important if an error occurs and the SERCOS III diagnosis locates this error to a certain structure instance. Then the virtual modules with smaller structure instances have to be taken into account too to find the corresponding defect module to the garbled structure instance.

Example III: (adding 2 virtual modules on the same SI)

In this example, an external tool has configured one additional virtual module with two function groups (one analog input and one analog output) to be placed into slot number 2.
As described in the example above, the bus coupler reads out the entries and insert the IDNs into the list of function groups S-0-1302.x.2 FunctionGroups.

List element	S-0-1302.x.2	S-0-1500.x.04	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1504.2.0	S-0-1500.0.0
2	S-0-1502.1.0	S-0-1505.2.0	S-0-1502.1.0
3	S-0-1503.2.0	--	**S-0-1504.2.0**

```
4            S-0-1506.3.0   --            S-0-1505.2.0
5            S-0-1505.4.0   --            S-0-1503.3.0
6            S-0-1503.5.0   --            S-0-1506.4.0
7            --             --            S-0-1505.5.0
8            --             --            S-0-1503.6.0
------------------------------------------------------------------

After the additional IDNs have been added by the bus coupler, the channel
quantity and width can be
configured by writing to the structure elements like shown in example II.
```

Error handling

- If an attempt is made to insert a passive or virtual module on structure instance '0', the error message 0x7008 "invalid operation data" will be created.

Name		List of inserted function groups
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4, OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.11 List of replaced function groups

- **Description**

The operation data of this IDN contains a list of function groups (added by a configuration tool) which replace the existing function groups detected by the bus coupler. Each function group shall be a member of IO function groups described within the FSP-IO. The format of the IDNs inserted to the list comprises the function group type and the desired SI of the new module. The structure element identifier (SE) is always zero (= 0).

The bus coupler shall update the IDNs like S-0-0017, S-0-1302.x.02, S-0-0187, S-0-0188, S-0-1300 according to IDN parameter changes. Changes shall take effect within the same SVC write access to S-0-1500.x.11 List of replaced function groups (refer to IO_state_machine for further information).

Since channel amount and width values are unknown for replaced parameters, the following structure elements shall initially be set to zero:

- IO_FG.x.3 Channel Amount PDOUT
- IO_FG.x.4 Channel Width PDOUT
- IO_FG.x.7 Channel Amount PDIN
- IO_FG.x.8 Channel Width PDIN
- IO_FG.x.11 Channel Amount DIAGOUT
- IO_FG.x.12 Channel Width DIAGOUT

- IO_FG.x.15 Channel Amount DIAGIN
- IO_FG.x.16 Channel Width DIAGIN

These *new* structure elements are write protected in (CP3/CP4) & OL.

The following examples show possible variants for replacing existing function group instances:

Example I: (one-to-one replacement)

In this example, an external configuration tool has replaced two function groups one by one.
Function group 1506 at SI 3 is replaced by function group 1507, and function group 1503 at SI 5 by function group 1513:

List element	S-0-1302.x.2	S-0-1500.x.11	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1507.3.0	S-0-1500.0.0
2	S-0-1502.1.0	S-0-1513.5.0	S-0-1502.1.0
3	S-0-1503.2.0	--	S-0-1503.2.0
4	S-0-1506.3.0	--	**S-0-1507.3.0**
5	S-0-1505.4.0	--	S-0-1505.4.0
6	S-0-1503.5.0	--	**S-0-1513.5.0**

Example II: (n-to-one replacement)

In this example, two different function groups on the same SI are replaced by one single function group:

List element	S-0-1302.x.2	S-0-1500.x.11	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1507.3.0	S-0-1500.0.0
2	S-0-1502.1.0	--	S-0-1502.1.0
3	S-0-1503.2.0	--	S-0-1503.2.0
4	S-0-1504.3.0	--	**S-0-1507.3.0**
5	S-0-1505.3.0	--	S-0-1503.4.0
6	S-0-1503.4.0	--	

Example III: (one-to-n replacement)

A reverse of the last example is a one-to-n replacement. In this example, an external configuration tool has replaced one single function group by two different ones.
Function group 1507 on SI 3 is replaced by function groups 1502 and 1503 with both on the same SI 3:

List element	S-0-1302.x.2	S-0-1500.x.11	S-0-1302.x.2 (new)
1	S-0-1500.0.0	S-0-1502.3.0	S-0-1500.0.0
2	S-0-1502.1.0	S-0-1503.3.0	S-0-1502.1.0
3	S-0-1503.2.0	--	S-0-1503.2.0
4	S-0-1507.3.0	--	**S-0-1502.3.0**

```
5              S-0-1503.4.0   --             S-0-1503.3.0
6              --             --             S-0-1503.4.0
-------------------------------------------------------------------
```

Example IV: (replacing spare modules)

```
Spare Modules ...

- are physical place holders,
- are physically present,
- participate in the local-bus communication,
- the bus coupler ident the spare modules with the Function group  Unknown
S-0-1501,
- have an own slot and structure-instance (SI),
- fulfill a placeholder function in the Container InputData or OutputData,
- their channel-amount and -width is adjustable with replacing the Function
group.

Replacing spare modules follows the same procedure as described above.
```

Error handling

- If an attempt is made to replace the bus coupler itself, the error message 0x7008 "invalid operation data" will be created.
- If an attempt is made to set not matching data (e.g. Channel Width or Channel Amount), the error message 0x7008 "invalid operation data" will be created.

Name		List of replaced function groups
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4, OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.19 Parameter Channel Receive

- **Description**

The parameter channel is an asynchronous communication mechanism for vendor-specific configuration and parametrization of bus coupler or I/O modules. The parameter channel is a simple mechanism which handles data in a transparent manner, implemented as a list (1 byte, hex). SERCOS max length indicates the internal buffer size.

Hints:

Polling via Parameter Channel Transmit is depreciated.

Name		Parameter Channel Receive
Attribute	Length (Octet)	1, variable

	Data Type and Display Format	hex
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.20 Parameter Channel Transmit

- **Description**

The parameter channel is an asynchronous communication mechanism for vendor specific configuration and parametrization of bus coupler or IO modules. The parameter channel is a simple transparent channel, implemented as a list (1 byte, hex). SERCOS max length indicates the internal buffer size. Data sized from 1 to max length bytes will be sent (via the service channel) to the Parameter Channel Transmit. The service channel deals with fragmentation. A successful transmission over the service channel will be used as a trigger for the command in the IO modules or in the bus coupler. This service channel write access will be confirmed to the master only after the action is completed. No further handshake is necessary. The answer can be fetched via the Parameter Channel Receive. The bus coupler or the IO modules will respond with variable length data.

Hints:

Polling via Parameter Channel Receive is depreciated.
The Parameter Channel Transmit is readable and writable.

Name		Parameter Channel Transmit
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hex
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	manufacturer-specific
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.23 Local Bus Cycle Time

- **Description**

The local bus cycle time is the time which is needed to exchange input and output data by the local bus.

Name		Local Bus Cycle Time

Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		µs
Minimum Value		1
Maximum Value		65 000
Scaling / Resolution of LSB		1µs

S-0-1500.x.30 List of monitored functions

- **Description**

According to the operation data of IO_FG.x.30 List of monitored functions the operation data of this IDN is a list. Each entry added to this list leads to a diagnosis class list entry within the operation data of S-0-1500.x.31 Monitored Diagnosis Classes.

Because the function group bus coupler has no own channels there is usually one list entry:

- number 255 points at the function group itself

Name		List of monitored functions
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	CP3, CP4, OL
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.31 Monitored Diagnosis Classes

- **Description**

The operation data of this IDN contains a list of diagnosis classes depending on S-0-1500.x.30 List of monitored functions and IO_FG.x.30 List of monitored functions. Each diagnosis class represents the current diagnosis state of the configured function group or channel.

All lists (S-0-1500.x.30 List of monitored functions and IO_FG.x.30 List of monitored functions) are evaluated in the following order:

- lists of a module or bus coupler with a smaller slot number (structure instance) first (IO_FG.1.30 before IO_FG.2.30)

- lists with a smaller data block number first (S-0-1500.0.30 before S-0-1503.0.30)
- in case of a sub bus the specific sub bus slaves are evaluated according to IO_FG.x.26 Available Slaves

If all lists contain no entries the operation data of S-0-1500.x.31 Monitored Diagnosis Classes is empty.

For cyclic monitoring purposes the S-0-1500.x.31 Monitored Diagnosis Classes can be mapped into the data of a connection using S-0-1050.x.6 Configuration List.

Table 4 describes the encoding of one list entry of this IDN:

Table 4: Monitored Diagnosis Classes

Bit No.	Value	Description	Comments
7-6		Diagnosis class of FG or channel	Diagnosis class of 4. monitored function group or channel (if configured)
	00	no diagnosis message	FG/Channel has no diagnosis message
	01	info	FG/Channel has informationen (e.g. Preventive maintenance required - condition monitoring)
	10	warning	FG/Channel has warning (e.g. Undervoltage inside the device) signaled with C2D Warnings disappear automatically, if reason is no longer present
	11	error	FG/Channel has error (e.g. Incorrect Systembus Slave present) signaled with C1D Errors have to be cleared with S-0-0099
5-4		Diagnosis class of FG or channel	Diagnosis class of 3. monitored function group or channel (if configured)
	00	no diagnosis message	see above
	01	info	see above
	10	warning	see above
	11	error	see above
3-2		Diagnosis class of FG or channel	Diagnosis class of 2. monitored function group or channel (if configured)
	00	no diagnosis message	see above
	01	info	see above
	10	warning	see above
	11	error	see above

1-0		Diagnosis class of FG or channel	Diagnosis class of 1. monitored function group or channel (if configured)
	00	no diagnosis message	see above
	01	info	see above
	10	warning	see above
	11	error	see above

Name		Monitored Diagnosis Classes
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1500.x.32 IO Diagnosis Message

- **Description**

The operation data of this IDN contains an IO diagnosis message consisting of an IO Status code, a diagnosis class and diagnosis localization information.

The IO diagnosis message shows always the IO diagnosis event with the highest priority within a device. A new IO diagnosis event with a specific priority overwrites an older event with the same priority (not in case of an uncleared error - C1D).

For cyclic monitoring purposes the S-0-1500.x.32 Diagnosis message can be mapped into the data of a connection using S-0-1050.x.6 Configuration List.

Bit No.	Value	Description	Comments
63		interpretation of status codes	
	0	standard IO status codes	Status codes are defined by TWG IO.
	1	product specific codes	Status codes are defined by the manufacturer.
62-58		reserved	

57-56		Diagnosis class	Diagnosis class of diagnosis message.
	00	no diagnostic message	FG/Sub bus slave/channel has no diagnostic message.
	01	info	FG/Sub bus slave/channel has informationen (e.g. Preventive maintenance required - condition monitoring).
	10	warning	FG/Sub bus slave/channel has warning (e.g. Undervoltage inside the device) signaled with C2D. Warnings disappear automatically, if reason is no longer present.
	11	error	FG/Sub bus slave/channel has error (e.g. Incorrect Systembus Slave present) signaled with C1D. Errors have to be cleared with S-0-0099.
55-40		Status code	16-Bit Status code either defined by TWG IO or by the manufacturer.
39-24		Number of IO function group	IO function group which causes the diagnosis message, e.g. 1503 - Digital Output or 1509 - Sub bus slave.
23-16		Slot number	Slot number of the module the function group is assigned to (structure instance).
15-8		Sub bus slave index	Sub bus slave which causes the diagnosis message (see IO_FG.x.25 Slave Index.). Only used in case of IO function group nr. 1509 - Sub bus slave.
7-0		Channel number	Channel which causes the diagnosis message. A channel number between 0 and 254 points at a specific channel of the function group (channel related diagnosis events, e.g. "Wire break"). Channel number 0 points at the function group itself (function group related diagnosis messages, e.g. Sub bus error - timeout).

Name		Diagnosis message
Attribute	Length (Octet)	8
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1501 Unknown

- *Description*

Modules described by function group Unknown could not be identified as any other SERCOS function group by the bus coupler, which is likely a lack of description in the bus coupler than an error on the local bus.

The channel quantity and width is shown by IO_FG.x.3 Channel quantity PDOUT, IO_FG.x.4 Channel width PDOUT, IO_FG.x.7 Channel quantity PDIN and IO_FG.x.8 Channel width PDIN.

These modules do not provide valid process data.

These modules can be mapped within S-0-1500.x.9 Container InputData and S-0-1500.x.5 Container OutputData . Nevertheless the process data will be invalid. Therefore, the input data is forced to zero and the output data will be ignored. The purpose of this procedure is to let all other modules function properly.

If a buscoupler shall work with this unknown module, information about the unknown module shall be loaded into the buscoupler using S-0-1500.x.11 List of replaced function groups.
All structure elements shown in Table 5 shall be present in this function group.

Table 5: IO Function Group Unknown

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory if available
4	Channel Width PDOUT		mandatory if available
7	Channel Quantity PDIN		mandatory if available
8	Channel Width PDIN		mandatory if available
19	Parameter Channel Receive		Optional
20	Parameter Channel Transmit		optional

S-0-1502 Digital Output

Description

Modules described by the digital output function group provide one or more binary output channels. These outputs can offer several features like switched or galvanic isolated gates.

In SERCOS, all available channels of a module are packed into full ensembles of bytes mapped within IO_FG.x.5 PDOUT where the complete ensemble is aligned to words. When configured in S-0-1500.x.5 Container OutputData the ensemble is aligned to bytes. See Appendix_D: Example Process Data and Appendix E: Example Channel Quantity - Channel Width for Inputs. The number of supported channels is available in IO_FG.x.3 Channel quantity PDOUT and the bit size of each single channel is accessable by IO_FG.x.4 Channel width PDOUT.

If the module provides additional diagnosis information, it can be accessed through IO_FG.x.17 DIAGIN. Optionally IO_FG.x.13 DIAGOUT can be used to acknowledge diagnosis information. Diagnosis input and output data of binary channels follow the same mapping rules as process data.

If a digital output module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 6 shall be present in this function group.

Table 6: IO Function Group digital output

SE	Name	Explanation	Condition
1	**Name of IO_FG**		**optional**
2	**Configuration of IO_FG**		**optional**
3	**Channel Quantity PDOUT**		**mandatory**
4	**Channel Width PDOUT**		**mandatory**
5	**PDOUT**		**optional**
11	**Channel Quantity DIAGOUT**		**mandatory if available**
12	**Channel Width DIAGOUT**		**mandatory if available**
13	**DIAGOUT**		**optional**
15	**Channel Quantity DIAGIN**		**mandatory if available**
16	**Channel Width DIAGIN**		**mandatory if available**
17	**DIAGIN**		**optional**
19	**Parameter Channel Receive**		**optional**
20	**Parameter Channel Transmit**		**optional**
22	**Fallback Value Output**		**optional**
23	**Min. Delay Time**		**optional**
24	**Max. Delay Time**		optional

S-0-1503 Digital Input

Description

Modules described by the digital input function group provide one or more binary input channels. These inputs can offer several features like switched or galvanic isolated gates.

In SERCOS, all available channels of a module are packed into full ensembles of bytes mapped within IO_FG.x.9 PDIN where the complete ensemble is aligned to words. When configured in S-0-1500.x.9 Container InputData the ensemble is aligned to bytes. See

Appendix_D: Example Process Data and Appendix E: Example Channel Quantity - Channel Width for Inputs. The number of supported channels is available in IO_FG.x.7 Channel quantity PDIN and the bit size of each single channel is accessable by IO_FG.x.8 Channel width PDIN.

If the module provides additional diagnosis information, it can be accessed through IO_FG.x.17 DIAGIN. Optionally IO_FG.x.13 DIAGOUT can be used to acknowledge diagnosis information. Diagnosis input and output data of binary channels follow the same mapping rules as process data.

If a digital input module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 7 shall be present in this function group.

Table 7: IO Function Group Digital Input

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
23	Min. Delay Time		optional

S-0-1504 Analog Output

Description

Modules described by the Analog Output function group provide one or more byte-oriented analog channels. These modules come with more or less complex functionality and are available for types of jobs like:

- Voltage interface (pos and/or neg)

- Current interface

In SERCOS, all available channels of an analog output module are mapped byte aligned and are arranged one after another within IO_FG.x.5 PDOUT, where the complete data is aligned to words. When configured in S-0-1500.x.5 Container OutputData , the data is aligned to bytes. See Appendix_D: Example Process Data and Appendix E: Example Channel Quantity - Channel Width for Inputs. The number of supported channels is available in IO_FG.x.3 Channel quantity PDOUT and the bit size of each single channel is accessable by IO_FG.x.4 Channel width PDOUT.

If the module provides additional diagnosis information, it can be accessed through IO_FG.x.17 DIAGIN. Optionally IO_FG.x.13 DIAGOUT can be used to acknowledge diagnosis information. Diagnosis input and output data of analog channels follow the same mapping rules as process data.

If an analog output module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 8 shall be present in this function group.

Table 8: IO Function Group Analog Output

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		Optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		Optional
19	Parameter Channel Receive		Optional
20	Parameter Channel Transmit		Optional
22	Fallback Value Output		Optional
23	Min. Delay Time		Optional
24	Max. Delay Time		optional

S-0-1505 Analog Input

Parameter V1.1.2.1.2

Description

Modules described by the analog input function group provide one or more byte-oriented analog channels. These modules come with more or less complex functionality and are available for a wide range of jobs, e.g.:

- Voltage interface (pos/neg)
- Current interface
- Sensors for resistance measurement
- Sensors for thermal elements

In SERCOS, all available channels of an analog input module are mapped byte aligned and are arranged one after another within IO_FG.x.9 PDIN, where the complete data is aligned to words. When configured in S-0-1500.x.9 Container InputData , the data is aligned to bytes. See Appendix_D: Example Process Data and Appendix E: Example Channel Quantity - Channel Width for Inputs. The number of supported channels is available in IO_FG.x.7 Channel quantity PDIN and the bit size of each single channel is accessable by IO_FG.x.8 Channel width PDIN.

If the module provides additional diagnosis information, it can be accessed through IO_FG.x.17 DIAGIN. Optionally IO_FG.x.13 DIAGOUT can be used to acknowledge diagnosis information. Diagnosis input and output data of analog channels follow the same mapping rules as process data.

If an analog input module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 9 shall be present in this function group.

Table 9: IO Function Group Analog Input

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional

19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1506 Counter

Description

Counter/Timer

Modules of the counter/timer function group are function modules with more or less complex functionality. They are able to monitor fast signals because of their high speed inputs. In addition, advanced versions do have control and state signals. Such modules can be operated in different operating modes, e.g.:

- Event counting
- Time measurement and
- Frequency measurement

In general, function modules have large complexity. Therefore, such modules typically could be variably configured (or configured in great detail) and may generate their own diagnostics.

From the SERCOS view, the Counter/Timer module supports the simple ProcessDataChannel and optionally a ParameterChannel. The meaning of data distribution of the ProcessDataChannel will be defined by the vendor. The process data is pre-processed. All structure elements shown in Table 1 shall be present in this function group. All structure elements shown in Table 10 shall be present in this function group.

Table 10: IO Function Group Counter

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional

15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1507 Complex Protocol

Description

Modules of type Complex Protocol are used to pass complex or asynchronous information (e.g. by a mailbox, a vendor specific protocol or process data itself) mapped into the process data and therefore into the S-0-1500.x.5 Container OutputData and S-0-1500.x.9 Container InputData .

The transfer of protocols and mailbox mechanism within process data shall be handled transparent. The meaning of protocol opcodes (protocol header) and values within process data is vendor specific. Basic applications are communication modules like serial interfaces, simple user handled subbusses and other asynchronous modules.

If a complex module supports advanced configuration, the optional parameter channel can be used to access the module (e.g., for configuration of mailbox or process data length). Parameter output data is provided by IO_FG.x.20 Parameter Channel Transmit and parameter input data IO_FG.x.19 Parameter Channel Receive. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 11 shall be present in this function group.

Table 11: IO Function Group Complex Protocol

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT	e.g. protocol transmit channel	optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN	e.g. protocol receive channel	optional
11	Channel Quantity DIAGOUT		mandatory if available

12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive	configuration of the module	optional
20	Parameter Channel Transmit	configuration of the module	optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1508 Sub bus Master

Description

This function group describes an I/O module which implements a sub bus master (like an AS-interface master).

This I/O module is used to implement a sub bus master module. The cyclic process and diagnosis data of all sub bus slaves are accessible through parameter S-0-1508.x.05 PDOUT and S-0-1508.x.09 PDIN with all sub bus slaves being arranged one after another.

Additionally, S-0-1508.x.05 PDOUT and S-0-1508.x.09 PDIN can be mapped into S-0-1500.x.05 Container OutputData and S-0-1500.x.09 Container InputData (see Appendix_J: Example data mapping for Sub bus Master and Slave).

The contents of channel quantity (e.g. S-0-1508.x.03 Channel quantity PDOUT) and channel width (e.g. S-0-1508.x.04 Channel width PDOUT) can be set manufacturer specific and shall represent the complete sub bus. This means, that the product of channel quantity and channel width shall fit to the real bit length of the complete sub bus including every connected sub bus slave.

Use cases

1. Channel quantity parameter is fixed to "1" and channel width parameter contains the complete bit length of the whole sub bus including all channels of every sub bus slave (and e.g. additional padding bits). This option is recommended for sub busses with an unequal channel bit size.
2. Channel quantity parameter contains the sum of all channels on the sub bus and channel width parameter contains the bit length of one single channel on the sub bus. This option is recommended for sub busses with an equal channel bit size.

If a sub bus master supports advanced configuration, the optional parameter channel can be used to access the module (e.g., for configuration of mailbox or process data length). Parameter output data is provided by S-0-1508.x.20 Parameter Channel Transmit and parameter input data by S-0-1508.x.19 Parameter Channel Receive. The parameter channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 12 shall be present in this function group.

Table 12: IO Function Group Sub bus Master

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional
26	Available Slaves		mandatory

S-0-1509 Sub bus Slave

Description

This function group describes an I/O module which implements a sub bus slave. The sub bus slave function group provides the functionality of one or more subordinate slaves of a sub bus master represented by the function group S-0-1508 Sub bus Master.

In order to access the element data of a specific sub bus slave, the user shall write the slave's index into S-0-1509.x.25 Slave Index and hence can access the referred slave process and diagnosis data by the structure elements of the same function group (e.g. S-0-1509.x.05 PDOUT etc.). The slave index is the equvalent entry position in the list of available slaves within S-0-1508.x.26 Available slaves. See S-0-1508.x.26 Available slaves and S-0-1509.x.25 Slave Index for further information.

Hints

The PDIN, PDOUT, DIAGIN and DIAGOUT of this function group contain the data of the selected sub bus slave and can not be configured on their own in any connection. These structure elements are only directly accessible via SVC.

The cyclic process and diagnoses data of one single sub bus slave is accessible through parameter S-0-1508.x.05 PDOUT and S-0-1508.x.09 PDIN.

Additionally, S-0-1508.x.05 PDOUT and S-0-1508.x.09 PDIN can be mapped into S-0-1500.x.05 Container OutputData and S-0-1500.x.09 Container InputData (see Appendix_J: Example data mapping for Sub bus Master and Slave).

Channel quantity parameter IDNs (e.g. S-0-1509.x.03 Channel quantity PDOUT) contain the number of signal channels available on the sub bus slave.

Channel width parameter IDNs (e.g. S-0-1509.x.04 Channel width PDOUT) contain the bit width of one single signal channel on the sub bus slave.

If a sub bus slave supports advanced configuration, the optional parameter channel can be used to access the module (e.g. for configuration of mailbox or process data length). Parameter output data is provided by S-0-1509.x.20 Parameter Channel Transmit and parameter input data by S-0-1509.x.19 Parameter Channel Receive. The parameter channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 13 shall be present in this function group.

Table 13: IO Function Group Sub bus Slave

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available

16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional
25	Slave Index		mandatory

S-0-1510 Safety Output

Description

Modules of the safety function group are function modules with safety functionality. The category (or SIL safety integrity level) is not specified. All safety modules use the CIP safety profile. All safe functionalities are handled via the S-0-1800 function groups. This function group has the ability to visualize safe process data as unsafe data in the S-0-1500.x.5 Container OutputData . Safety relevant digital output data will be served by CIP Safety on SERCOS. So S-0-1510 provides the ability to show the output data in the PDIN.

If the safety module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 14 shall be present in this function group.

Table 14: IO Function Group Safety Output

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
7	Channel Quantity PDIN	read back PDOUT	optional
8	Channel Width PDIN	read back PDOUT	optional
9	PDIN	read back PDOUT	optional
11	Channel Quantity DIAGOUT		optional
12	Channel Width DIAGOUT		optional
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		optional
16	Channel Width DIAGIN		optional
17	DIAGIN		optional

19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional

S-0-1511 Safety Input

Description

Modules of the safety function group are function modules with safety functionality. The category (or SIL safety integrity level) is not specified. All safety modules use the CIP safety profile. All safe functionalities are handled via the S-0-1800 function groups. This function group has the ability to visualize safe process data as unsafe data in the S-0-1500.x.9 Container InputData . Safety relevant digital input data will be served by CIP Safety on SERCOS. So S-0-1510 provides the ability to show the input data in the PDIN.

If the safety module supports advanced configuration, the optional parameter channel can be used to access the module. Parameter output data is provided by IO_FG.x.20 Parameter channel Output and parameter input data IO_FG.x.19 Parameter channel Input. This channel is manufacturer specific and therefore may be used for different services. All structure elements shown in Table 15 shall be present in this function group.

Table 15: IO Function Group Safety Input

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
7	Channel Quantity PDIN		optional
8	Channel Width PDIN		optional
9	PDIN		optional
11	Channel Quantity DIAGOUT		optional
12	Channel Width DIAGOUT		optional
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		optional
16	Channel Width DIAGIN		optional
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional

S-0-1512 PLC-Module

Description

Such PLC modules are autonomous PLC's, which are typically programmed and engineered via a private interface. This PLC-Module could have on-board I/O's or it's own (field-) sub bus system. In this case it is also a Sub bus master S-0-1508. Typically it is used for data preprocessing, technology functions or high availability control functions. In general the PLC function is finished und tested before the complete application is assembled.

From the SERCOS view, the PLC-Module supports the simple ProcessDataChannel and optionally a ParameterChannel. The meaning of data distribution of the ProcessDataChannel will be defined by the PLC engineering.

If necessary, the PLC could also be programmed and engineered via the ParameterChannel, e.g., the complete program could be downloaded. Therefore, the ParameterChannel must be available in phase 2. All structure elements shown in Table 16 shall be present in this function group.

Table 16: IO Function Group PLC-Module

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1513 Motorstarter

Description

Modules of the motor starter function group are function modules with more or less complex functionality. They are able to manage standard motor functions. The motor could be switched, protected (overcurrent) and monitored. Advanced versions do have control and state signals in addition for the surrounded sensors

In general, Function modules have large complexity. Therefore such modules typically could be varied configured and own a complete diagnosis.

From the SERCOS view, the motor starter module supports the simple ProcessDataChannel and optionally a ParameterChannel. The meaning of data distribution of the ProcessDataChannel will be defined by the vendor. All structure elements shown in Table 17 shall be present in this function group.

Table 17: IO Function Group Motorstarter

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1514 PWM

Description

Modules of the Pulse Width Modulation (PWM) function group are function modules with more or less complex functionality. They are able to generate different kind of pulses and/or frequencies, such as:

- PWM (pulse width modulation)
- Frequency generator
- Single shot (single pulse generator)
- Pulse direction signal

From the SERCOS view, PWM modules support the simple ProcessDataChannel and optionally a ParameterChannel. The meaning of data distribution of the ProcessDataChannel will be defined by the vendor. All structure elements shown in Table 18 shall be present in this function group.

Table 18: IO Function Group Pulse Width Modulation

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional
22	Fallback Value Output		optional
23	Min. Delay Time		optional
24	Max. Delay Time		optional

S-0-1515 Positioning

Description

Modules of the positioning function group are function modules with more or less complex functionality. They are able to monitor signals with a position background. The module's special positioning inputs are typically interfaces for absolute encoders (single-turn or multi-turn) or incremental encoders (symmetrical or asymmetrical encoders). Additionally, advanced versions do have control and state signals.

From the SERCOS view, the Positioning module supports the simple ProcessDataChannel and optionally a ParameterChannel. The meaning of data distribution of the ProcessDataChannel will be defined by the vendor. The process data is pre-processed. All structure elements shown in Table 19 shall be present in this function group.

Table 19: IO Function Group Positioning

SE	Name	Explanation	Condition
1	Name of IO_FG		optional
2	Configuration of IO_FG		optional
3	Channel Quantity PDOUT		mandatory
4	Channel Width PDOUT		mandatory
5	PDOUT		optional
7	Channel Quantity PDIN		mandatory
8	Channel Width PDIN		mandatory
9	PDIN		optional
11	Channel Quantity DIAGOUT		mandatory if available
12	Channel Width DIAGOUT		mandatory if available
13	DIAGOUT		optional
15	Channel Quantity DIAGIN		mandatory if available
16	Channel Width DIAGIN		mandatory if available
17	DIAGIN		optional
19	Parameter Channel Receive		optional
20	Parameter Channel Transmit		optional

S-0-1516 Passive

Description

Modules belonging to the passive function group are physical modules connected to the local bus but do not supply any kind of process or diagnosis data (e.g. power supply/injection or local bus extension).

As these modules do not participate in local bus communication, length information is not available and therefore indicated by zero. In some cases, the bus coupler has no ability of getting informations about passive modules. In this case, it is possible to manually inform the bus coupler about existing modules by editing S-0-1500.x.04 List of inserted function groups.

Example

Writing the IDN S-0-1516.3.0 into S-0-1500.x.04 List of inserted function groups will indicate a passive module on position 3 after the bus coupler.

Please note that the original position(SI) of all modules positioned past the passive module will be increased by one.

This proceeding is intended to be optional and is useful to guarantee a consistent addressing of SIs in relation to physical modules. In the other case, if the bus coupler can achieve informations about passive modules connected to the local bus, this procedure is not necessary. Additionally, the name of the module can be assigned by writing to S-0-1516.x.1 Name of IO FG. All structure elements shown in Table 20 shall be present in this function group.

Table 20: IO Function Group Passive

SE	Name	Explanation	Condition
1	**Name of IO_FG**		optional

S-0-1800 Safety Device

The function group **S-0-1800 Safety Device** is the core function group of a CIP Safety on SERCOS device. It contains parameters to access the attributes of the (sub-)device's Safety Supervisor object and additional (sub-)device related information.

This function group includes the following IDNs:

- S-0-1800.x.01 List of applications
- S-0-1800.x.02 List of validator objects
- S-0-1800.x.03 SV Safety connection fault count
- S-0-1800.x.10 SSO Device Status
- S-0-1800.x.11 SSO Exception status
- S-0-1800.x.12 SSO Alarm enable
- S-0-1800.x.13 SSO Warning enable
- S-0-1800.x.14 SSO Output connection point owners
- S-0-1800.x.15 SSO Safety configuration identifier
- S-0-1800.x.16 SSO Configuration lock
- S-0-1800.x.17 SSO Configuration UNID
- S-0-1800.x.18 SSO Proposed TUNID
- S-0-1800.x.19 SSO Target UNID

S-0-1800.x.01 List of applications

- **Description**

This parameter contains a list of all SERCOS safety applications contained in the safety device referenced by the given SI. For example, in a modular IO device, S-0-1800.42.1 contains a list of all SERCOS safety applications of the module in slot 42.

Name		List of applications
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.02 List of validator objects

- **Description**

Contains a list of all validator objects available in the safety device referenced by the parameter's SI.

Name		List of validator objects
Attribute	Length (Octet)	4
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.03 SV Safety connection fault count

- **Description**

This parameter contains the safety validator fault counter for the safety device referenced by the parameter's SI. It corresponds to the Safety connection fault count attribute of the CIP Safety Validator Object (refer to CIP Vol. 5). This counter will be incremented each time one of the safety validators in this safety device flags a connection as "faulted" for any reason. The counter will auto-rollover.

Name		Safety connection fault count
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.10 SSO Device Status

- **Description**

The "device status" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.5 for details.

Name		Device status
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.11 SSO Exception status

- **Description**

The "Exception status" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.6 for details.

Name		Exception status
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.12 SSO Alarm enable

- **Description**

The "Alarm enable" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.8 for details.

Name		Alarm enable
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.13 SSO Warning enable

- **Description**

The "Warning enable" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.8 for details.

Name		Warning enable
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.14 SSO Output connection point owners

- **Description**

The "Output connection point owners" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.16 for details.

Name		Output connection point owners
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-1800.x.15 SSO Safety configuration identifier

- **Description**

The "Safety configuration identifier" (SCID) attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.13 for details.

Name		Safety configuration identifier
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.16 SSO Configuration lock

- **Description**

The "Configuration lock" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.11 for details.

Name		Configuration lock
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.17 SSO Configuration UNID

- **Description**

The "Configuration UNID" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.12 for details.

Name		Configuration UNID
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.18 SSO Proposed TUNID

- **Description**

The "Proposed TUNID" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.15 for details.

Name		Proposed TUNID
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1800.x.19 SSO Target UNID

- **Description**

The "Target UNID" attribute of the safety device's Safety Supervisor object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-4.2.1.14 for details.

Name		Target UNID
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	hexadecimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a

Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-1810

- The function group **S-0-1810 Safety Validator** implements the CIP Safety Validator object.

This function group includes the following IDNs:

- S-0-1810.x.01 SV Max data age
- S-0-1810.x.02 Safety Validator state
- S-0-1810.x.03 SV Error code
- S-0-1810.x.04 Safety Validator type
- S-0-1810.x.05 SV Time coord msg min multiplier
- S-0-1810.x.06 SV Max consumer number
- S-0-1810.x.07 SV Timeout multiplier
- S-0-1810.x.08 SV Ping interval EPI multiplier
- S-0-1810.x.09 SV Network time expectation multiplier

S-0-1810.x.01 SV Max data age

- **Description**

The "Max data age" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.12 for details.

Name		Max data age
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.02 Safety Validator state

- **Description**

The "Safety Validator state" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.1 for details.

Name		Safety Validator state
Attribute	Length (Octet)	2

	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.03 SV Error code

- **Description**

The "Error code" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3 for details.

Name		Error code
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.04 Safety Validator type

- **Description**

The "Safety Validator type" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.2 for details.

Name		Safety Validator type
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.05 SV Time coord msg min multiplier

- **Description**

The "Time coord msg min multiplier" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.4 for details.

Name		Time coord msg min multiplier
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.06 SV Max consumer number

- **Description**

The "Max consumer number" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.7 for details.

Name		Max consumer number
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.07 SV Timeout multiplier

- **Description**

The "Timeout multiplier" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.6 for details.

Name		Timeout multiplier
Attribute	Length (Octet)	1, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter

	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.08 SV Ping interval EPI multiplier

- **Description**

The "Ping interval EPI multiplier" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.3 for details.

Name		Ping interval EPI multiplier
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1810.x.09 SV Network time expectation multiplier

- **Description**

The "Network time expectation multiplier" attribute of the Safety Validator object as defined by the CIP Safety specification. Refer to CIP Vol. 5 edition 1.3 section 5-5.3.5 for details.

Name		Network time expectation multiplier
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1820

- The function group **S-0-1820 Safety Application** defines the application interface of a Safety Validator Object.

This function group includes the following IDNs:

- S-0-1820.x.01 CIP assembly instance number
- S-0-1820.x.02 Safety Validator instance number
- S-0-1820.x.03 Safety Application type
- S-0-1820.x.04 Safety Application data size

- **----- S-0-1820.x.01 CIP assembly instance number -----**
  Description

In CIP Safety the endpoint of a connection is defined by specifying the object class code, object instance number and the object attribute which shall provide or receive the data sent across the connection. For cylic I/O data connections, this so-called "path" usually references an instance of the assembly object (class code 0x04).

The structure element 1 of FG Safety Application displays the (usually hard-coded) instance number of the assembly that represents this Safety Application.

Name		CIP assembly instance number
Attribute	Length (Octet)	4
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1820.x.02 Safety Validator instance number

- **Description**

This parameter contains the CIP Safety object instance number of the Safety Validator object that is assigned to this safety application.

The Safety Validator instance number connects an instance of the function group *S-0-1820 Safety Application* to an instance of the function group *S-0-1810 Safety Validator Object*. A value of *S-0-0000.0.0* indicates that this safety application is currently not connected to a validator and neither produces nor consumes safety data.

Name		Safety Validator instance number
Attribute	Length (Octet)	4
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0

	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1820.x.03 Safety Application type

- **Description**

This parameter displays the type of safety application defined by this function group instance. At this time, the data direction is the only difference between applications.

Bit No.	Value	Description	Comments
15-1		reserved	
0		Data direction	
	0	producing application (client)	
	1	consuming application (server)	

Name		Safety Application type
Attribute	Length (Octet)	2
	Data Type and Display Format	binary
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1820.x.04 Safety Application data size

- **Description**

This parameter displays the data size of the associated Safety Application Instance in octets.

Name		Safety Application data size
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a

Minimum Value	1
Maximum Value	250
Scaling / Resolution of LSB	1

S-0-1830

- Each instance of FG Safety Connection defines a set of black channel communication channels used by a Safety Validator Object. The instance number corresponds to the instance number of the Safety Validator Object that uses this set of connections, e.g. the Safety Validator Object instance 7 (S-0-1810.7.SE) uses the connections defined in Safety Connection instance 7 (S-0-1830.7.SE).

This function group includes the following IDNs:

- S-0-1830.x.01 Cyclic SMP container (out)
- S-0-1830.x.02 Cyclic SMP Session ID (out)
- S-0-1830.x.03 List of cyclic SMP containers (in)
- S-0-1830.x.04 List of cyclic SMP Session IDs (in)
- S-0-1830.x.05 List of UCM SMP containers (in)
- S-0-1830.x.06 List of UCM SMP SessionIDs (in)
- S-0-1830.x.07 List of UCM SMP containers (out)
- S-0-1830.x.08 List of UCM SMP Session IDs (out)
- S-0-1830.x.09 List of consumer numbers

All IDNs of this function group except S-0-1830.x.1 and S-0-1830.x.2 form a table which lists the statically configured connections to a Safety Validator Object, and, in case of a producing connection, the dynamically assigned consumer number for a connection.

Index	**S-0-1830.x.3 Cyc.Cont.In**	**S-0-1830.x.4 Cyc.Sess.In**	**S-0-1830.x.5 UCM Cont.In**	**S-0-1830.x.6 UCM Sess.In**	**S-0-1830.x.7 UCM Cont.Out**	**S-0-1830.x.8 UCM Sess.Out**	**S-0-1830.x.9 Cons.Num**
1							
2							
...	...	...	...	...	...	...	...
15							

S-0-1830.x.1 and S-0-1830.x.2 are fixed-length parameters, because they describe the single cyclic output connection of a Safety Validator Object. This may be the data connection of either a singlecast or a multicast producer, or it may be the time coordination connection of a single- or multicast consumer. There's no type of safety connection that requires a Safety Validator Object to establish more than one cyclic output connections.

- **----- S-0-1830.x.01 Cyclic SMP container (out) -----**
  Description

This parameter contains the IDN of the SMP data container that is used to transmit cyclic data from this device for the given connection.

Name		Cyclic SMP container (out)
Attribute	Length (Octet)	4
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.02 Cyclic SMP Session ID (out)

- **Description**

This parameter contains the SMP Session Identifier of the session that is used to transmit cyclic data from this device for the given connection.

Name		Cyclic SMP Session ID (out)
Attribute	Length (Octet)	2
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.03 List of cyclic SMP containers (in)

- **Description**

This list contains the IDNs of SMP data containers that are used to transmit cyclic data to this device for the given connection. The list can contain up to 15 entries (for a multi cast producer that receives Time Coordination data from 15 different consumers).

Name		List of cyclic SMP containers (in)
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1

Unit	n/a
Minimum Value	n/a
Maximum Value	n/a
Scaling / Resolution of LSB	1

S-0-1830.x.04 List of cyclic SMP Session IDs (in)

- **Description**

This list contains the SMP Session Identifiers of the sessions that are used to transmit cyclic data to this device for the given connection. The list can contain up to 15 entries (for a multi cast producer that receives Time Coordination data from 15 different consumers). The position on the list corresponds to the List of cyclic SMP containers (in): a session in S-0-1830.x.4 will be set up in the container on the same list position in S-0-1830.x.3.

Name		List of cyclic SMP Session IDs (in)
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.05 List of UCM SMP containers (in)

- **Description**

This list contains the IDNs of SMP data containers that are used to transmit unconnected acyclic data to this device for the given connection. The list can contain up to 15 entries (for a multi cast producer target that receives requests from up to 15 different originators).

Name		List of UCM SMP containers (in)
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.06 List of UCM SMP SessionIDs (in)

- **Description**

This list contains the SMP Session Identifiers of the sessions that are used to transmit unconnected acyclic data to this device for the given connection. The list can contain up to 15 entries (for a multi cast producer target that receives requests from 15 different originators). The position on the list corresponds to the List of UCM SMP containers (in): a session in S-0-1830.x.6 will be set up in the container on the same list position in S-0-1830.x.5.

Name		List of UCM SMP Session IDs (in)
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.07 List of UCM SMP containers (out)

- **Description**

This list contains the IDNs of SMP data containers that are used to transmit unconnected acyclic data from this device for the given connection. The list can contain up to 15 entries (for a multi cast producer target that sends replies to up to 15 different originators).

Name		List of UCM SMP containers (out)
Attribute	Length (Octet)	4, variable
	Data Type and Display Format	IDN
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.08 List of UCM SMP Session IDs (out)

- **Description**

This list contains the SMP Session Identifiers of the sessions that are used to transmit unconnected acyclic data from this device for the given connection. The list can contain up to 15 entries (for a multi cast producer target that sends replies to 15 different originators). The position on the list corresponds to the List of UCM SMP containers (out): a session in S-0-1830.x.8 will be set up in the container on the same list position in S-0-1830.x.7.

Name		List of UCM SMP Session IDs (out)
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	never
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1

S-0-1830.x.09 List of consumer numbers

- **Description**

A Safety Validator Client maintains up to 15 simultaneous connections to multicast consumers. The client/producer assigns a unique consumer number to each consumer. This list contains the consumer numbers assigned to the consumers that are using the SMP sessions defined in this instance of S-0-1830.

For producing singlecast connections, the list contains a single entry with the value "1". For consuming connections all consumer numbers will be set to "0".

Name		List of Consumer Numbers
Attribute	Length (Octet)	2, variable
	Data Type and Display Format	unsigned decimal
	Function	Parameter
	Places after Decimal Point	0
	Write Protection	always
	Conversion Factor	1
Unit		n/a
Minimum Value		n/a
Maximum Value		n/a
Scaling / Resolution of LSB		1